



Plant : CCM 3

Machine : General PLC - Hydraulic & Discharge RIO (MCC Room)

Customer : TATA STEEL LIMITED, GAMHARIA

Specification : General PLC - Hydraulic & Discharge RIO (MCC Room)

Order number : 2300075403/J11, 09.07.2025

CAD - Number : ADPL/25-26/045/TSLG/01

Job number : 25-26/045/PJ/TSLG

Client documents no. :

Status V1 - 01

Preliminary

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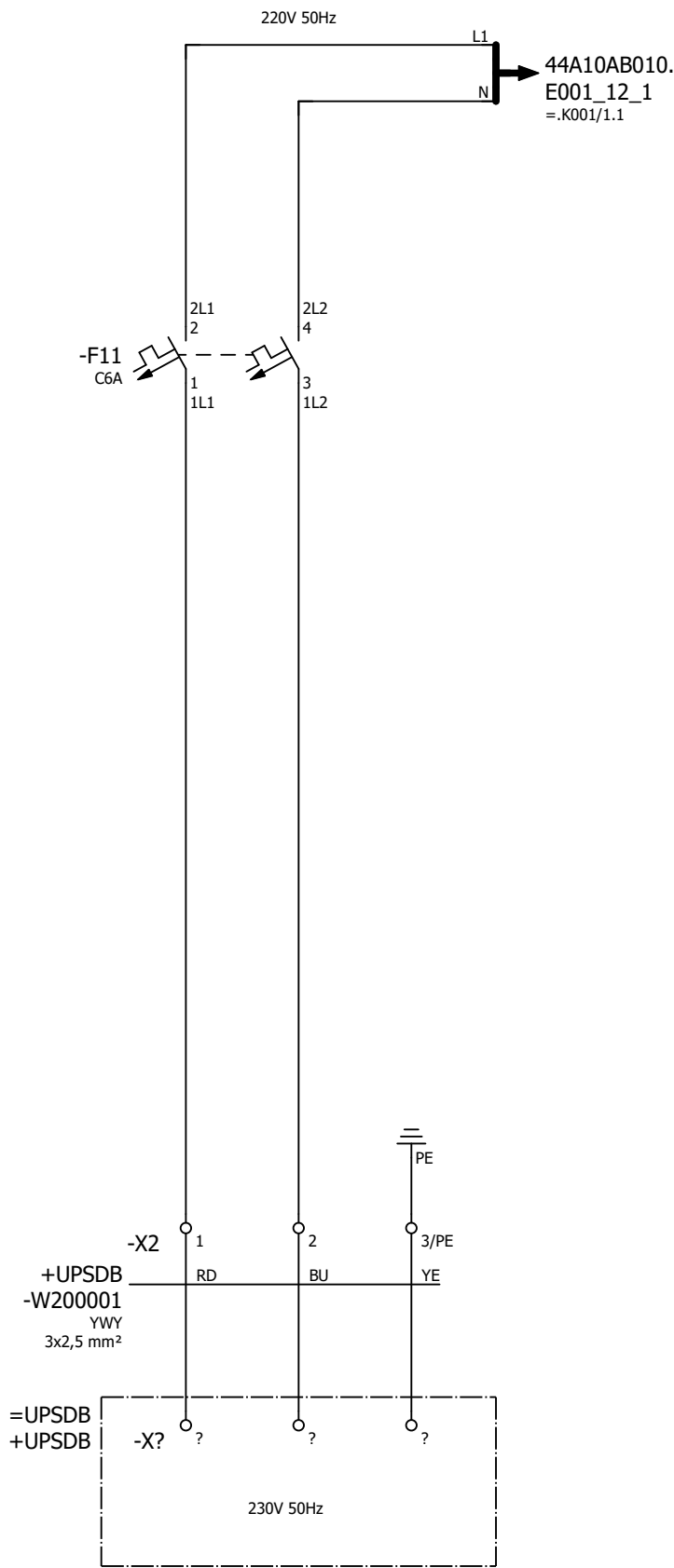
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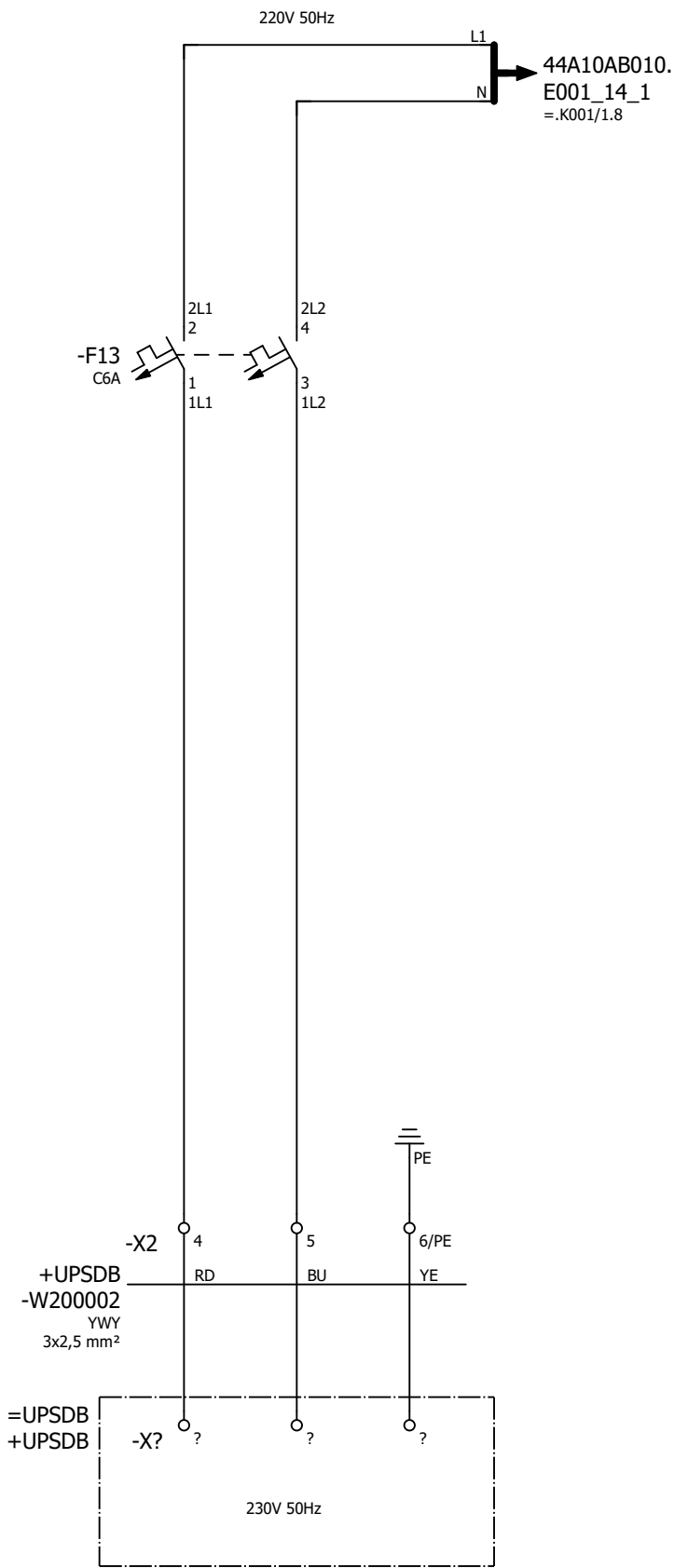
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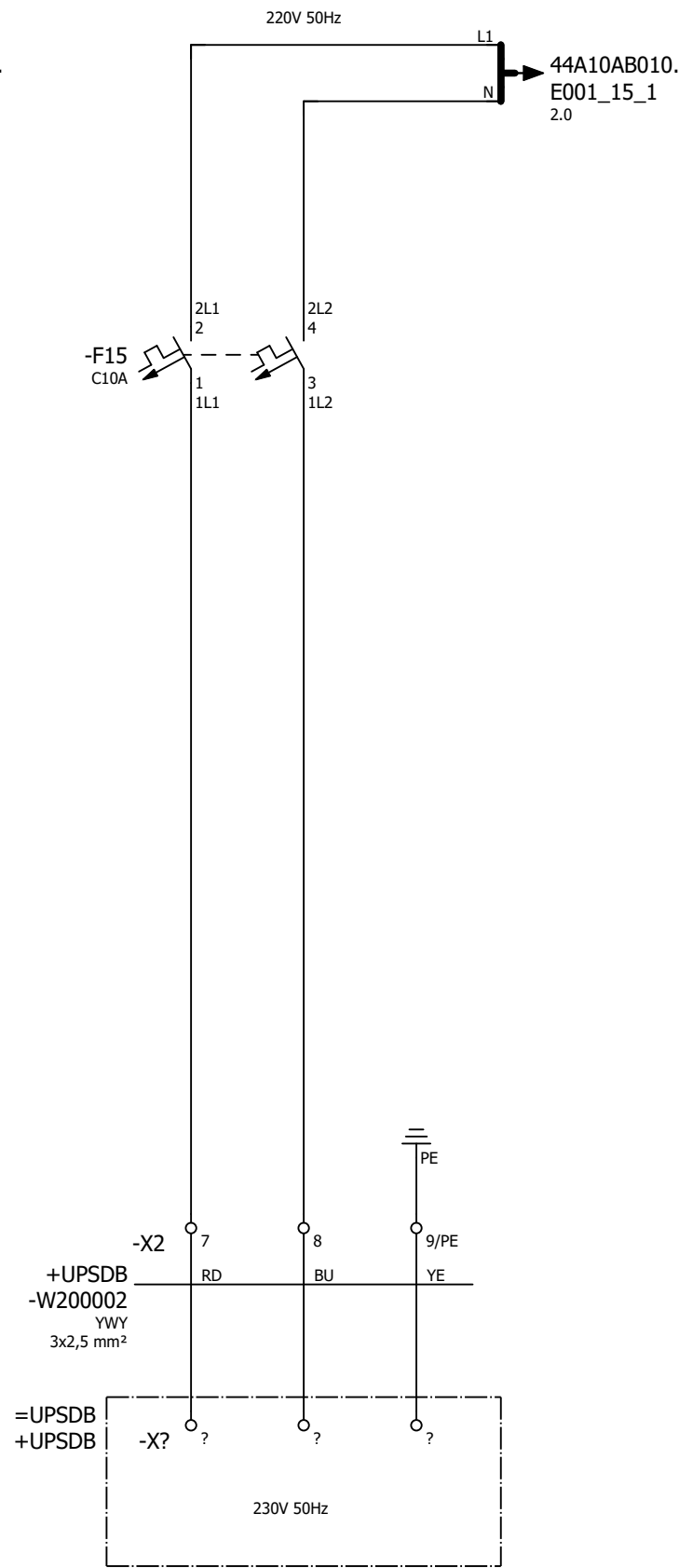
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UPS-1 Supply
230V 50Hz
power supply-1



UPS-2 Supply
230V 50Hz
power supply-2



UPS Supply-1
230V 50Hz
For IO Cards

=44A10AA000.A003/1

Revision	V1 -01	Approved	
Department	E&I	Checked	
Edited by	CHRU02	Designed	
Date	29.08.2025		



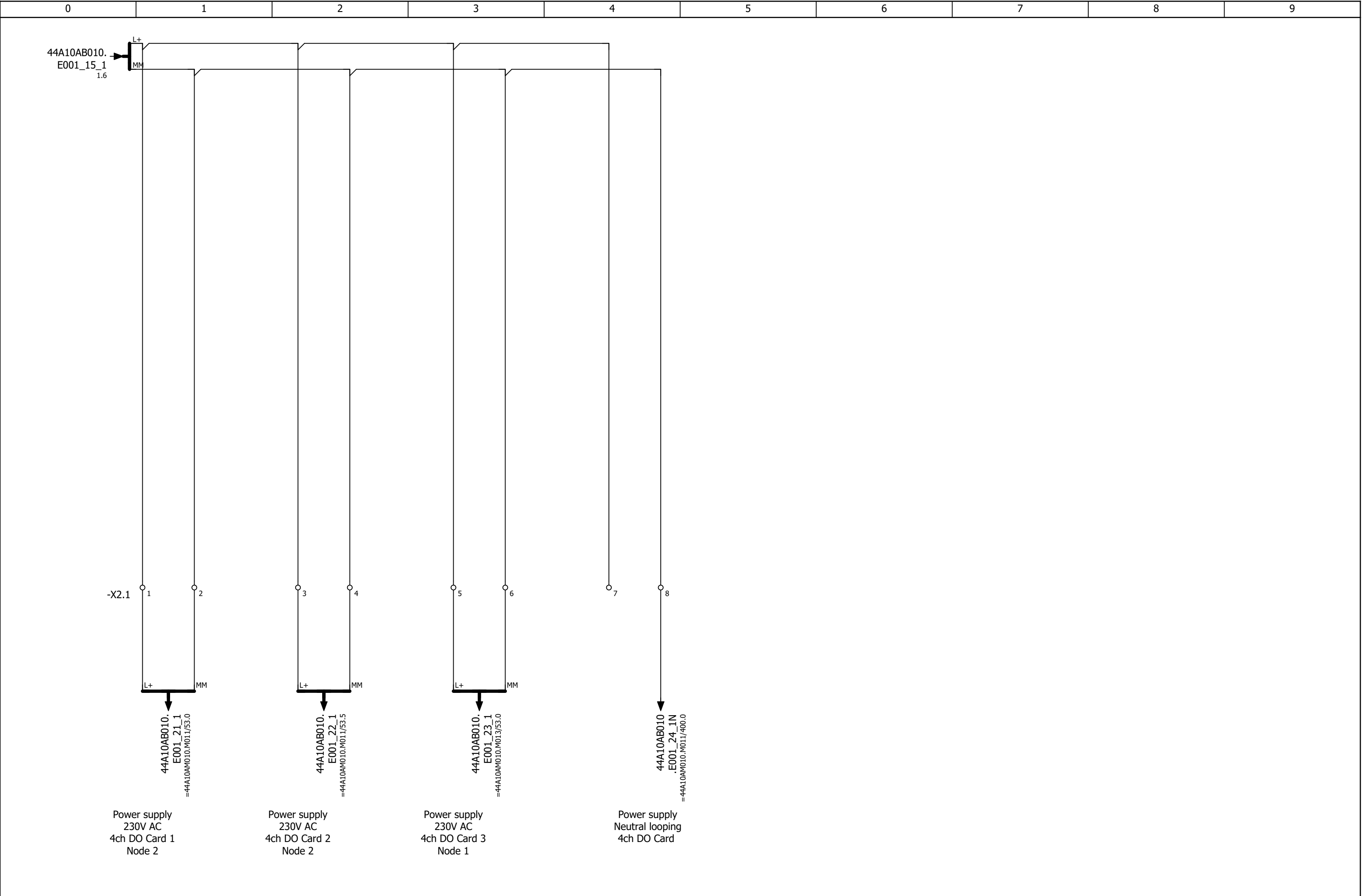
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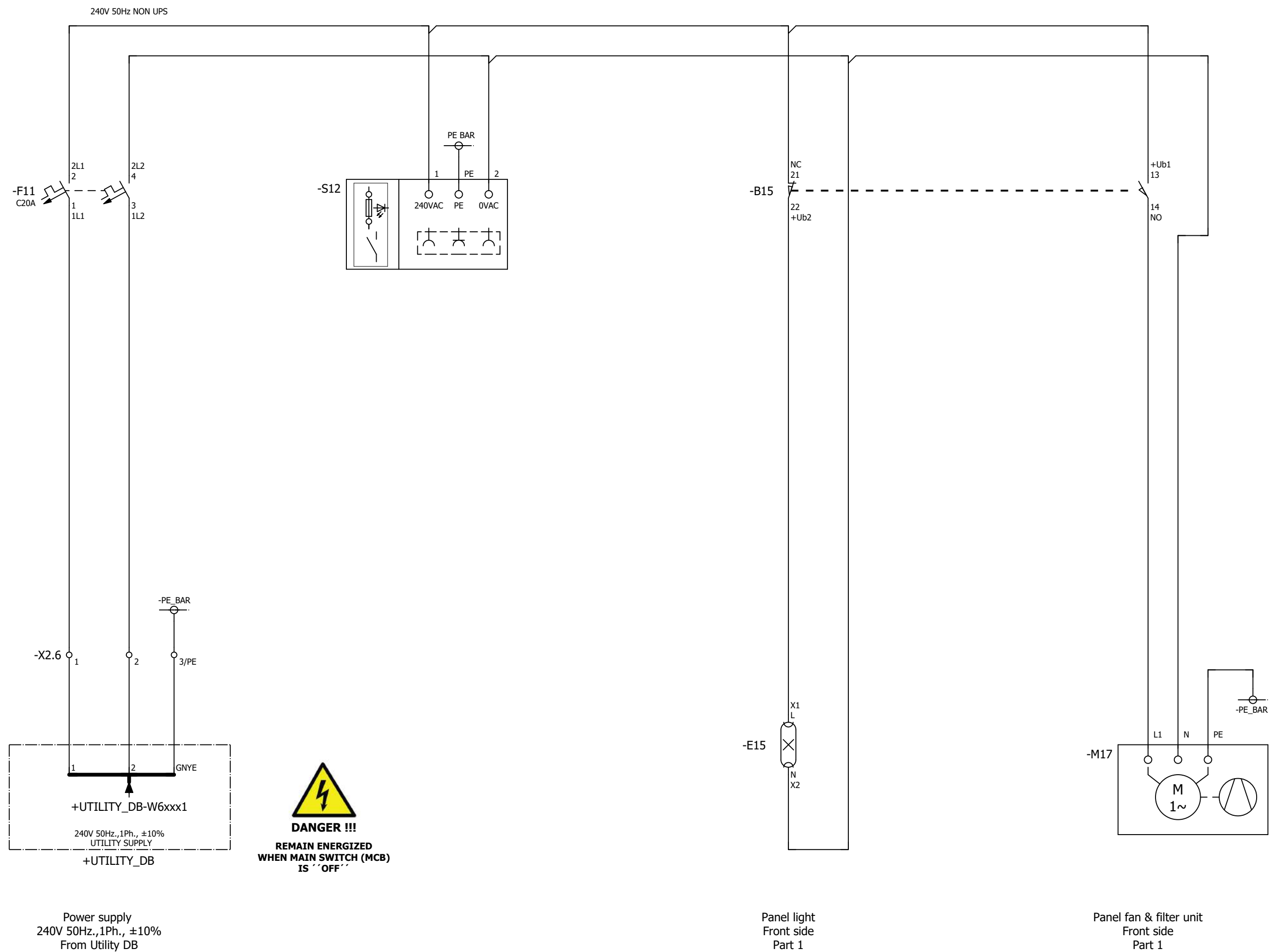
TATA STEEL LIMITED, GAMHARIA

General PLC - Hydraulic & Discharge RIO (MCC Room)

UPS Supply incomer
230V 50Hz

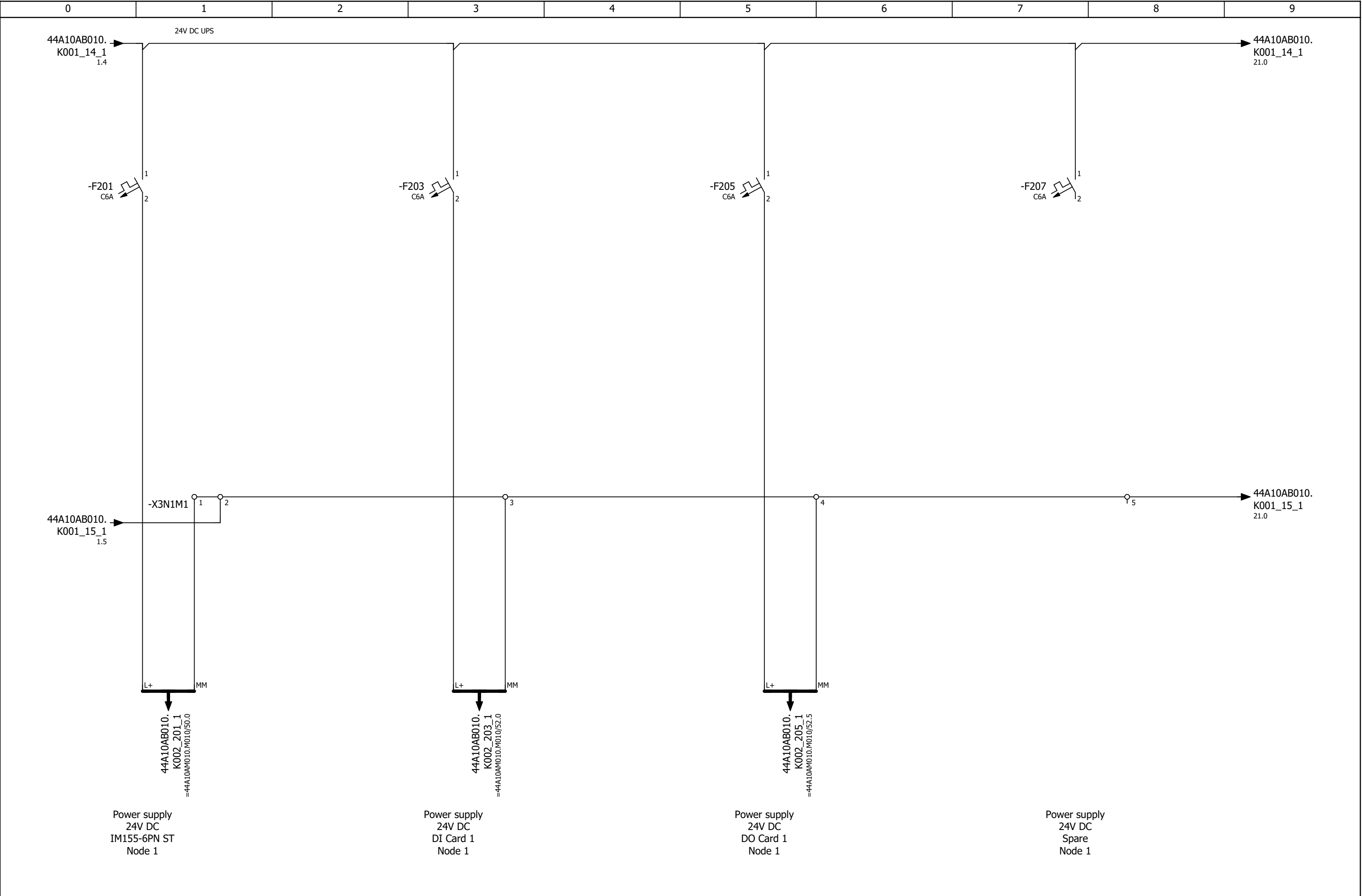
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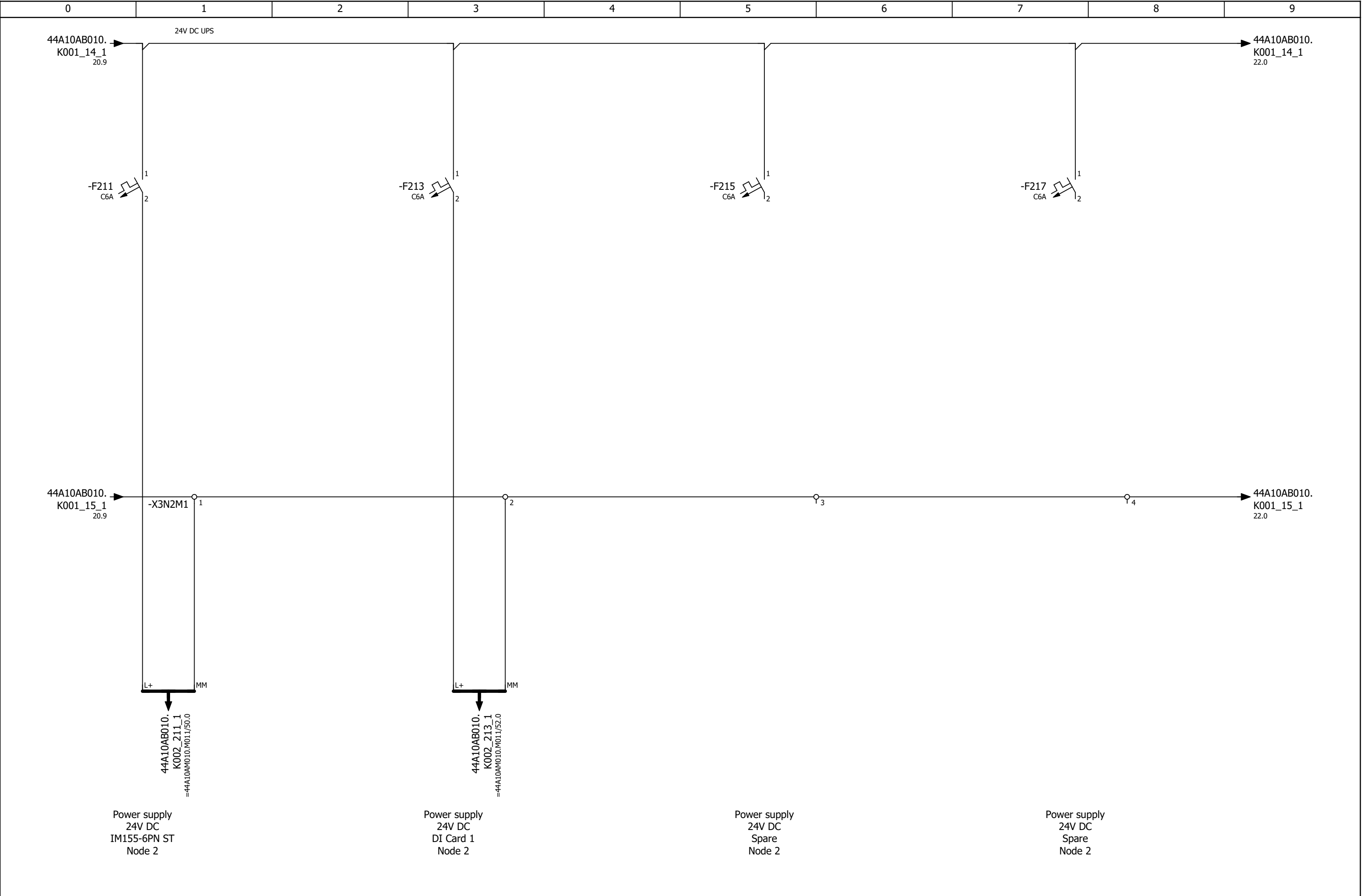


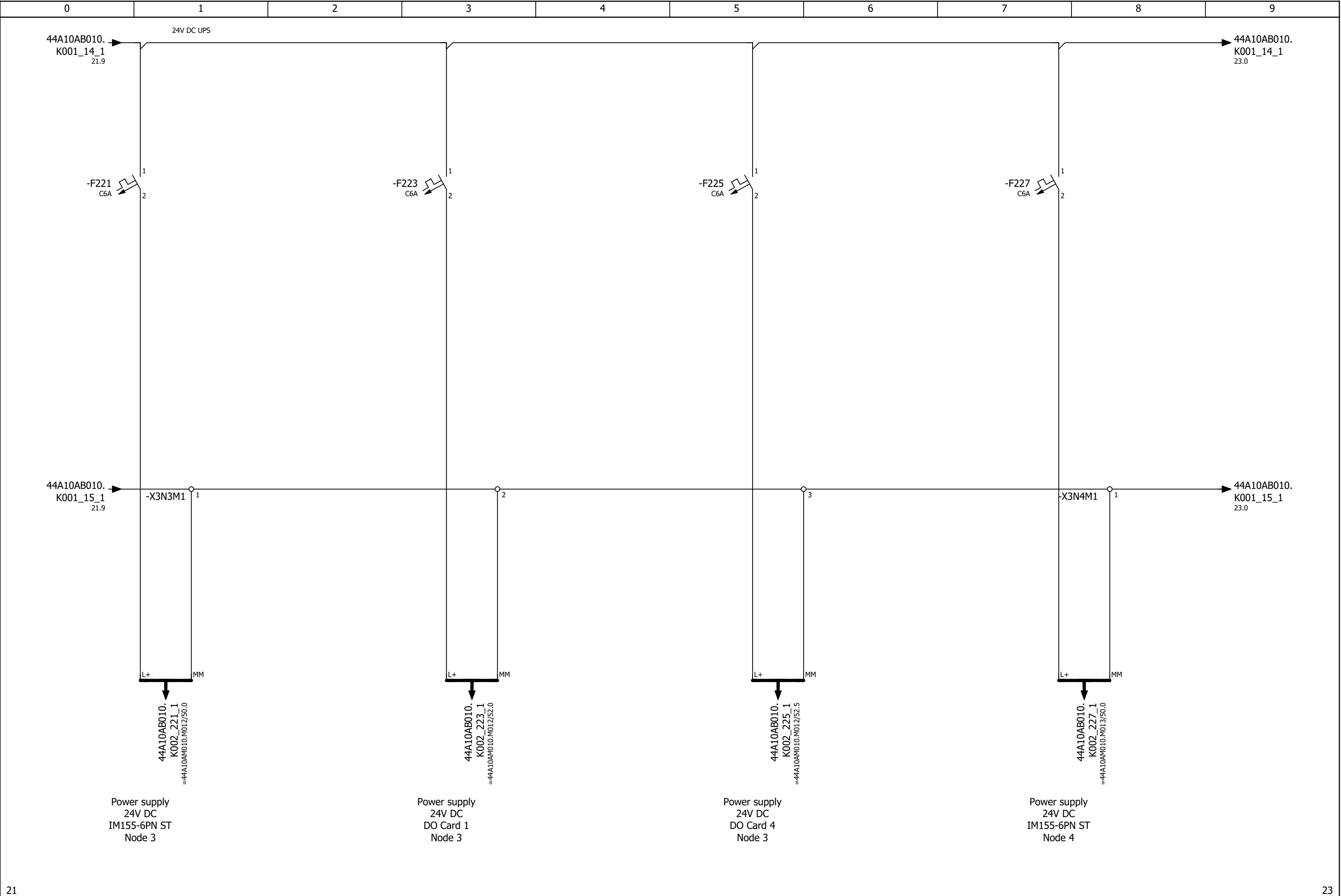


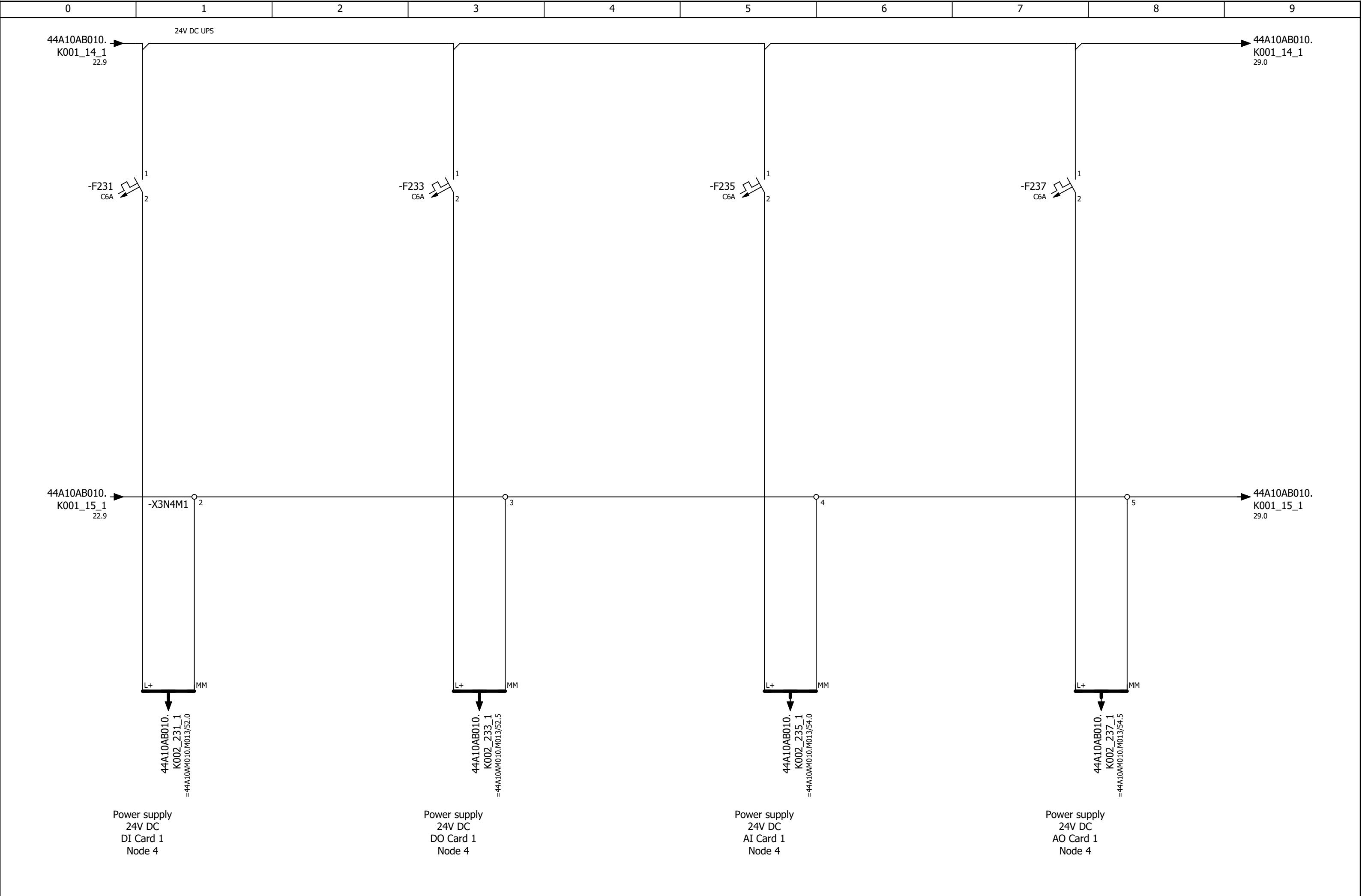
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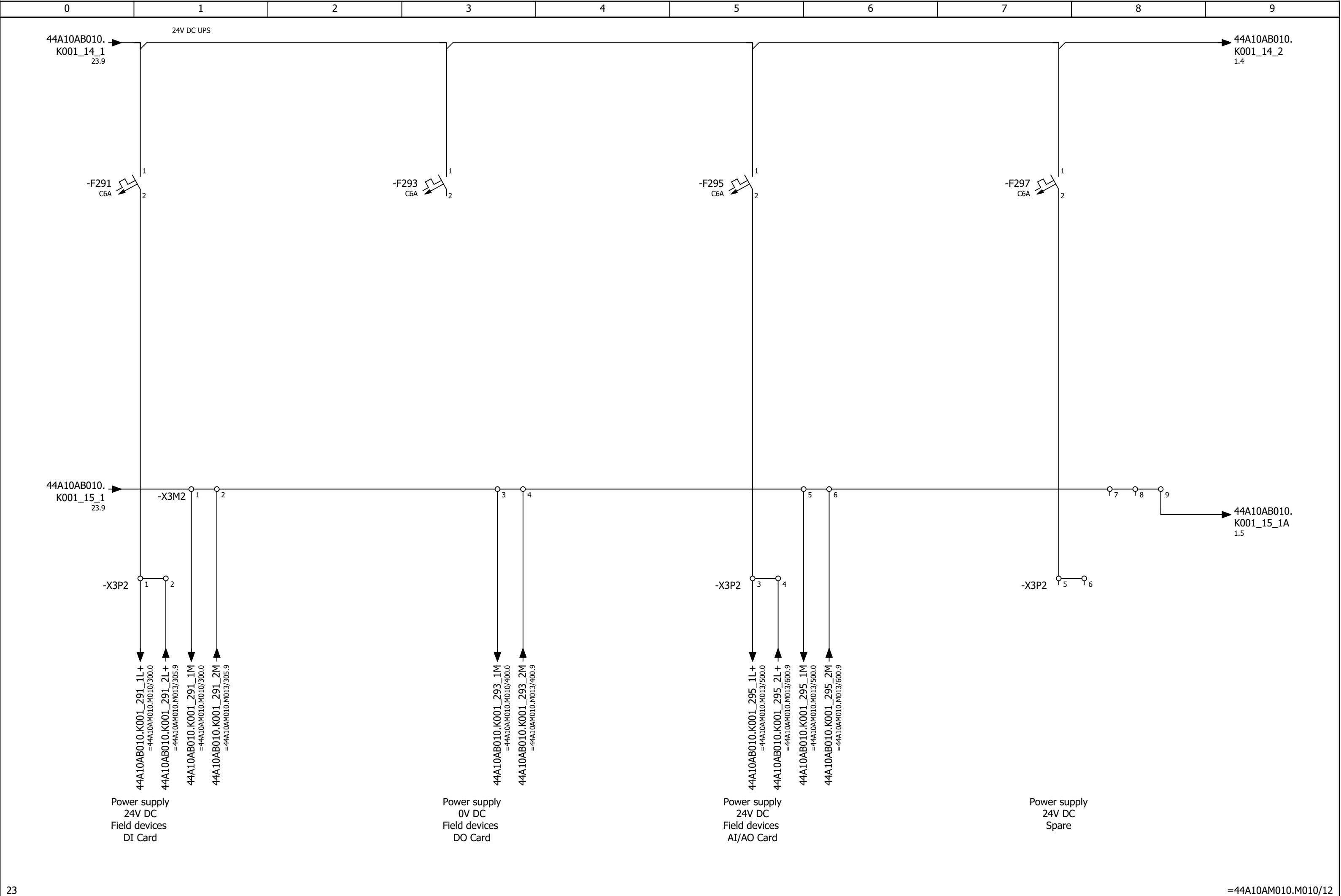
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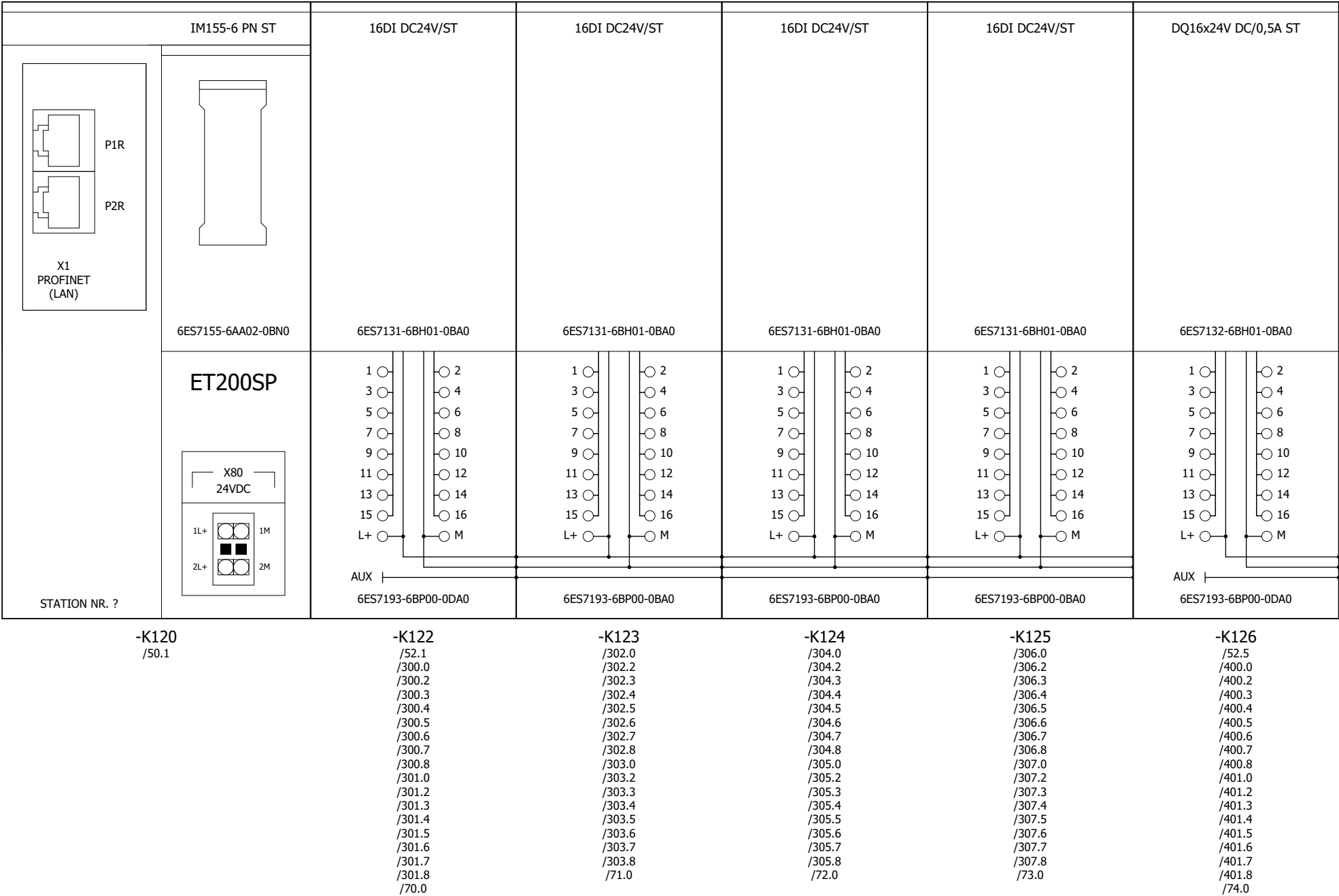


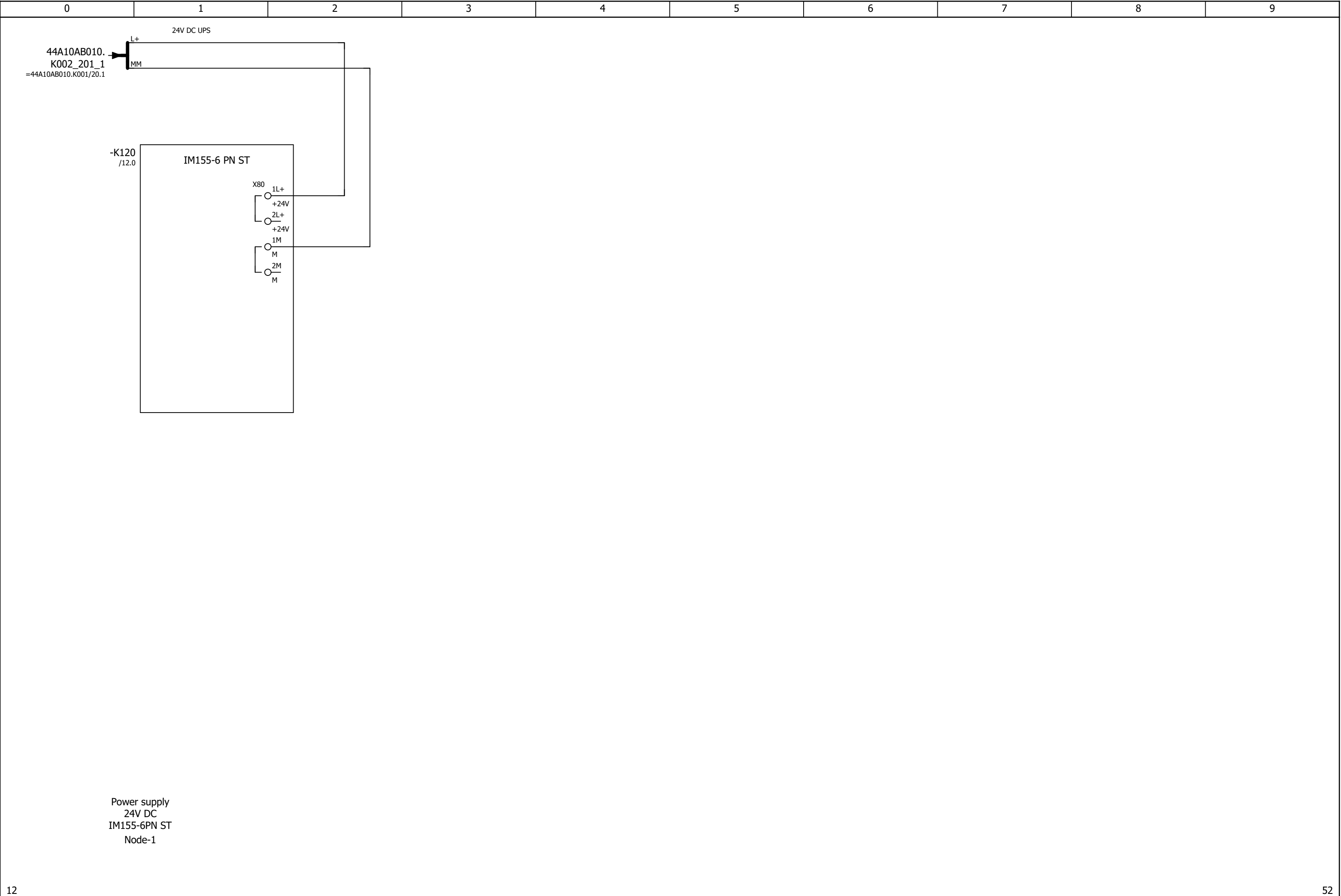


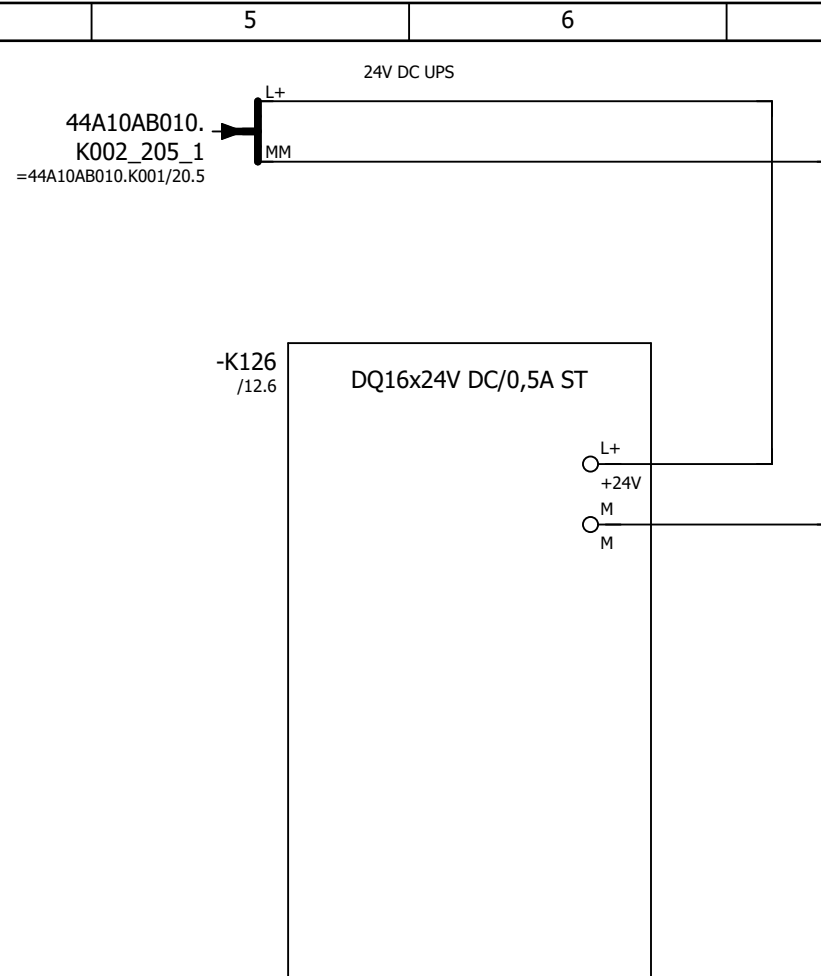
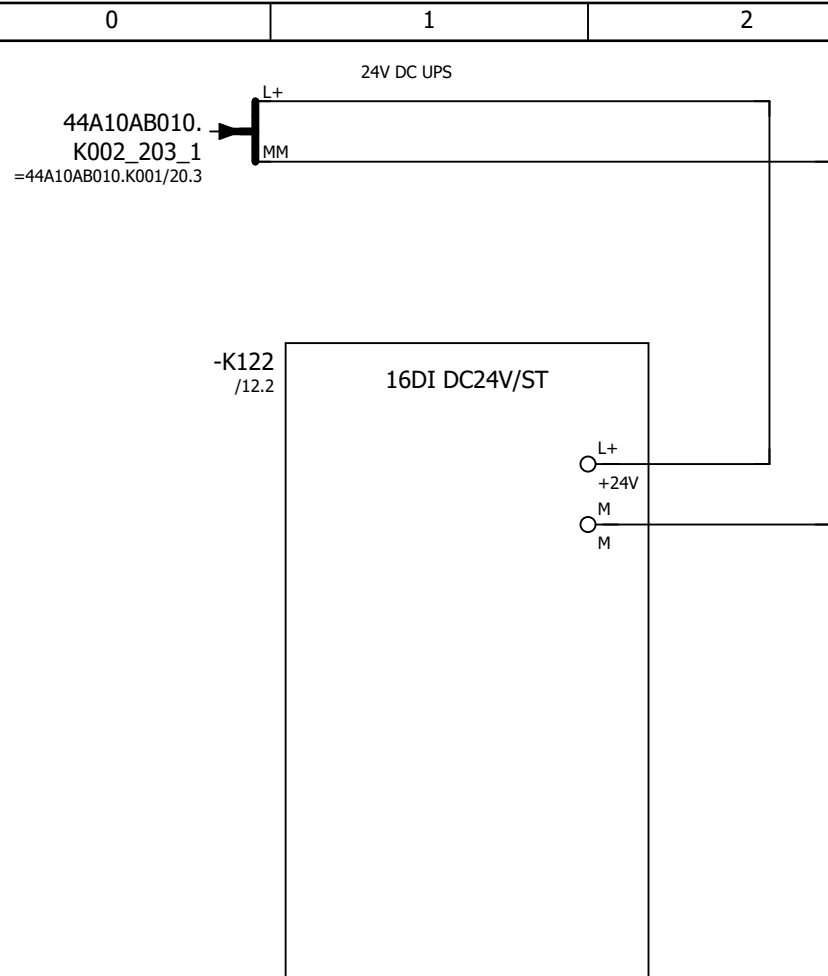


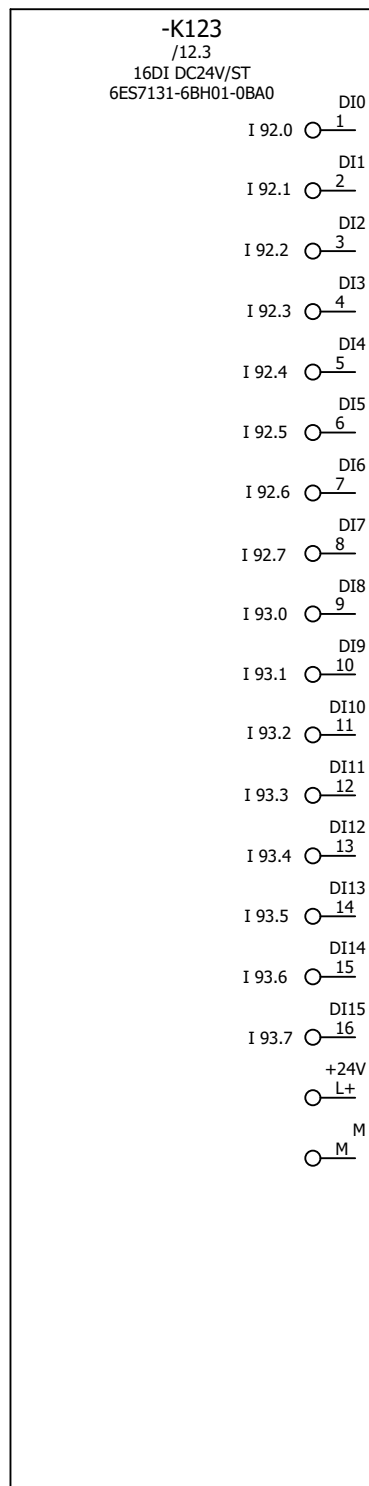




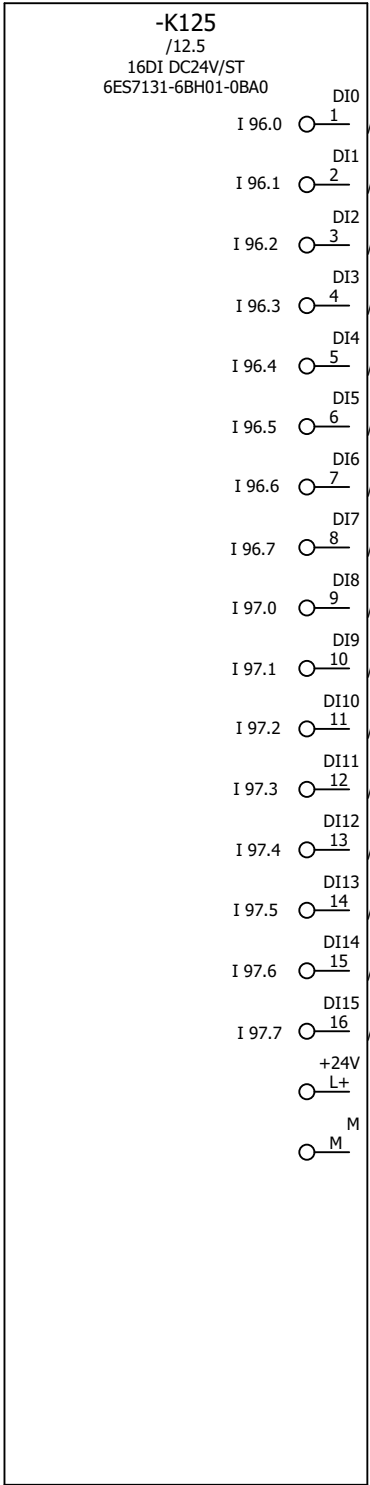








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COM_OC_LCS74_LAMP_TEST	Local control station LCS74 lamp test
COM_OC_LCS74_LOC_ENAB	Local control station LCS74 enable
X90_OC_P1_SEL	Lateral Liftable Transfer H.U. pump#1 select
X90_OC_P2_SEL	Lateral Liftable Transfer H.U. pump#2 select
X90_OC_P_STA	Lateral Liftable Transfer H.U. pump start
X90_OC_P_STP	Lateral Liftable Transfer H.U. pump stop
COM_OC_LCS75_EMG_STP	Local control station LCS75 emergency stop (0=stop)
COM_OC_LCS75_LAMP_TEST	Local control station LCS75 lamp test
COM_OC_LCS75_LOC_ENAB	Local control station LCS75 enable
X80_OC_P1_SEL	Run out area H.U. pump#1 select
X80_OC_P2_SEL	Run out area H.U. pump#2 select
X80_OC_P3_SEL	Run out area H.U. pump#3 select
X80_OC_P4_SEL	Run out area H.U. pump#4 select
X80_OC_P5_SEL	Run out area H.U. pump#5 select

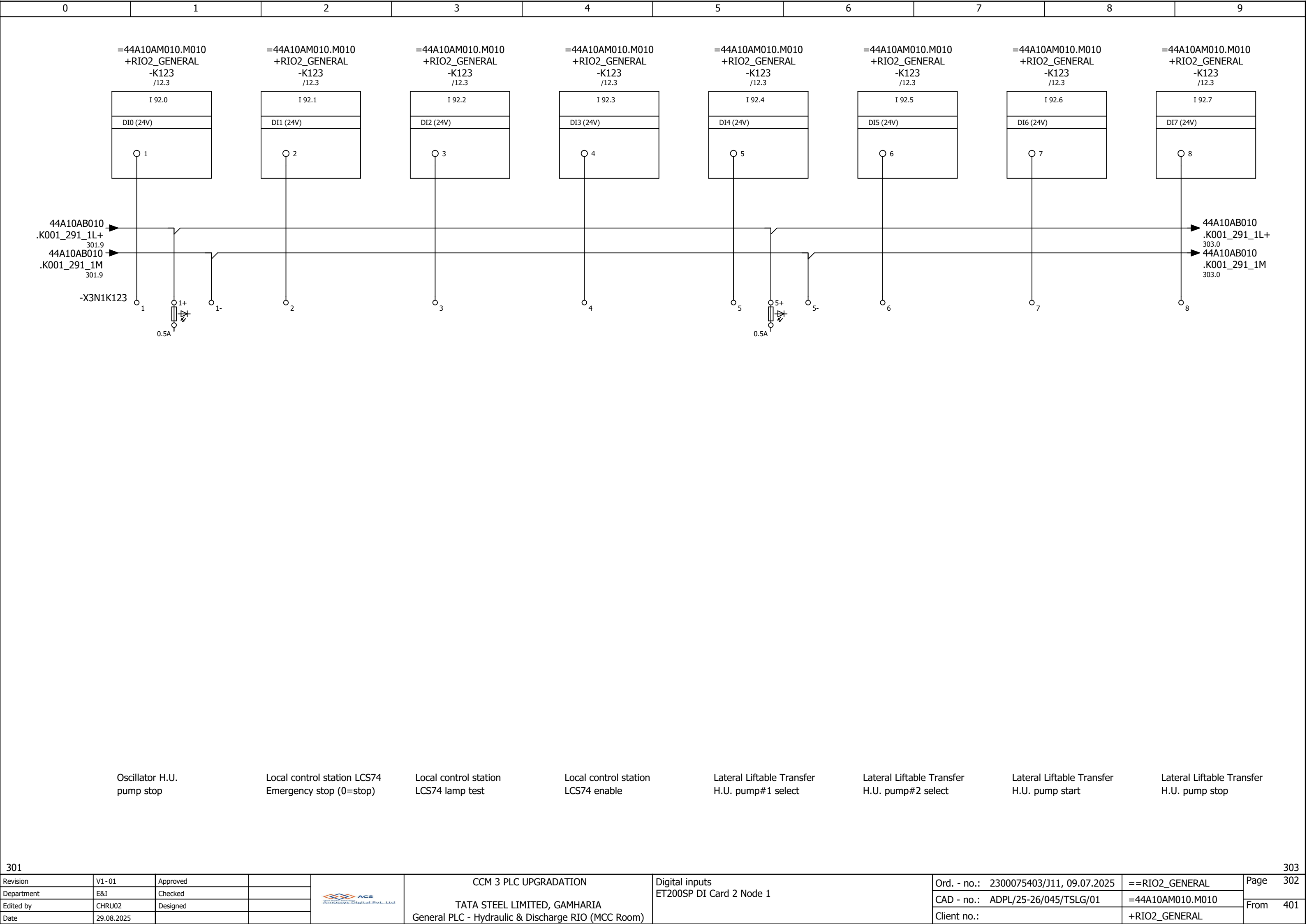


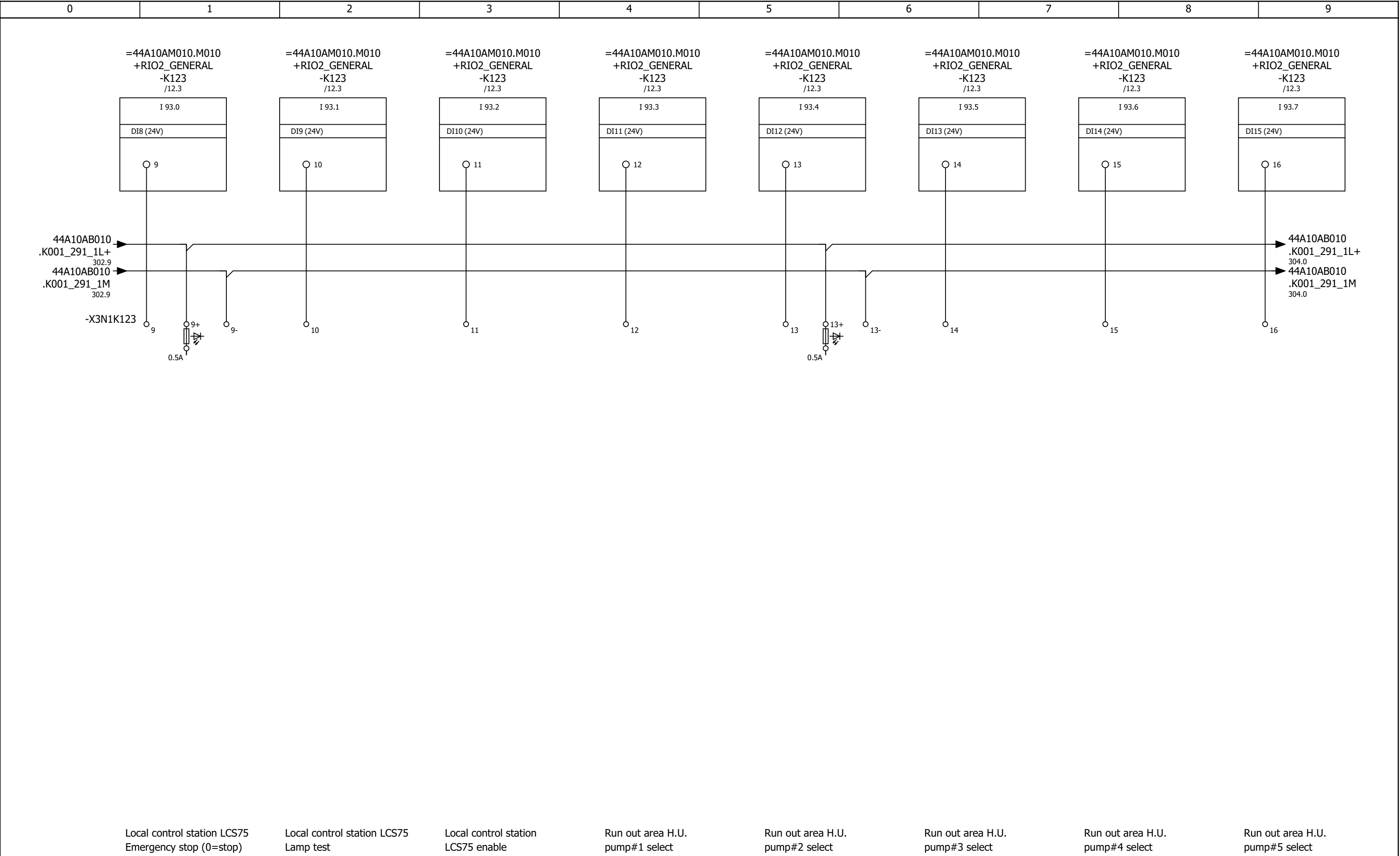
TLS_SI_PLIER2_OPN	Transfer car hoist plier 2 opened
TLS_SI_PLIER3_OPN	Transfer car hoist plier 3 opened
TLS_SI_PLIER4_OPN	Transfer car hoist plier 4 opened
TLS_SI_PLIER5_OPN	Transfer car hoist plier 5 opened
TLS_SI_PLIER6_OPN	Transfer car hoist plier 6 opened
TLS_SI_PLIER7_OPN	Transfer car hoist plier 7 opened
TLS_SI_PLIER1_CLS	Transfer car hoist plier 1 closed
TLS_SI_PLIER2_CLS	Transfer car hoist plier 2 closed
TLS_SI_PLIER3_CLS	Transfer car hoist plier 3 closed
TLS_SI_PLIER4_CLS	Transfer car hoist plier 4 closed
TLS_SI_PLIER5_CLS	Transfer car hoist plier 5 closed
TLS_SI_PLIER6_CLS	Transfer car hoist plier 6 closed
TLS_SI_PLIER7_CLS	Transfer car hoist plier 7 closed
Spare	Spare
STR1_POS_PS	LLT over Strand 1 proximity swich
STR2_POS_PS	LLT over Strand 2 proximity swich

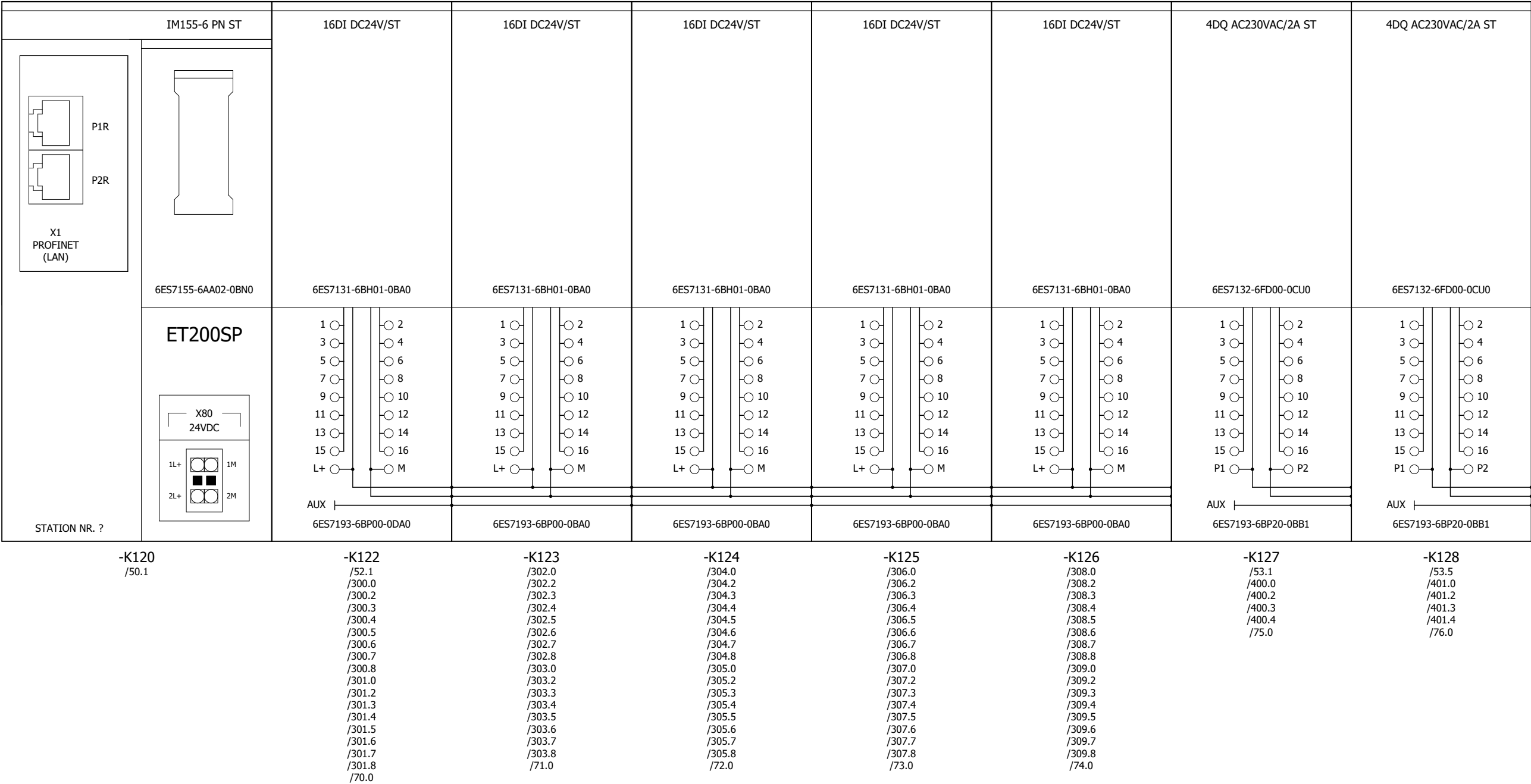


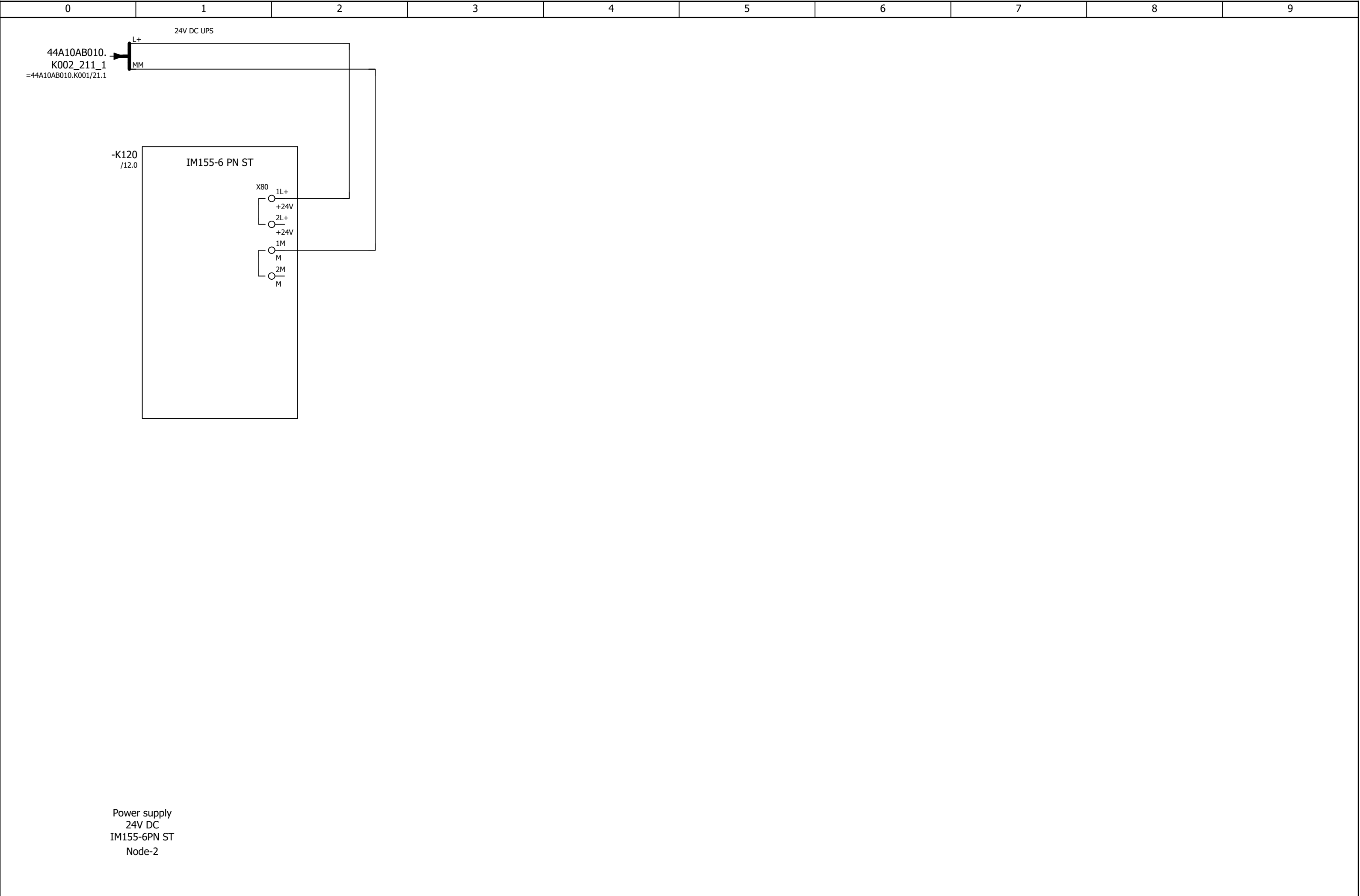
0	1	2	3	4	5	6	7	8	9
-K126 /12.6 DQ16x24V DC/0,5A ST 6ES7132-6BH01-0BA0 DQ0 Q 80.0 1 /400.1 DQ1 Q 80.1 2 /400.2 DQ2 Q 80.2 3 /400.3 DQ3 Q 80.3 4 /400.4 DQ4 Q 80.4 5 /400.5 DQ5 Q 80.5 6 /400.6 DQ6 Q 80.6 7 /400.7 DQ7 Q 80.7 8 /400.9 DQ8 Q 81.0 9 /401.1 DQ9 Q 81.1 10 /401.2 DQ10 Q 81.2 11 /401.3 DQ11 Q 81.3 12 /401.4 DQ12 Q 81.4 13 /401.5 DQ13 Q 81.5 14 /401.6 DQ14 Q 81.6 15 /401.7 DQ15 Q 81.7 16 /401.9 +24V L+ /52.6 M M /52.6 COM_OS_LCS72_EMG_STP Local control station LCS72 emergency stop active COM_OS_LCS72_LOC_ENAB Local control station LCS72 enable X70_OS_P_RUN Main H.U. pump running X70_OS_P_STP Main H.U. pump stopped COM_OS_LCS73_EMG_STP Local control station LCS73 emergency stop active COM_OS_LCS73_LOC_ENAB Local control station LCS73 enable X65_OS_P_RUN Oscillator H.U. pump running X65_OS_P_STP Oscillator H.U. pump stopped COM_OS_LCS74_EMG_STP Local control station LCS74 emergency stop active COM_OS_LCS74_LOC_ENAB Local control station LCS74 enable X90_OS_P_RUN Lateral Lifiable Transfer H.U. pump running X90_OS_P_STP Lateral Lifiable Transfer H.U. pump stopped COM_OS_LCS75_EMG_STP Local control station LCS75 emergency stop active COM_OS_LCS75_LOC_ENAB Local control station LCS75 enable X80_OS_P_RUN Run out area H.U. pump running X80_OS_P_STP Run out area H.U. pump stopped									

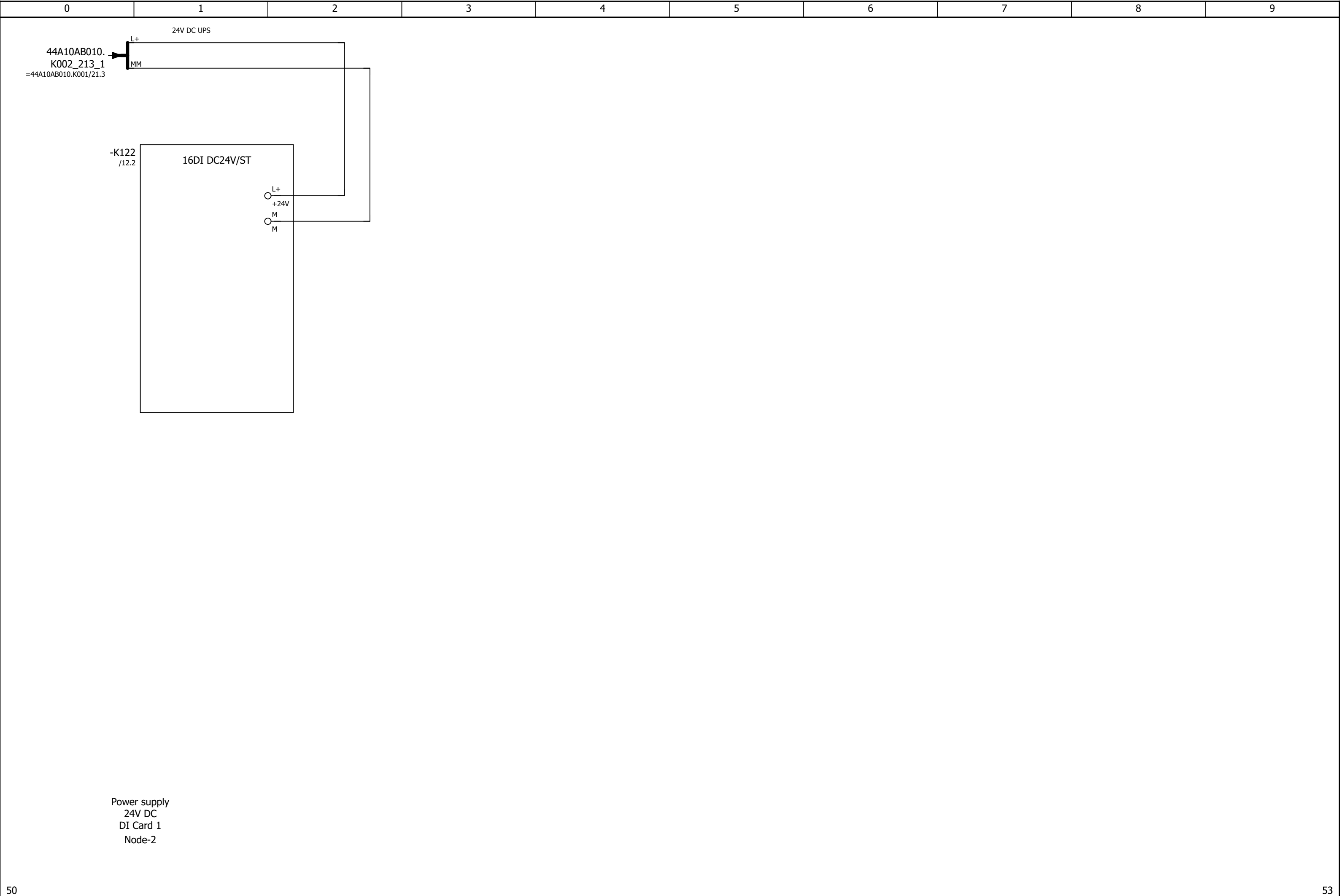
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Revision	V1 -01	Approved			CCM 3 PLC UPGRADATION TATA STEEL LIMITED, GAMHARIA General PLC - Hydraulic & Discharge RIO (MCC Room)	Overview ET200SP DO Cards	Ord. - no.:	2300075403/J11, 09.07.2025	==RIO2_GENERAL	Page	74
Department	E&I	Checked					CAD - no.:	ADPL/25-26/045/TSLG/01	=44A10AM010.M010	From	401
Edited by	CHRU02	Designed					Client no.:				
Date	29.08.2025										

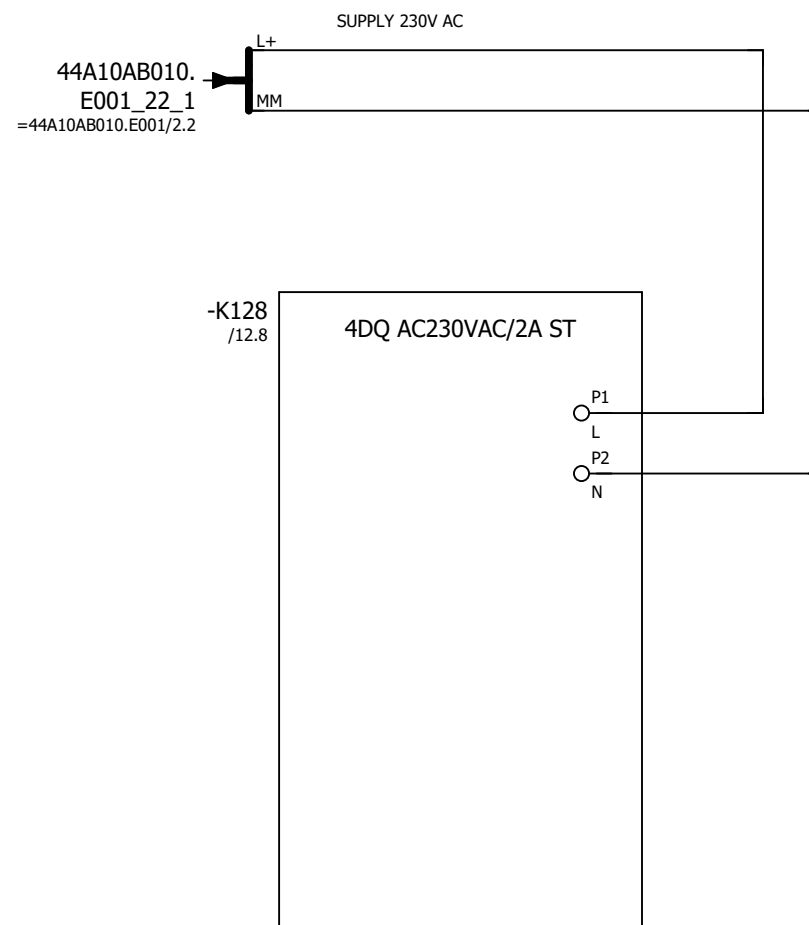
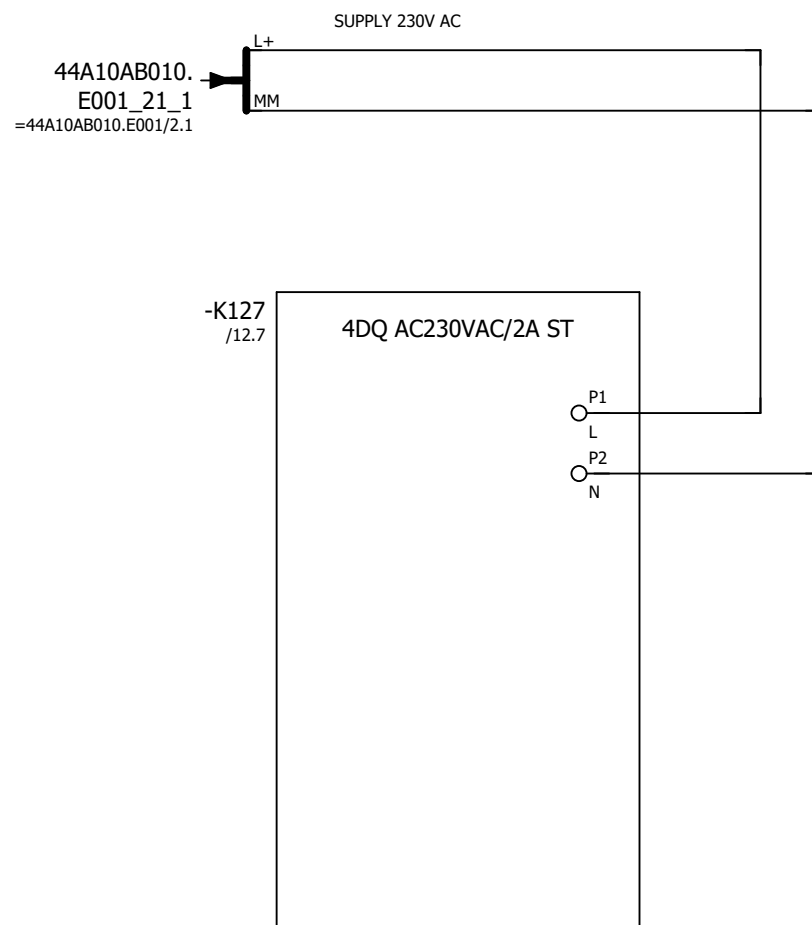


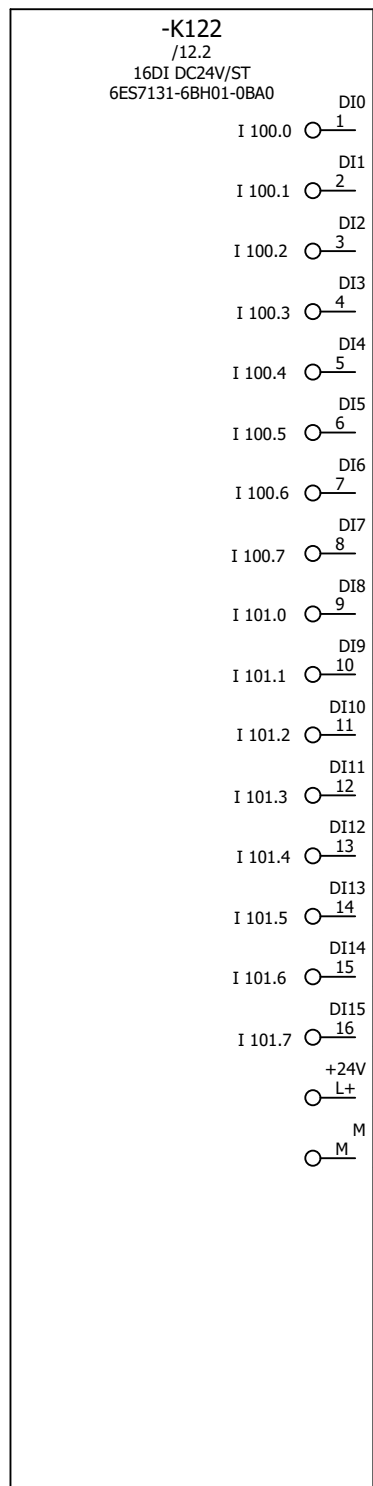




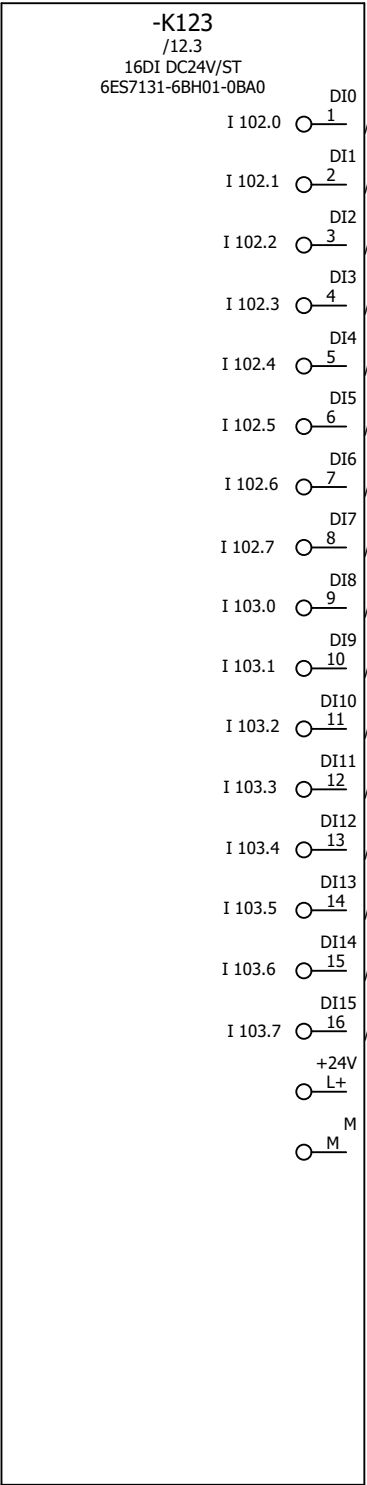








F10PWI_SP_OIL_F_CLOG	Oil filter clogged (0=clogged)
F10PWI_SI_CYL_A1_TOP_PO	Cyl.#A1 top(high) position
F10PWI_SI_CYL_A1_BOT_PO	Cyl.#A1 bottom(low) position
F10PWI_SI_CYL_A4_TOP_PO	Cyl.#A4 top(high) position
F10PWI_SI_CYL_A4_BOT_PO	Cyl.#A4 bottom(low) position
F10PWI_SI_CYL_A_SDWN	Lowering slowdown - Cyl.#4
F10PWI_SI_CYL_B1_TOP_PO	Cyl.#B1 top(high) position
F10PWI_SI_CYL_B1_BOT_PO	Cyl.#B1 bottom(low) position
F10PWI_SI_CYL_B4_TOP_PO	Cyl.#B4 top(high) position
F10PWI_SI_CYL_B4_BOT_PO	Cyl.#B4 bottom(low) position
F10PWI_SI_CYL_B_SDWN	Lowering slowdown - Cyl.#2
F10PWI_SP_MAT_DET_ETR	Material detection on w.b. entry side
F10PWI_SP_MAT_DET_EXT	Material detection on w.b. exit side
F10PWI_SP_FDIV_HIGH	Walking beam flow divider end stroke (W.B. high)
F10PWI_SP_FDIV_LOW	Walking beam flow divider end stroke (W.B. low)
F10PWI_SI_FD_FWD_POS	Feeding device in advance position

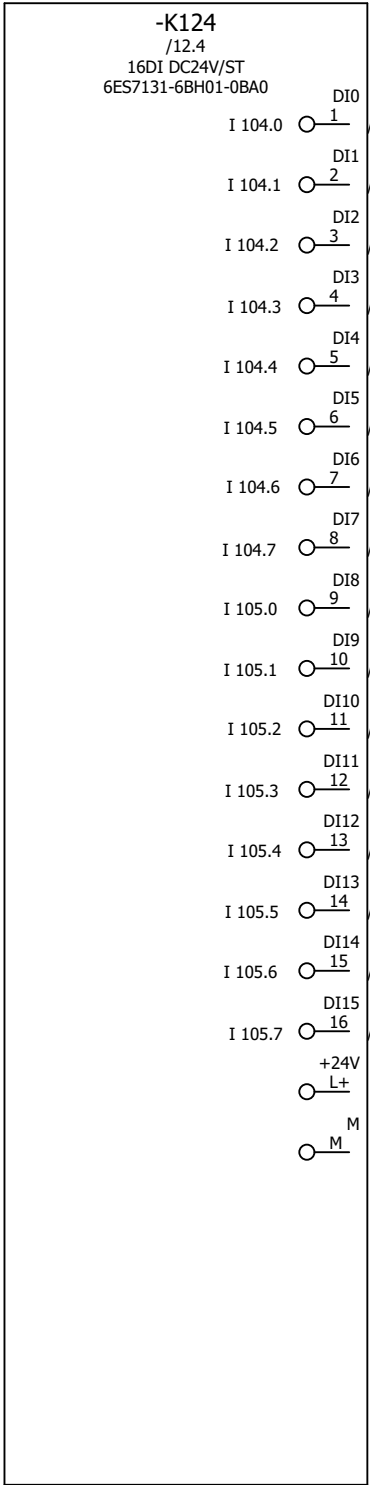


F10PWI_SI_FD_REV_POS	Feeding device in reverse position
F10PRS_SI_FWD_POS	Pusher Collecting Table forward position
F10PRS_SI_REV_POS	Pusher Collecting Table reverse position
F10PRC_SI_FWD_POS	Pawls Collecting Table forward position
F10PRC_SI_REV_POS	Pawls Collecting Table reverse position
F10PRC_SI_INTM_POS	Pawls Collecting Table intermediate position
F10PRC_SI_MAT_DET1	Pawls Collecting Table material present entry side
F10PRC_SI_MAT_PACK_RDY	Pawls Collecting Table material ready for pack
X30_SL_TK_LV_MINMIN	Ladle Turret & Tundish Car H.U. main tank oil MINMIN level
X30_SI_P1_SC_VV_OPN	Ladle Turret & Tundish Car H.U. pump 1 suction valve open
X30_SI_P2_SC_VV_OPN	Ladle Turret & Tundish Car H.U. pump 2 suction valve open
X30_SP_REC_F_CLOG	Ladle Turret & Tundish Car H.U. recirculation filter clogged
X30_SP_RET_F1_CLOG	Ladle Turret & Tundish Car H.U. return filter 1 clogged
X30_SP_RET_F2_CLOG	Ladle Turret & Tundish Car H.U. return filter 2 clogged
X65_SL_TK_LV_MINMIN	Oscillator H.U. main tank oil MINMIN level
X65_SI_P1_SC_VV_OPN	Oscillator H.U. pump 1 suction valve open

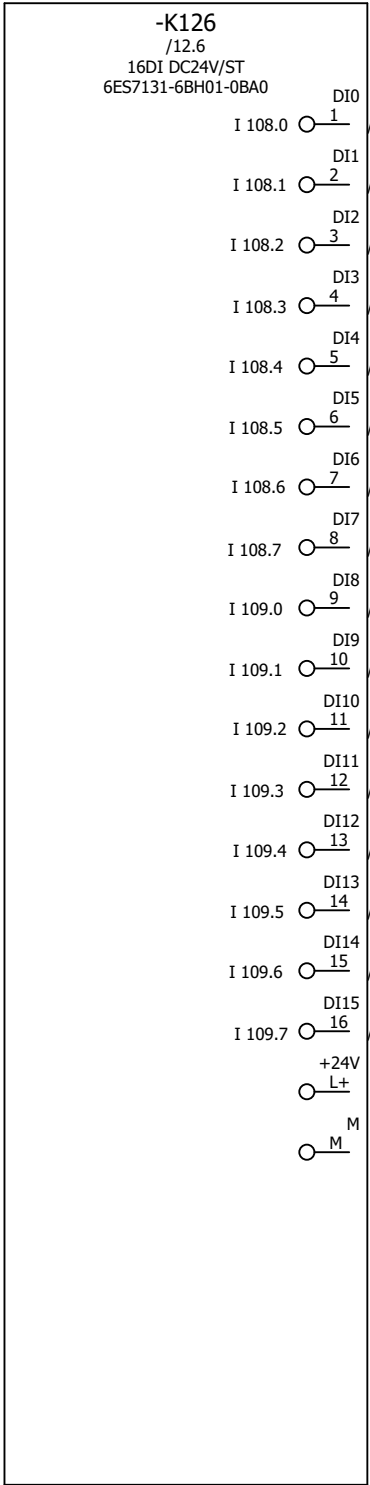
Revision	V1 -01	Approved	
Department	E&I	Checked	
Edited by	CHRU02	Designed	
Date	29.08.2025		



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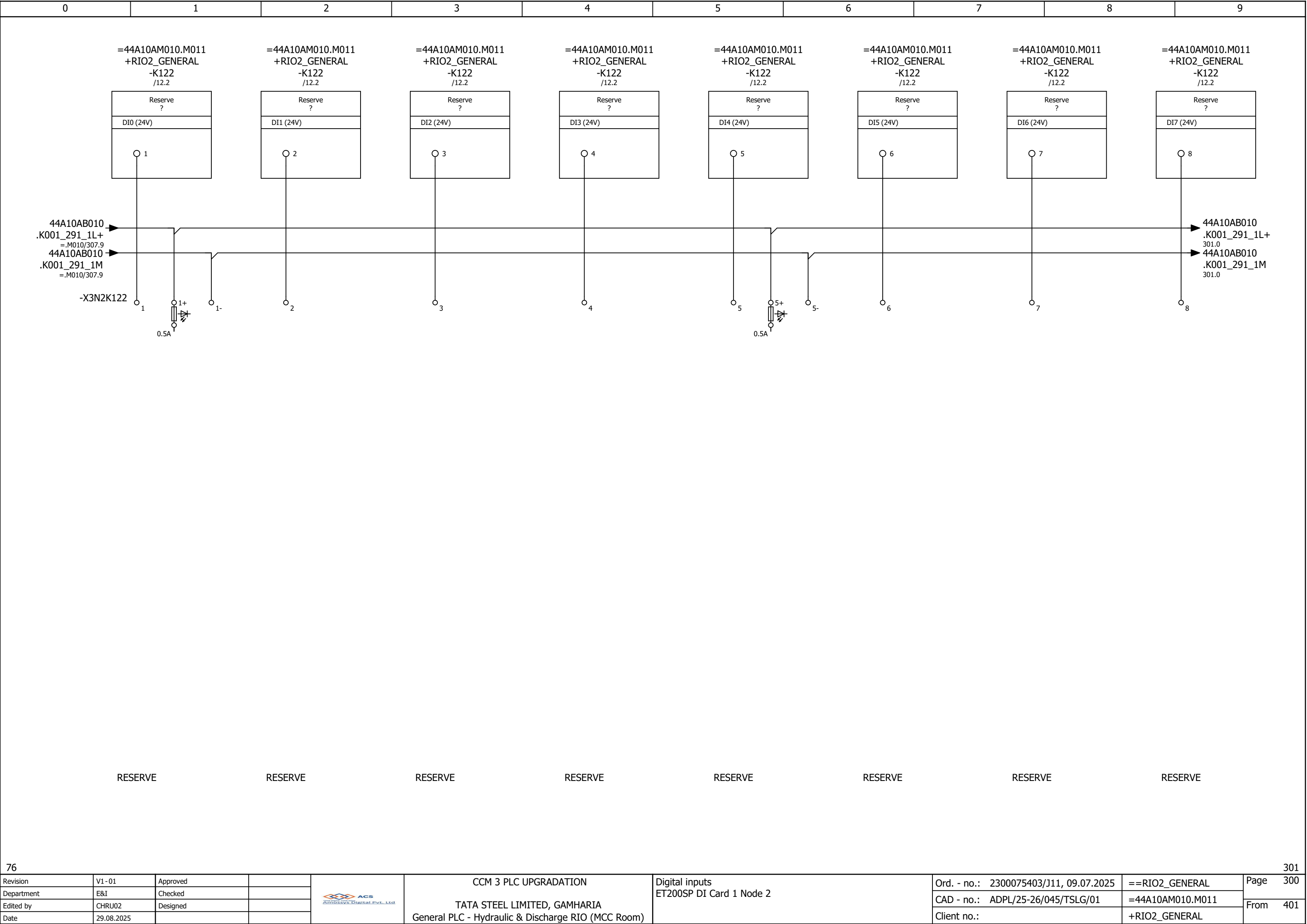


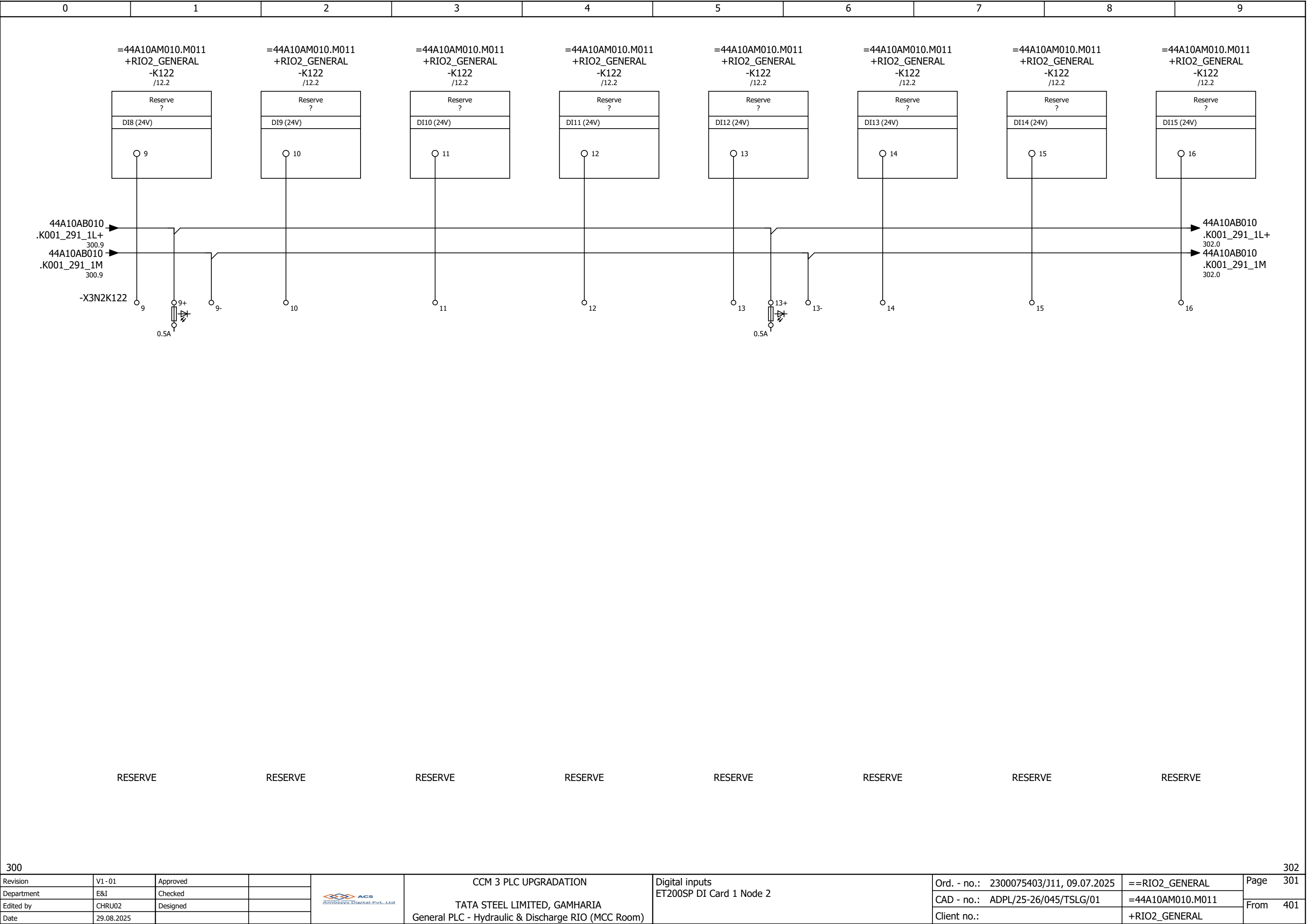
X65_SI_P2_SC_VV_OPN	Oscillator H.U. pump 2 suction valve open
X65_SI_P3_SC_VV_OPN	Oscillator H.U. pump 3 suction valve open
X65_SP_PIL_F_CLOG	Oscillator H.U. pilot line filter clogged
X65_SP_REC_F_CLOG	Oscillator H.U. recirculation filter clogged
X65_SP_RET_F1_CLOG	Oscillator H.U. return filter 1 clogged
X65_SP_RET_F2_CLOG	Oscillator H.U. return filter 2 clogged
X70_SL_TK_LV_MINMIN	Main H.U. main tank oil MINMIN level
X70_SI_P1_SC_VV_OPN	Main H.U. pump 1 suction valve open
X70_SI_P2_SC_VV_OPN	Main H.U. pump 2 suction valve open
X70_SP_REC_F_CLOG	Main H.U. recirculation filter clogged
X70_SP_P1_F_CLOG	Main H.U. pump 1 filter clogged
X70_SP_P2_F_CLOG	Main H.U. pump 2 filter clogged
X70_SP_RET_F1_CLOG	Main H.U. return filter 1 clogged
X70_SP_RET_F2_CLOG	Main H.U. return filter 2 clogged
X80_SL_TK_LV_MINMIN	Run out area H.U. main tank oil MINMIN level
X80_SI_P1_SC_VV_OPN	Run out area H.U. pump 1 suction valve open

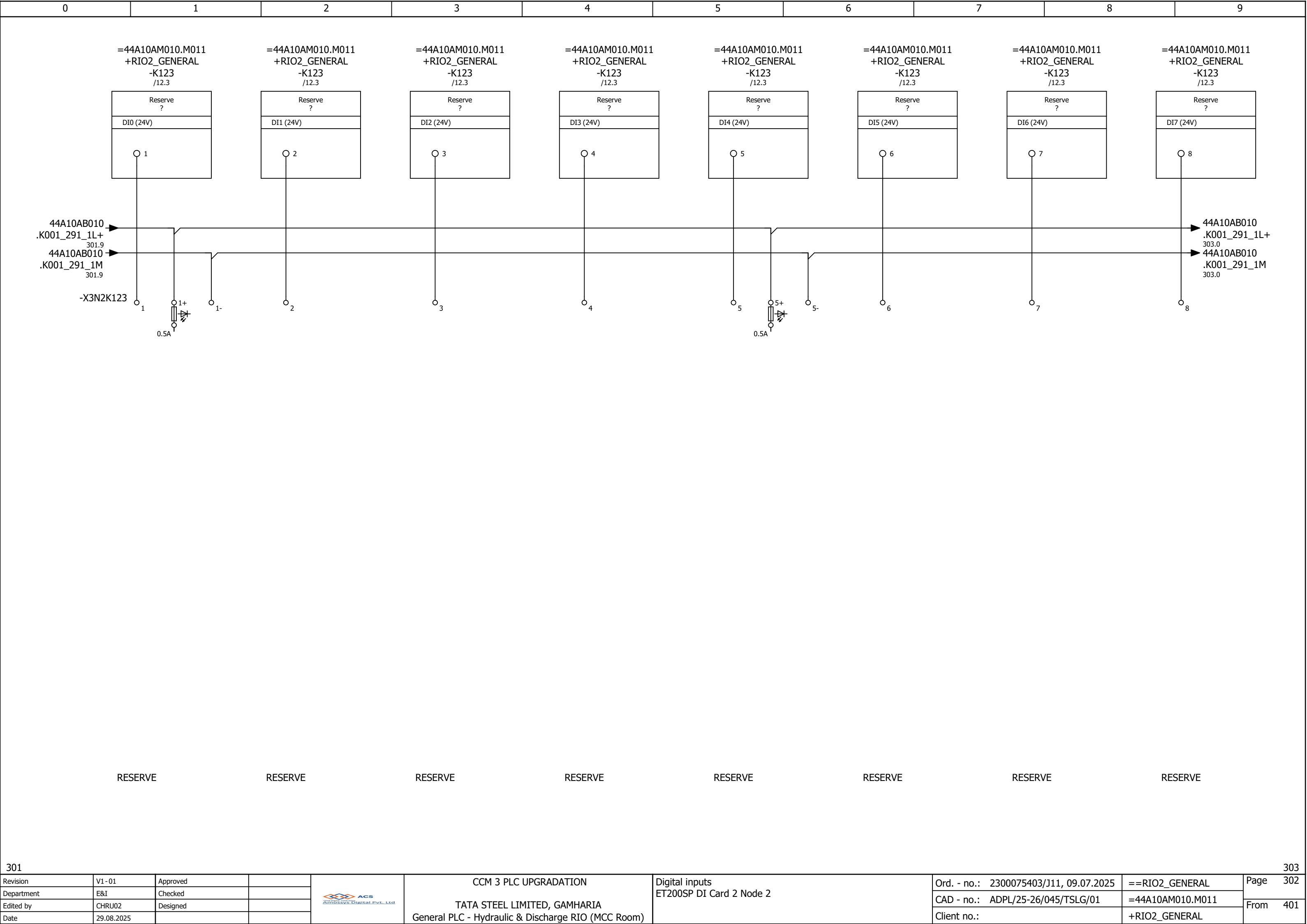


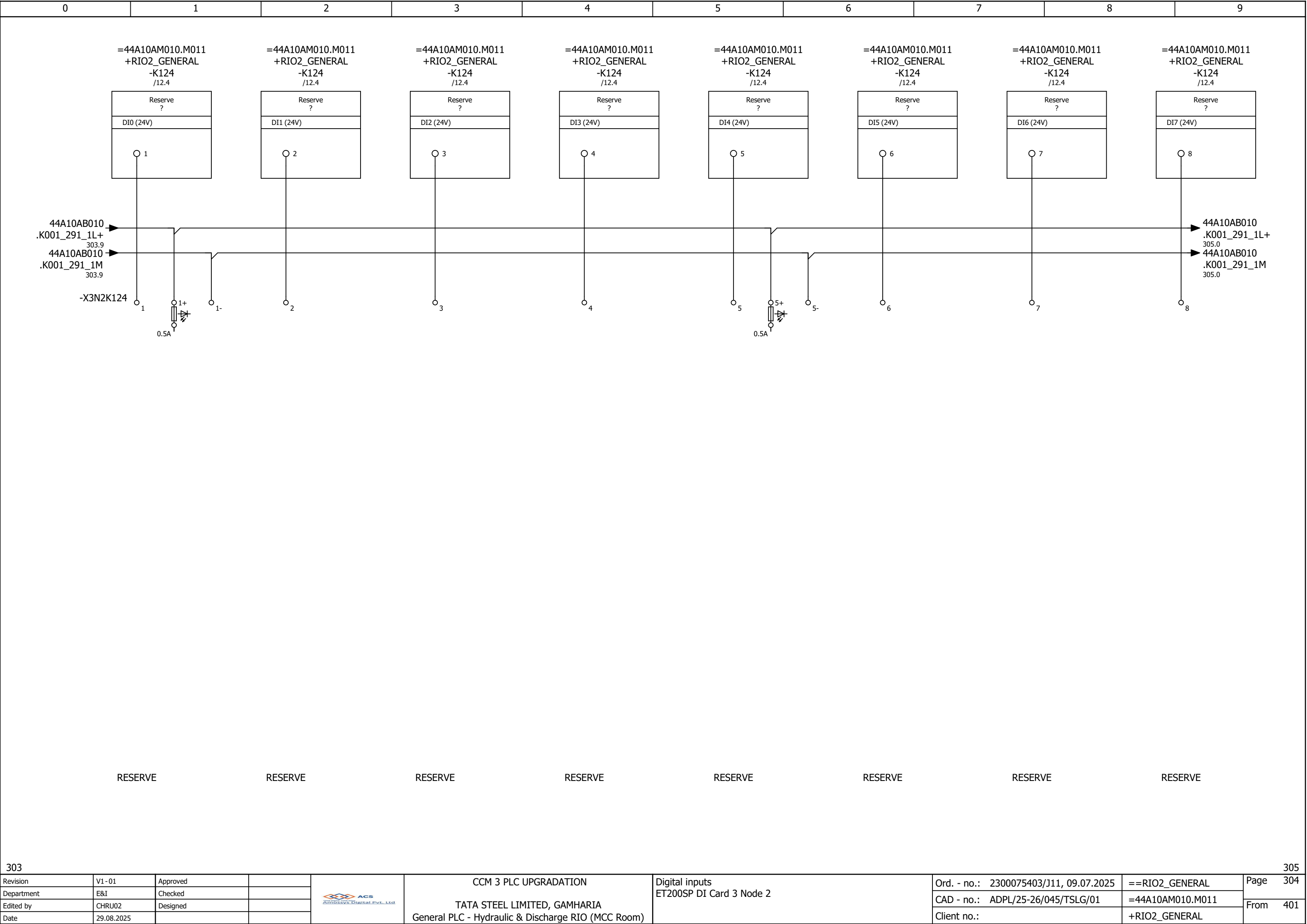
Y03CLG_SL_MIN	Grease tank MIN level (1=level above MIN)
Y03CLG_SL_MAX	Grease tank MAX level (1=level above MAX)
Y03CLG_SP_L1_END	End of line 1 pressure reached
Y03CLG_SP_L2_END	End of line 2 pressure reached
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare
Spare	Spare

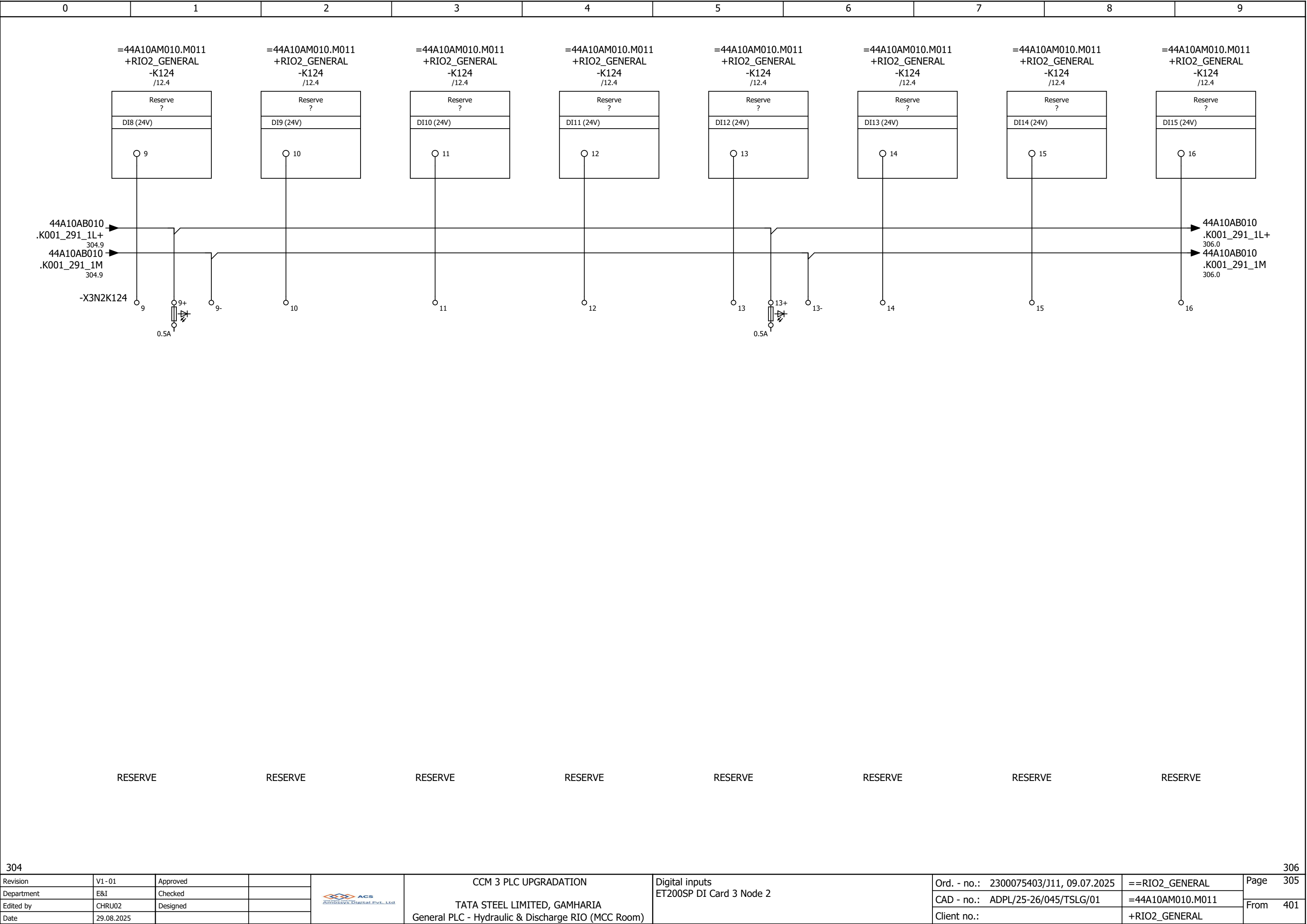
0	1	2	3	4	5	6	7	8	9
-K127 /12.7 4DQ AC230VAC/2A ST 6ES7 132-6FD00-0BB1 DQ0 1 Q 90.0 /400.1 Spare Spare N1 9 /400.1 DQ1 2 Q 90.1 /400.2 Spare Spare N2 10 /400.2 DQ2 3 Q 90.2 /400.3 Spare Spare N3 11 /400.3 DQ3 4 Q 90.3 /400.4 Spare Spare N4 12 /400.4 DQ0' 5 N1' 13 DQ1' 6 N2' 14 DQ2' 7 N3' 15 DQ3' 8 N4' 16 L P1 /53.2 N P2 /53.2									
74									76
Revision	V1 -01	Approved		ACS Automation & Control Systems Digital Pvt. Ltd.	CCM 3 PLC UPGRADATION TATA STEEL LIMITED, GAMHARIA General PLC - Hydraulic & Discharge RIO (MCC Room)	Overview ET200SP DO Cards	Ord. - no.: 2300075403/J11, 09.07.2025	==RIO2_GENERAL	Page 75
Department	E&I	Checked					CAD - no.: ADPL/25-26/045/TSLG/01	=44A10AM010.M011	From 401
Edited by	CHRU02	Designed							
Date	29.08.2025						Client no.:	+RIO2_GENERAL	

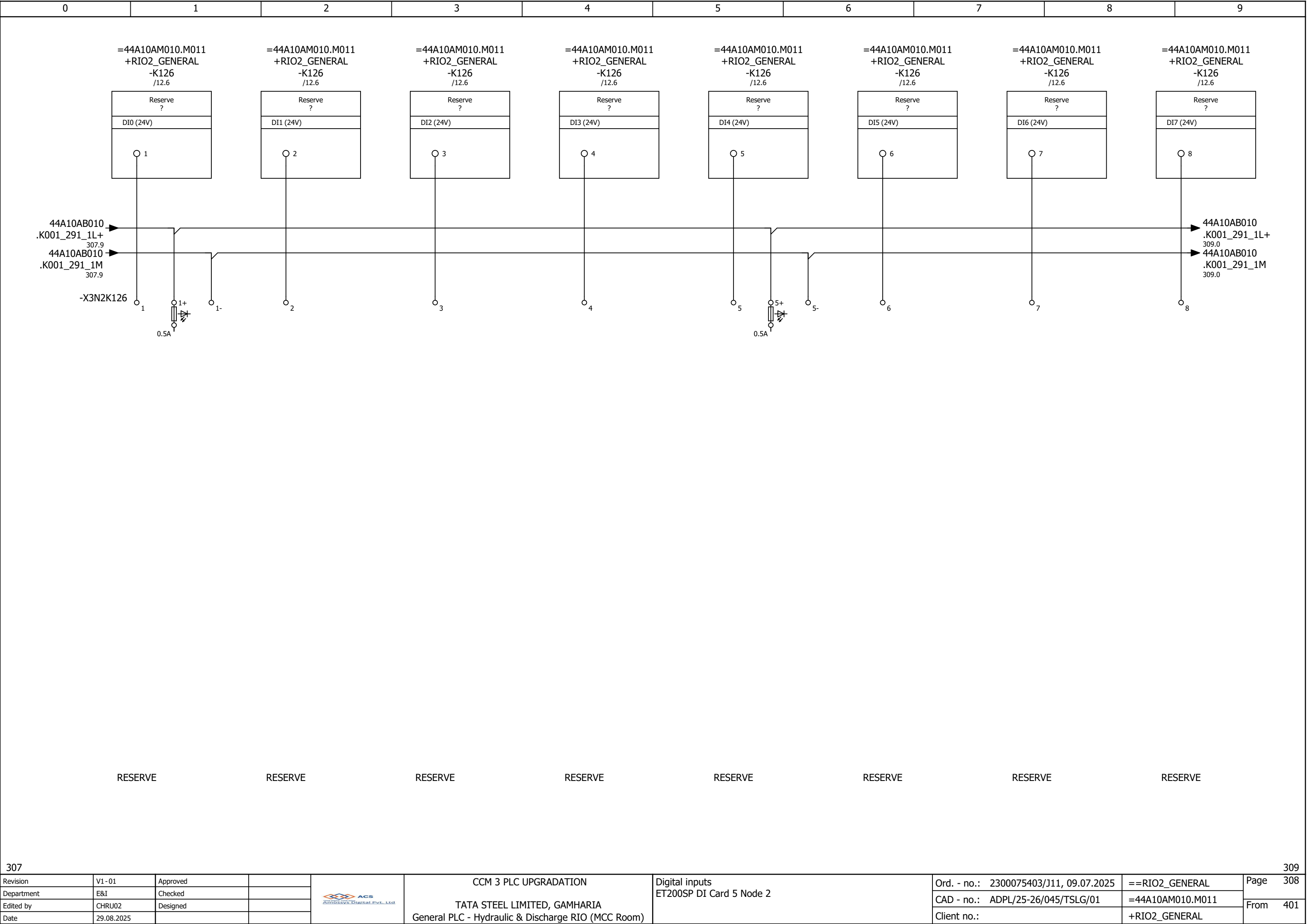


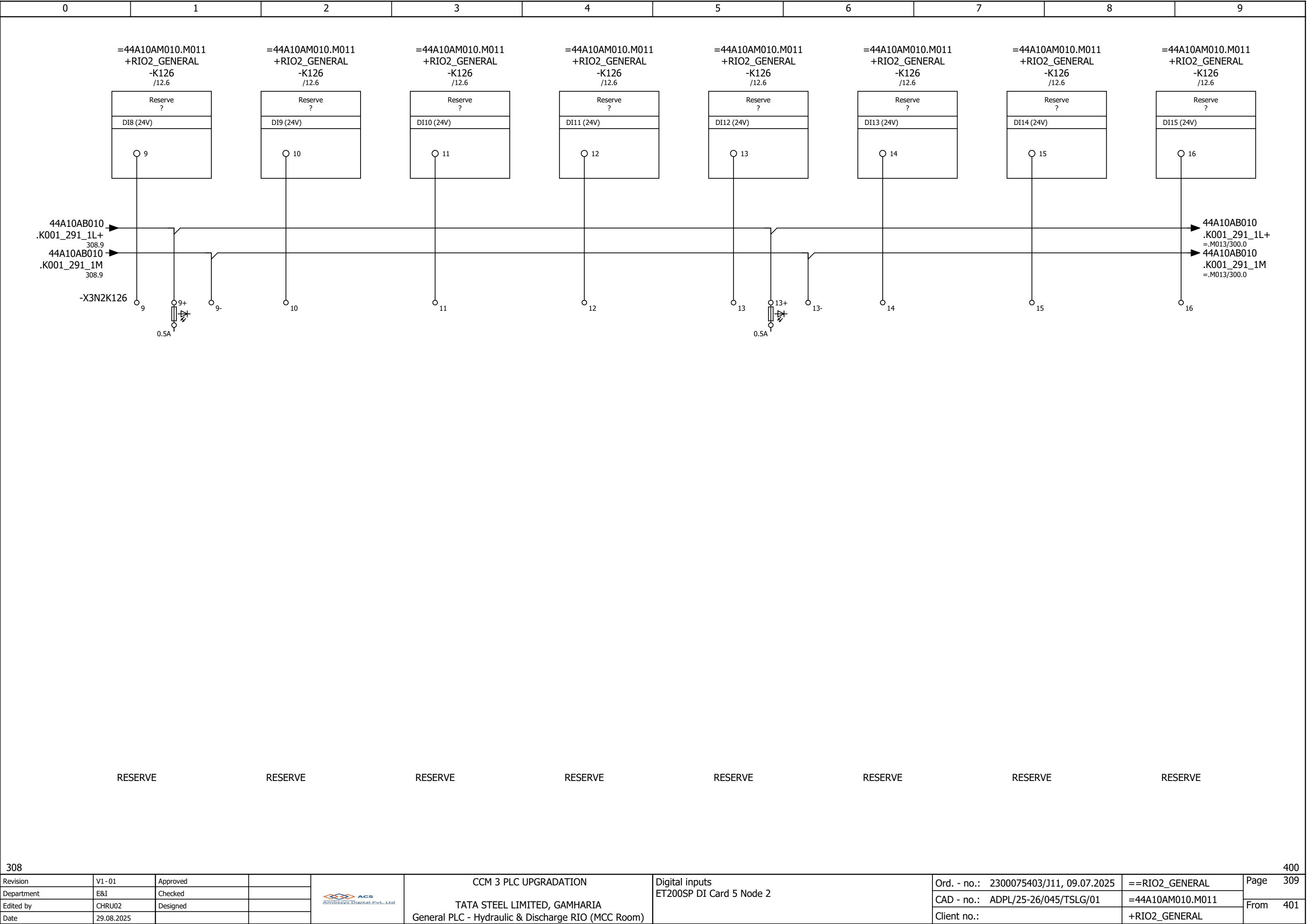


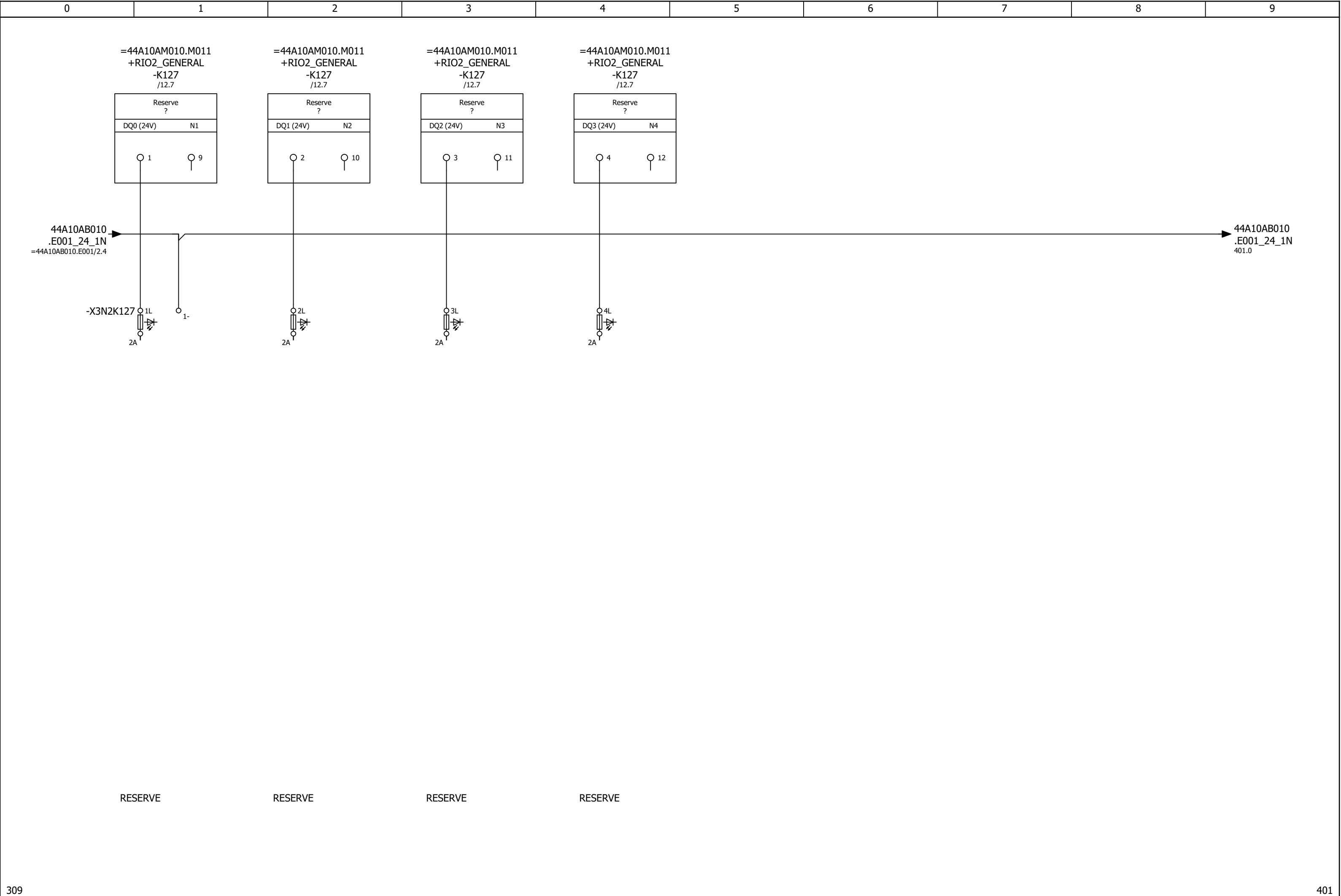


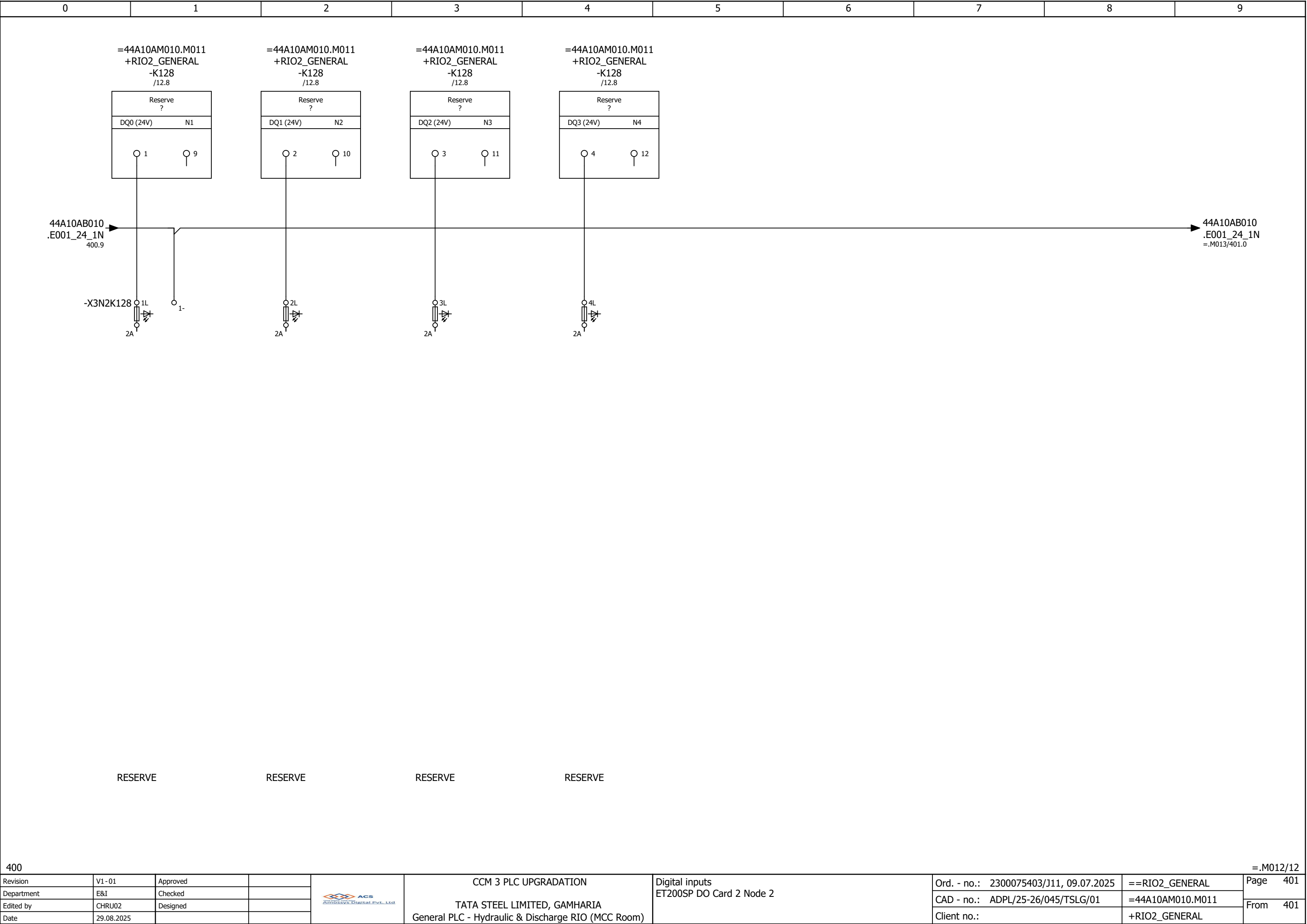


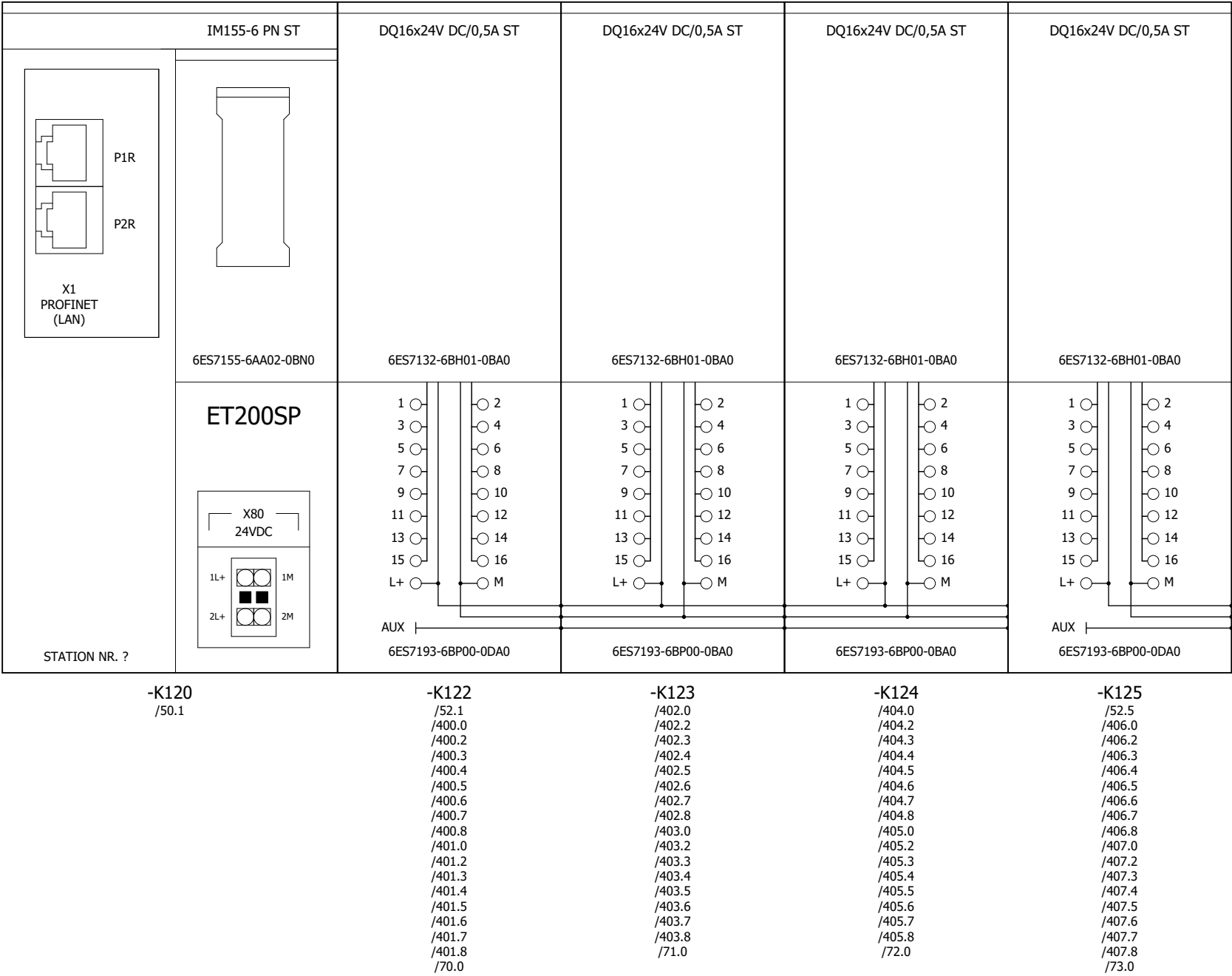


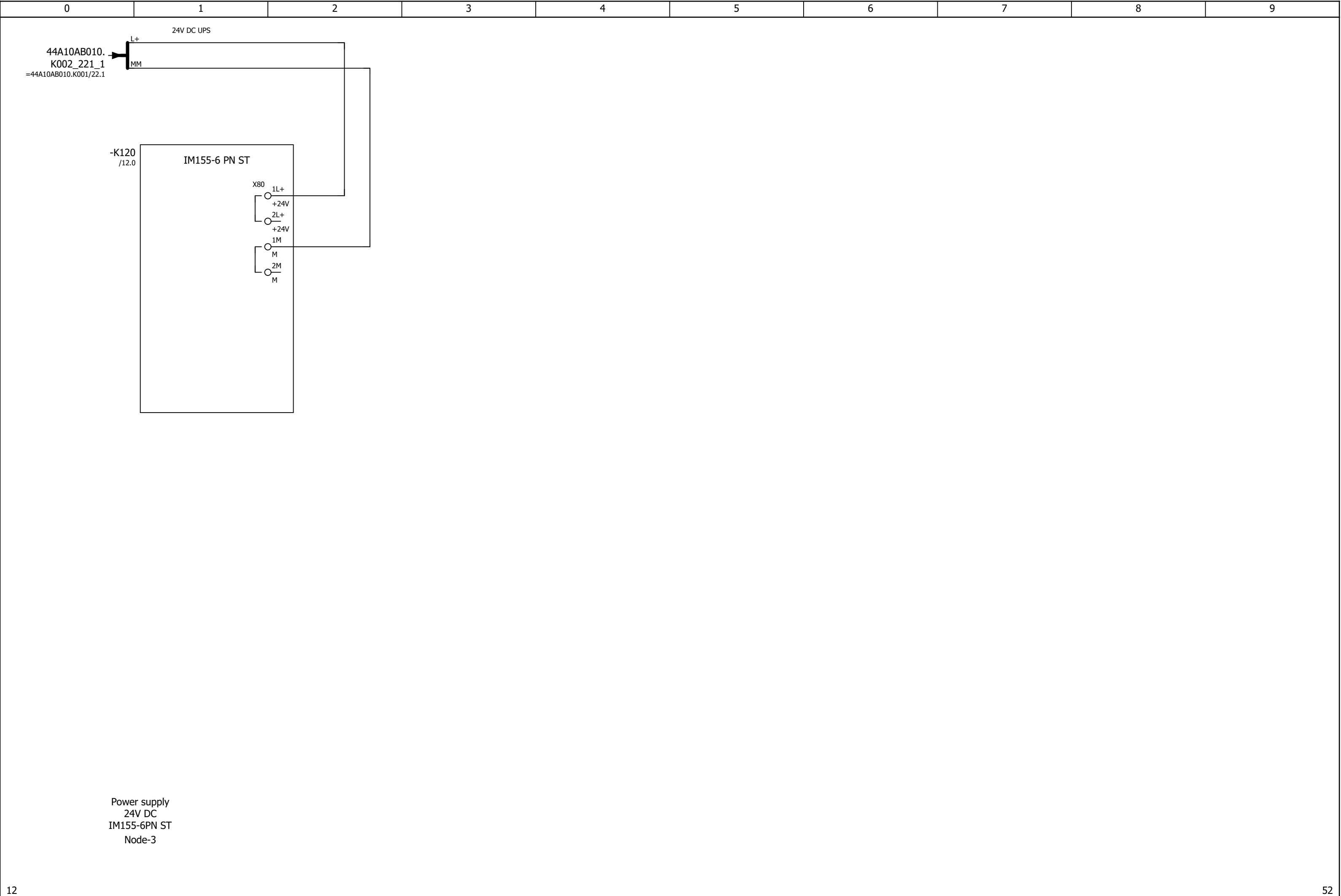


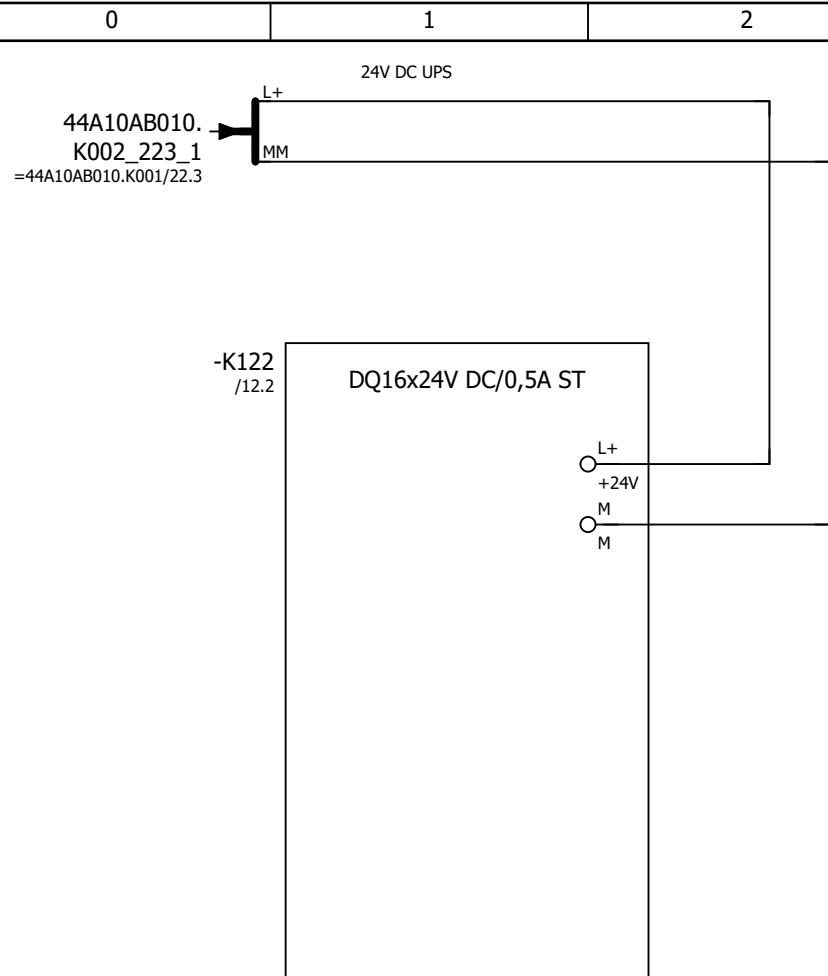




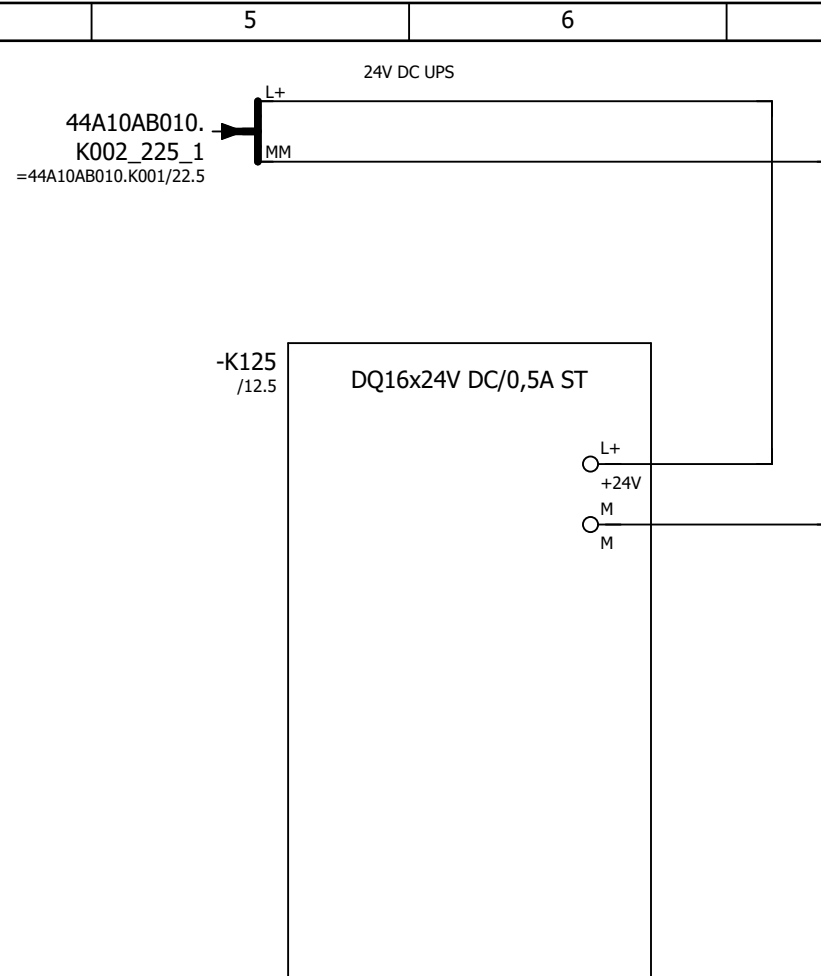




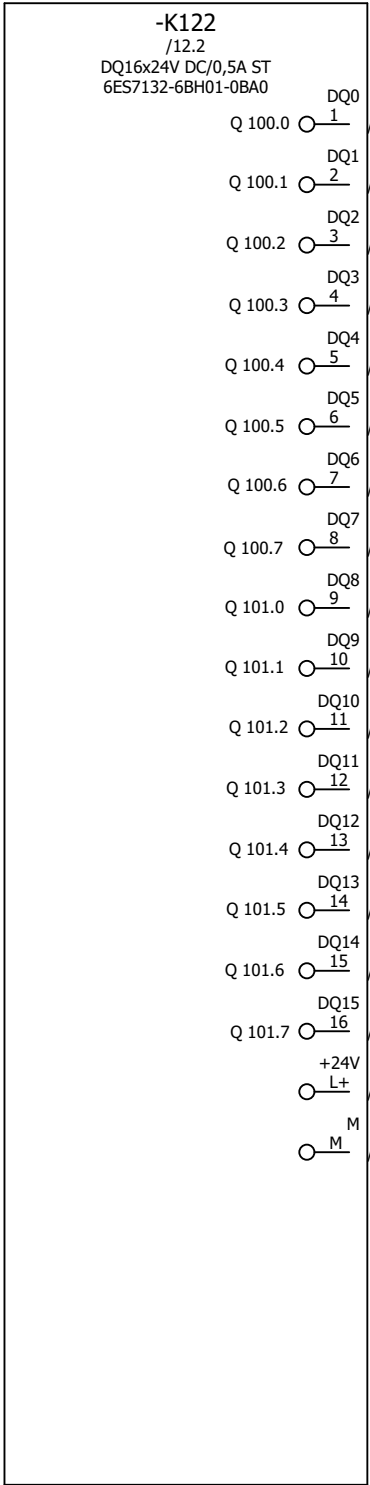




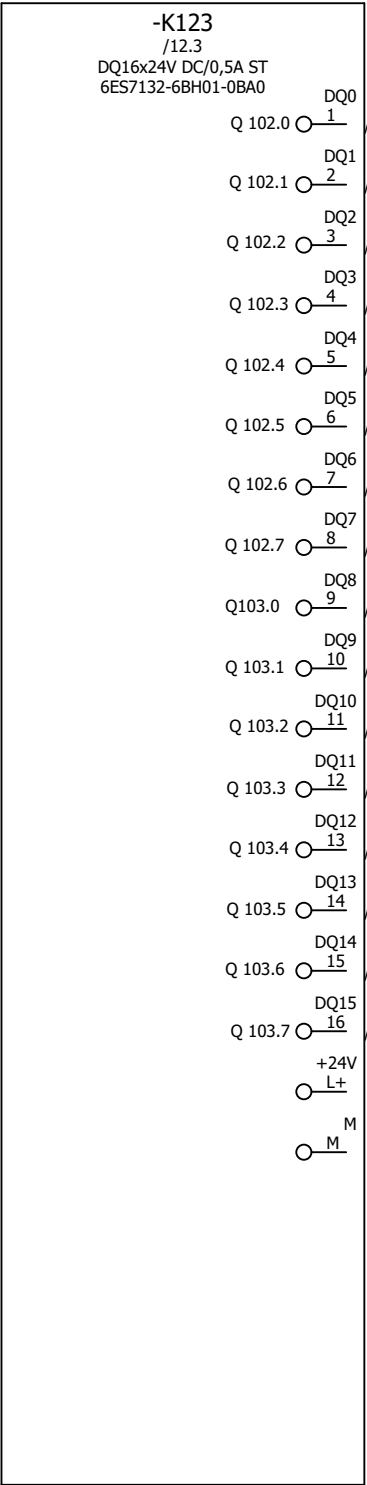
Power supply
24V DC
DO Card 1
Node-3



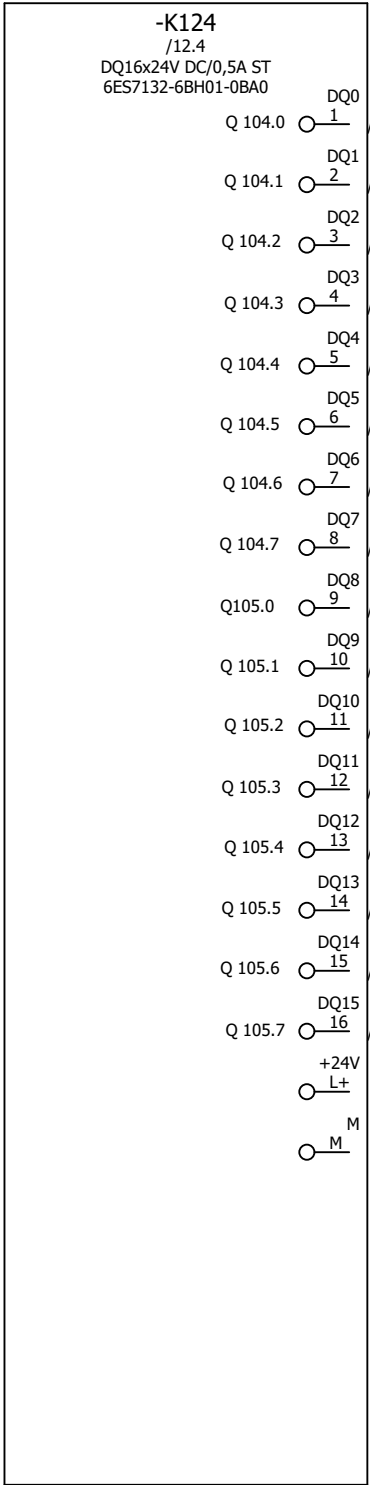
Power supply
24V DC
DO Card 4
Node-3



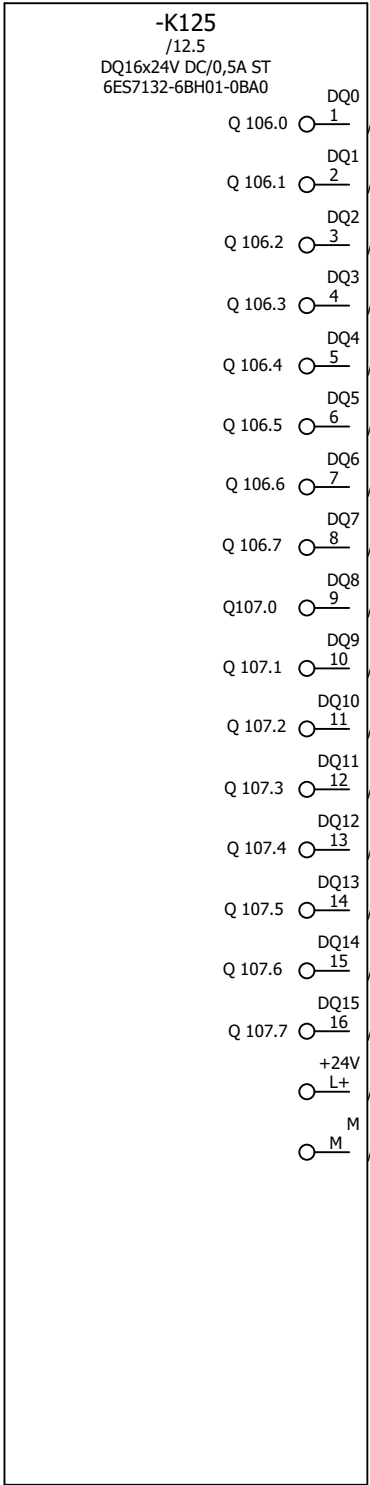
TLS_DO_H_LWR_UNLCK	Transfer car hoist unblocking for lowering
TLS_DO_H_PLIERS_OPN	Transfer car hoist pliers opening
TLS_DO_H_PLIERS_CLS	Transfer car hoist pliers closing
Spare	Spare
F10PWI_VV1_LFT	First valve (A")Lifting"
F10PWI_VV1_LWR	First valve (A")Lowering"
F10PWI_VV2_LFT	First valve (B")Lifting"
F10PWI_VV2_LWR	First valve (B")Lowering"
F10PWI_VV1_ULCK	Horizontal movements unlock valve
F10PWI_VV_WB_FILL	Walking beam filling valve
F10PWI_VV_WB_FLUSH1	Walking beam flushing valve
Spare	Spare
F10PWI_VV_WB_FLUSH_A1	Walking beam lifting cylinder #1 flushing
F10PWI_VV_WB_FLUSH_A2	Walking beam ifting cylinder #2 flushing
F10PWI_VV_WB_FLUSH_A3	Walking beam ifting cylinder #3 flushing
F10PWI_VV_WB_FLUSH_A4	Walking beam ifting cylinder #4 flushing

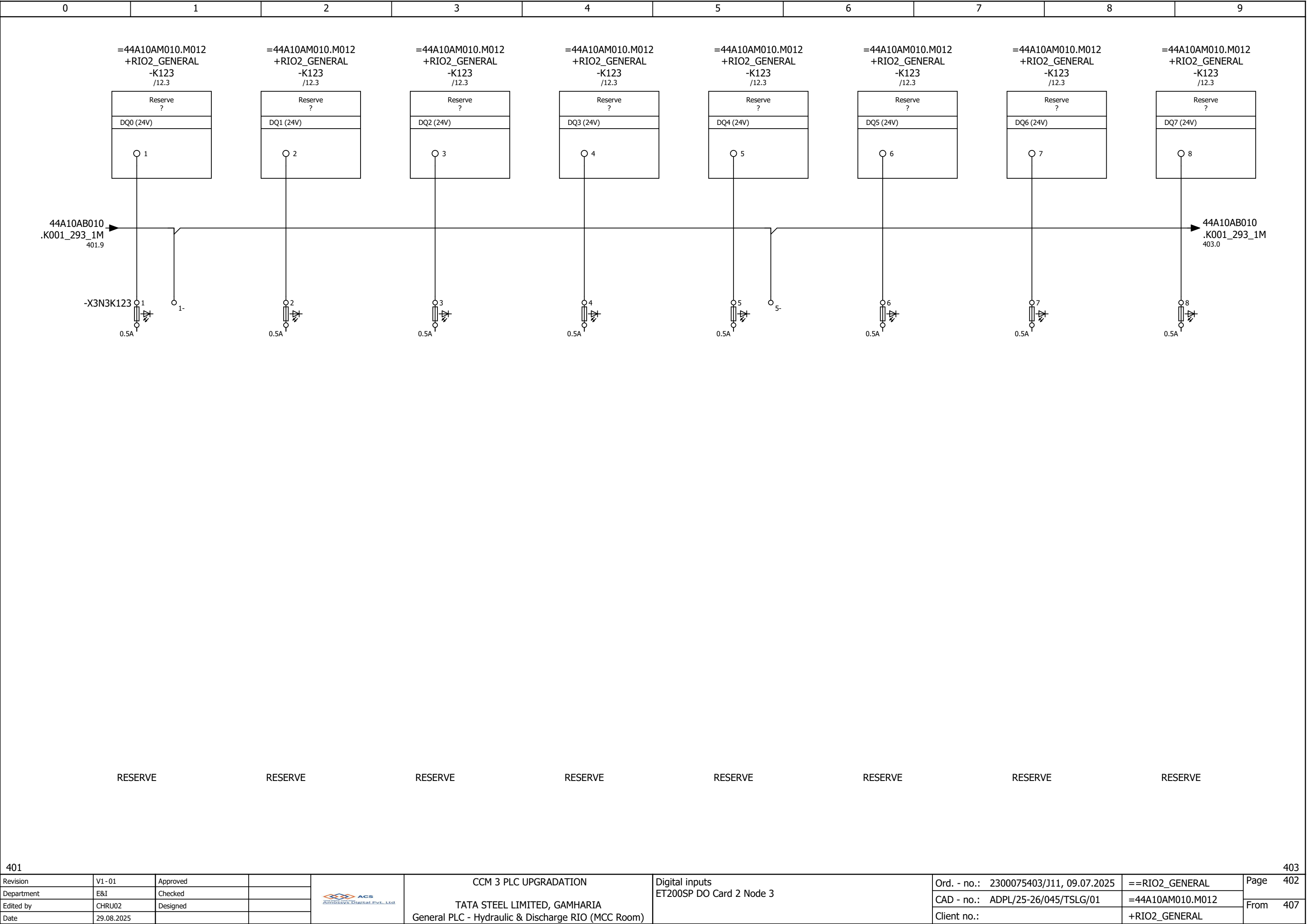


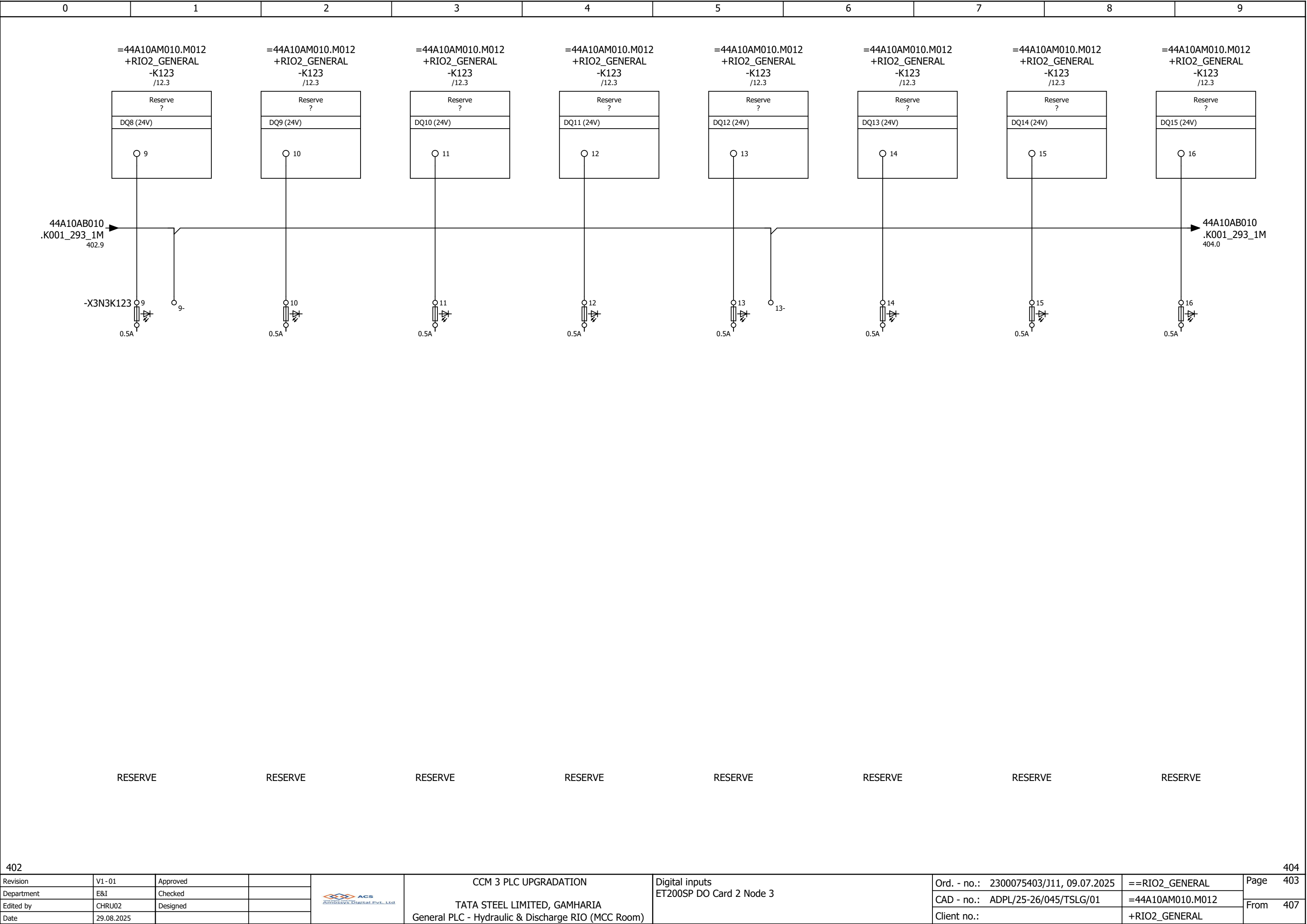
F10PWI_VV_WB_FLUSH_B1	Walking beam lifting cylinder #1 flushing
F10PWI_VV_WB_FLUSH_B2	Walking beam ifting cylinder #2 flushing
F10PWI_VV_WB_FLUSH_B3	Walking beam ifting cylinder #3 flushing
F10PWI_VV_WB_FLUSH_B4	Walking beam ifting cylinder #4 flushing
F10PWI_VV_FD_FWD	Feeding device advancing
F10PWI_VV_FD_REV	Feeding device advancing
F10PRS_VV_FWD	Pusher Collecting Table forward control
F10PRS_VV_REV	Pusher Collecting Table reverse control
F10PRC_VV_FWD	Pawls Collecting Table forward control
F10PRC_VV_REV	Pawls Collecting Table reverse control
TOCB_LEFT_FRD_CMD	TOCB LEFT FRD CMD
TOCB_LEFT_REV_CMD	TOCB LEFT REV CMD
Y01CLG_VV_LIN_INV	Ladle Turret grease Line inversion control valve
Y02CLG_VV_LIN_INV	Line inversion control valve
Y03CLG_VV_LIN_INV	Line inversion control valve
Spare	Spare

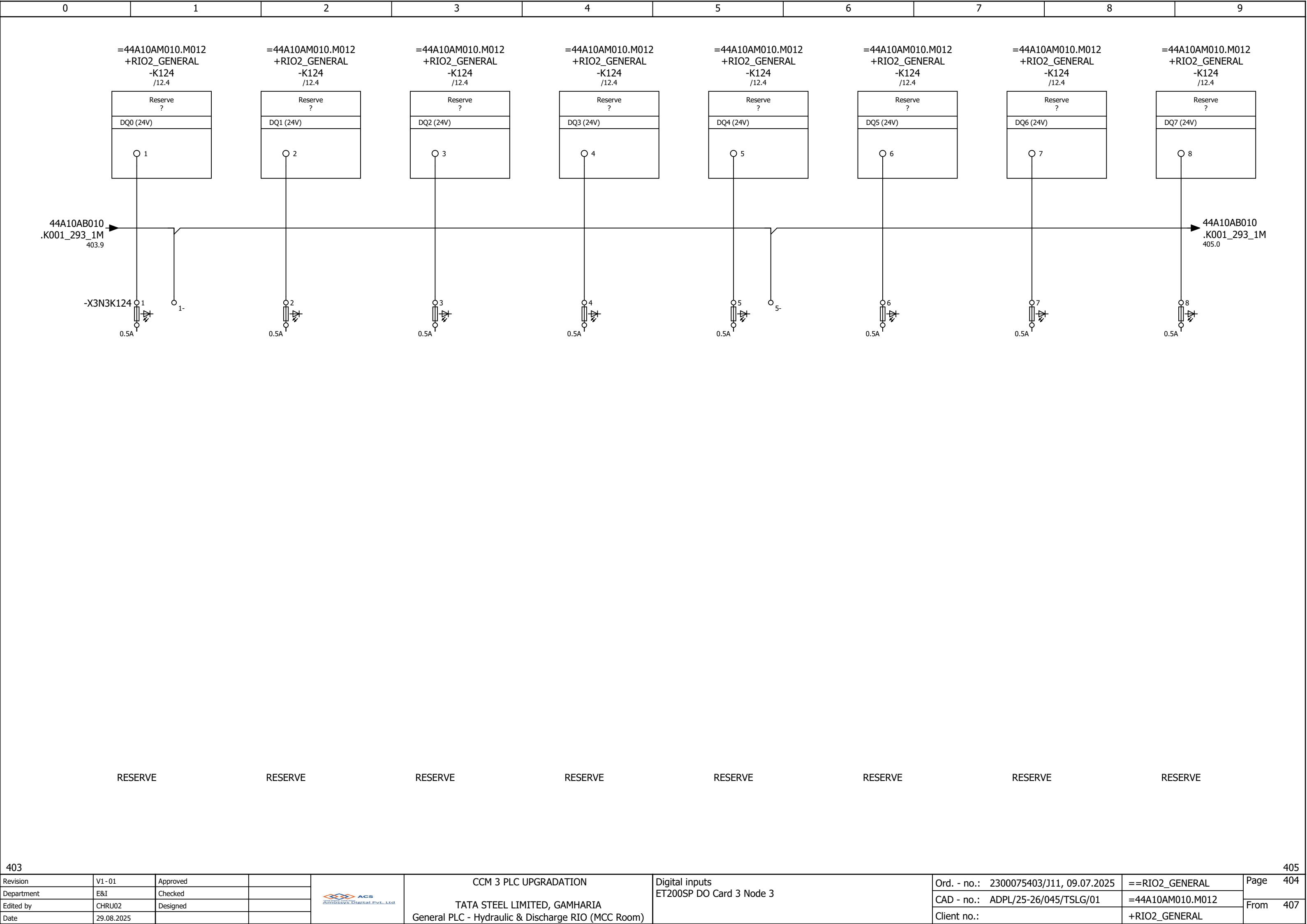


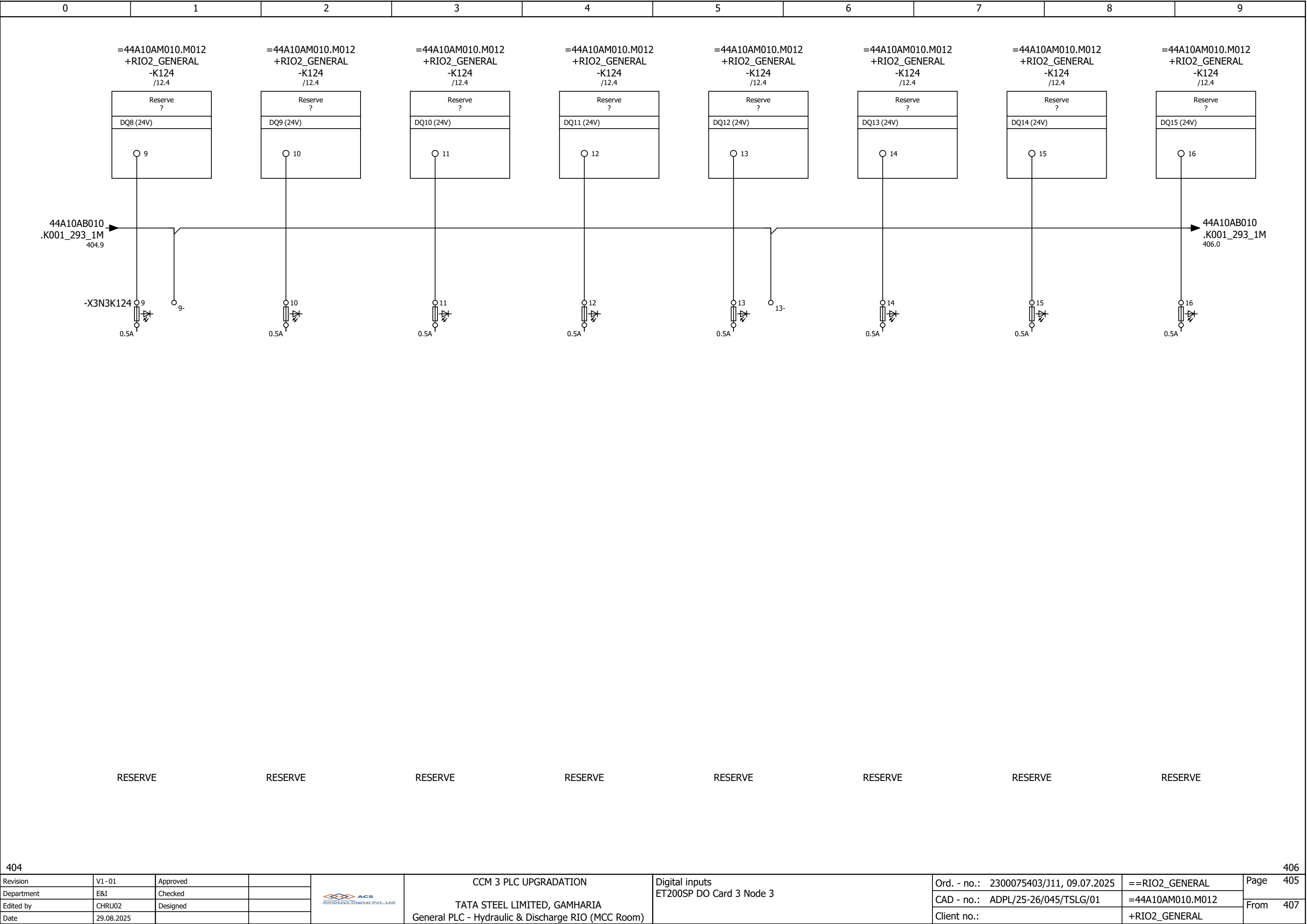
X30_VV_COOLING_OPN	Ladle Turret & Tundish Car H.U. heat exchanger water open valve
X30_VV_GAP_RESET	Ladle Arm 1&2 slide gate pressure selection valve for gap reset
Spare	Spare
Spare	Spare
X70_VV_COOLING_OPN	Main H.U. heat exchanger water open valve
X70_VV_ACC1_BEN1_VV_CHG	Main H.U. acc. 1 bench 1 pressure discharge valve
X70_VV_ACC2_BEN1_VV_CHG	Main H.U. acc. 2 bench 1 pressure discharge valve
Spare	Spare
X65_VV_COOLING_OPN	Oscillator H.U. heat exchanger water open valve
X65_VV_ACC_BEN_DISC_CLS	Oscillator H.U. accumulators pressure discharge valve
Spare	Spare
Spare	Spare
X80_VV_COOLING_OPN	Run out area H.U. heat exchanger water open valve
X80_VV_ACC1_BEN1_VV_CHG	Run out area H.U. acc. 1 bench 1 pressure discharge valve
X80_VV_ACC1_BEN2_VV_CHG	Run out area H.U. acc. 1 bench 2 pressure discharge valve
X80_VV_ACC2_BEN1_VV_CHG	Run out area H.U. acc. 2 bench 1 pressure discharge valve

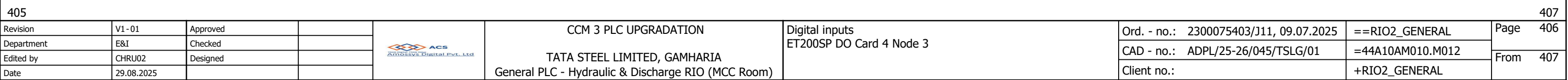


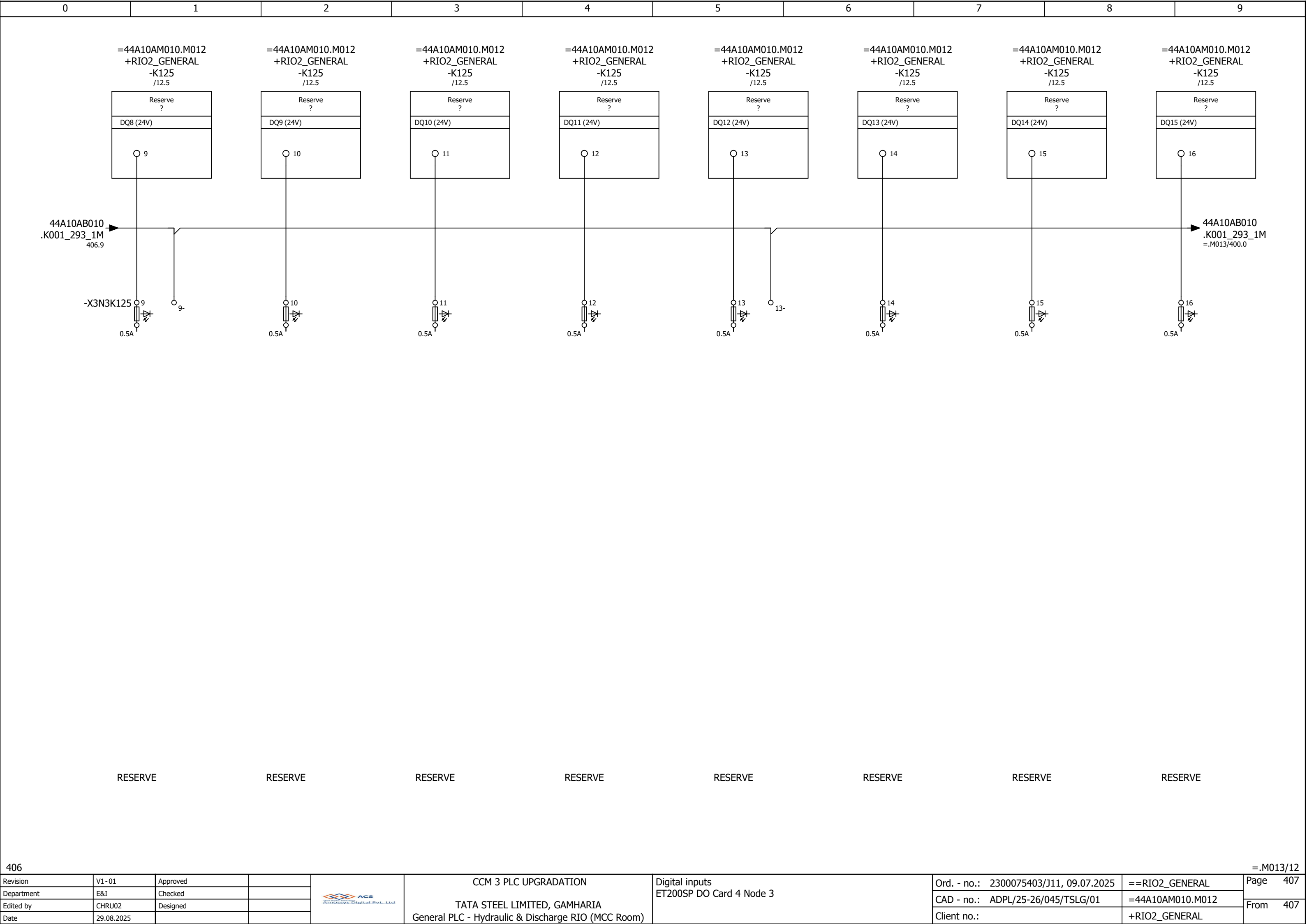


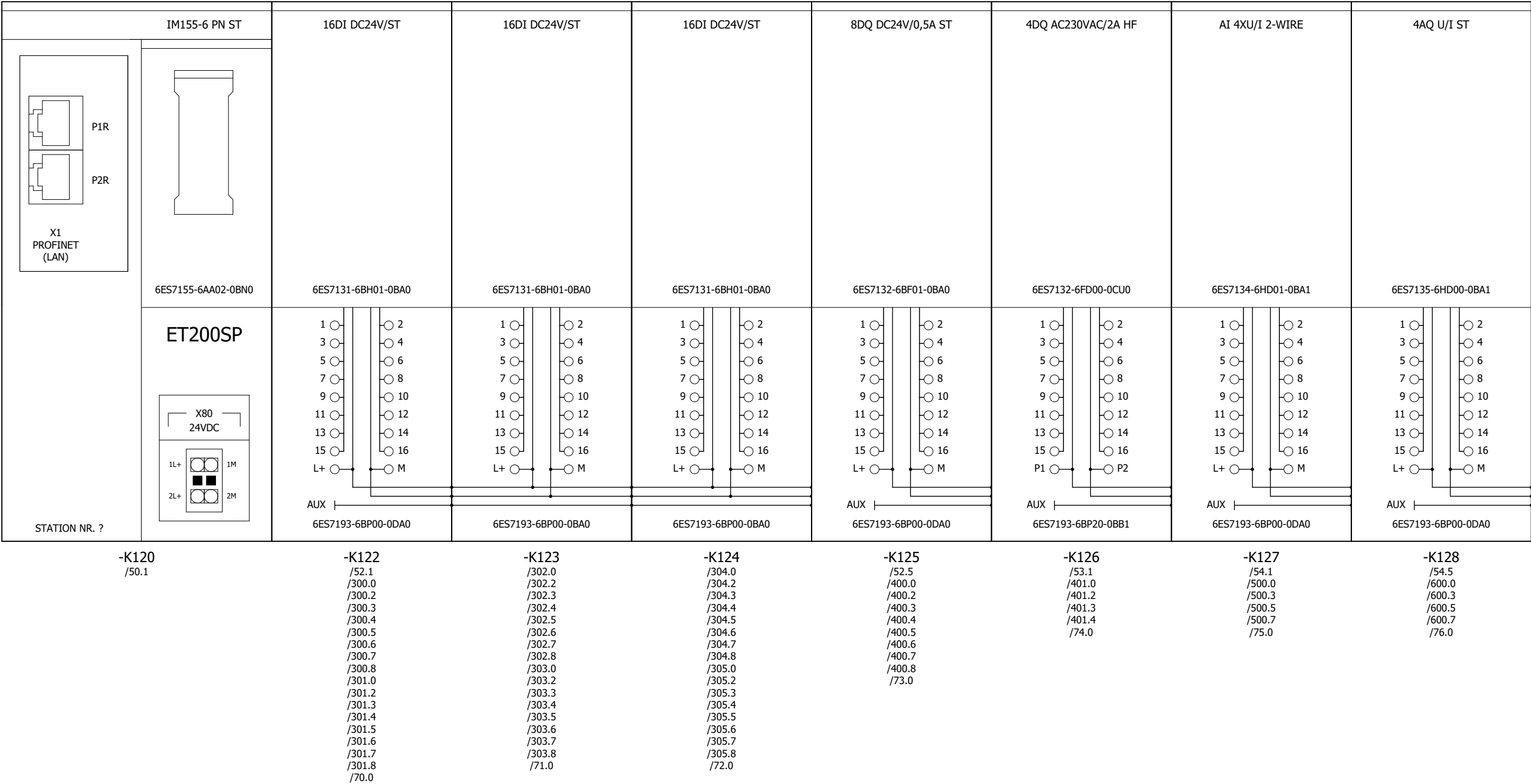


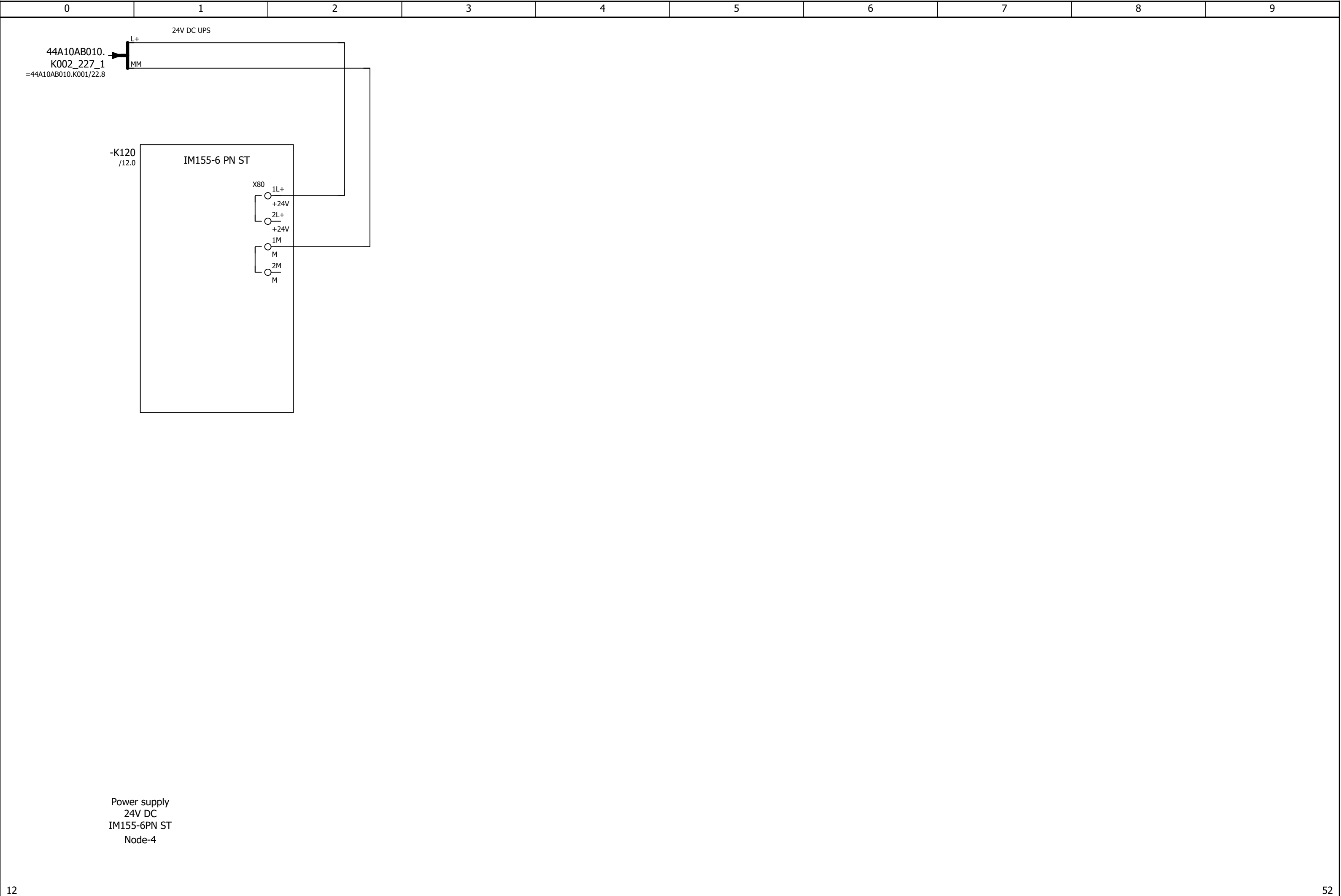


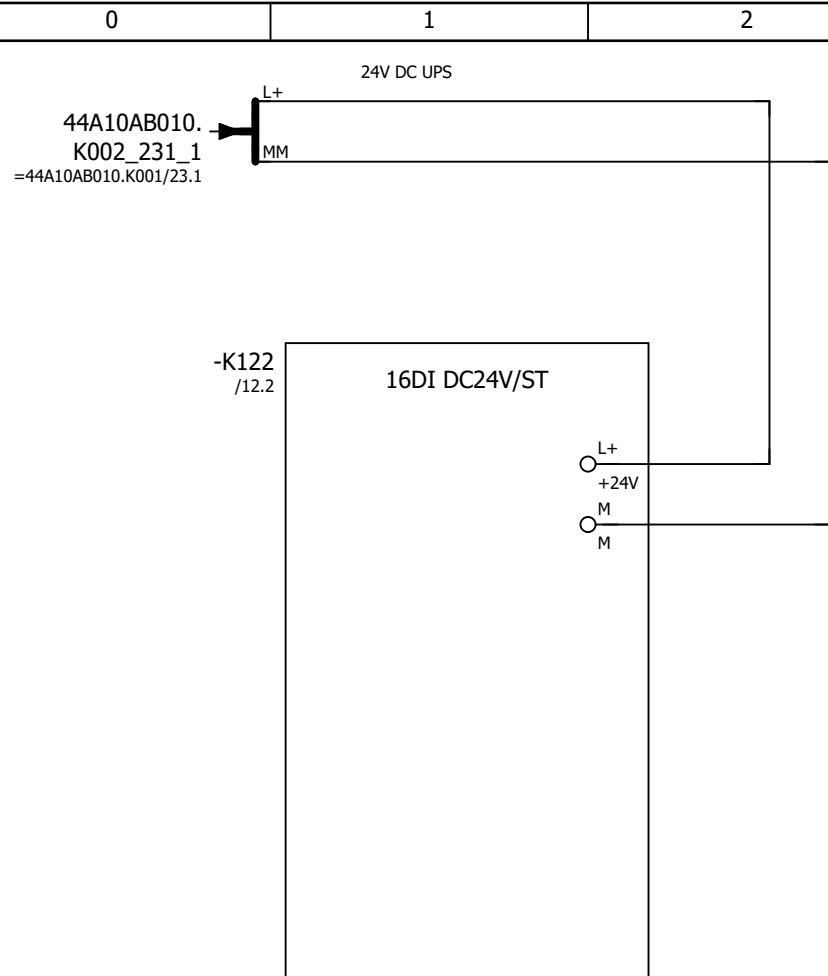




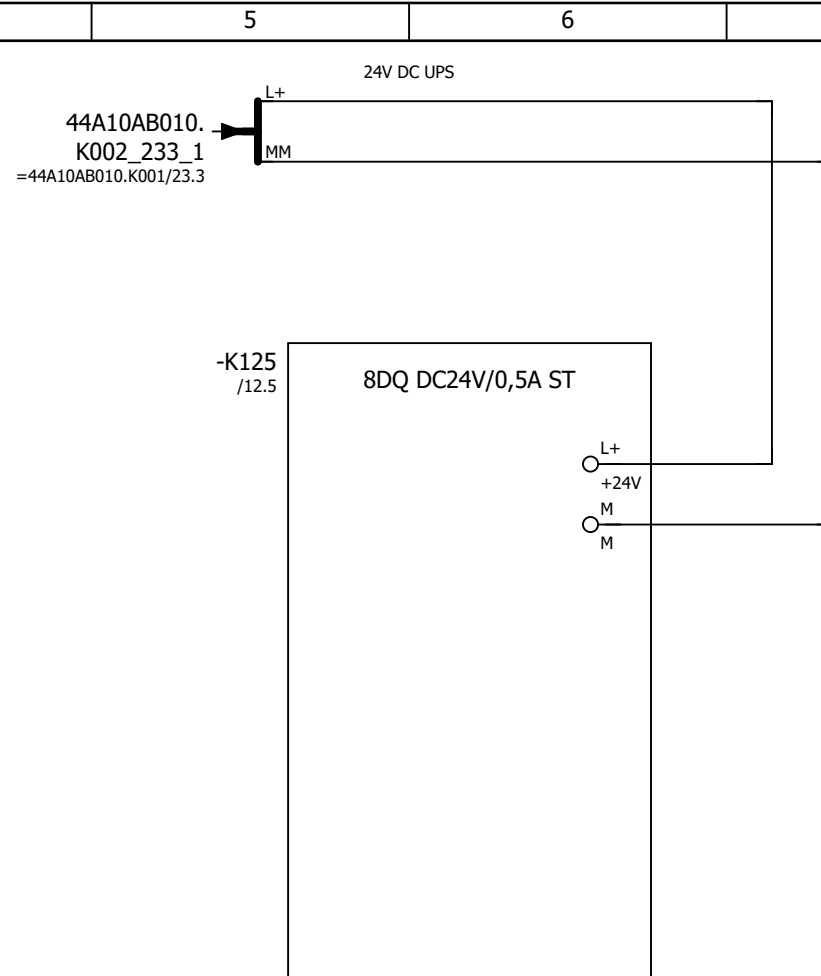




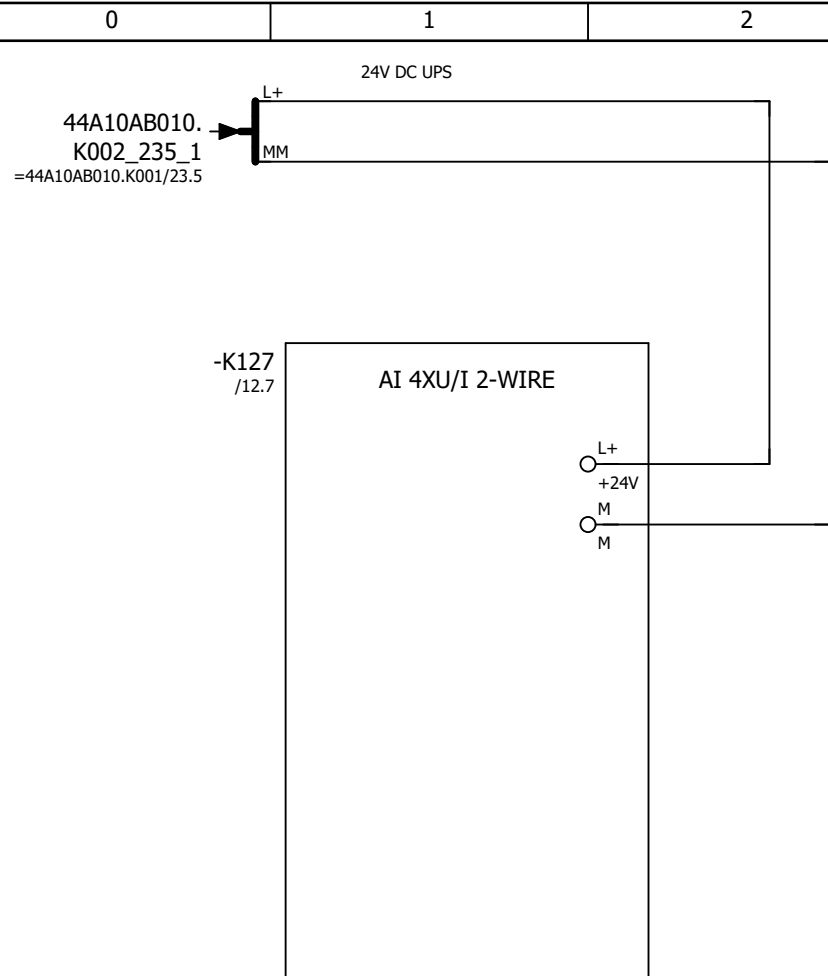




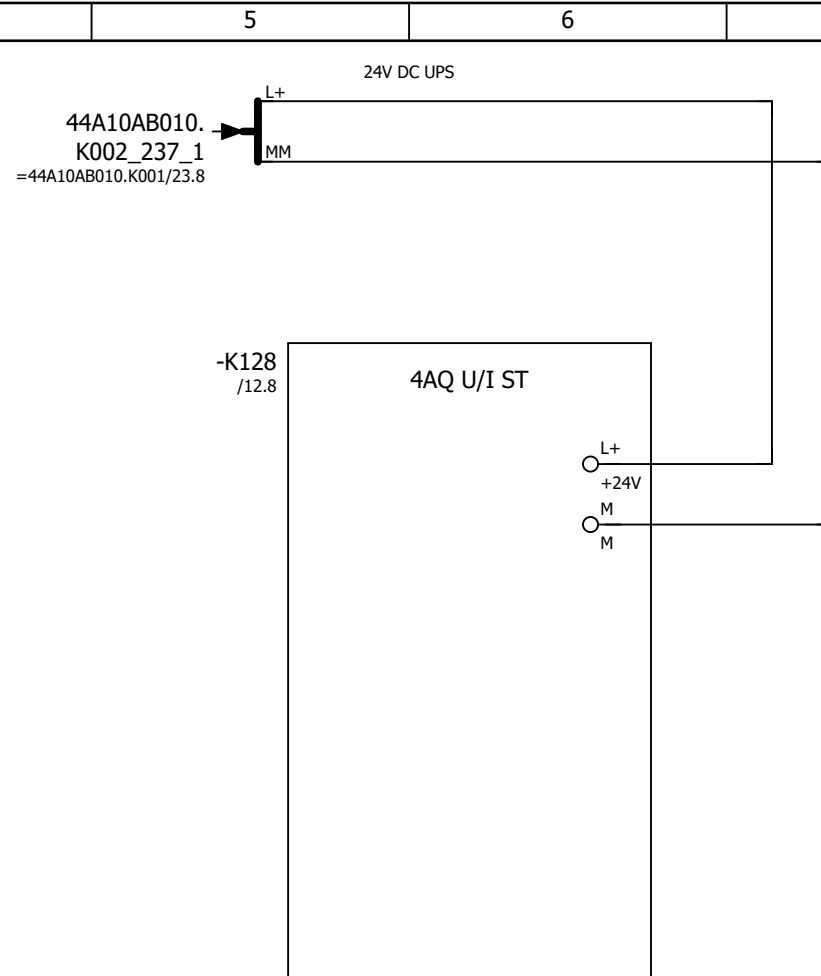
Power supply
24V DC
DI Card 1
Node-4



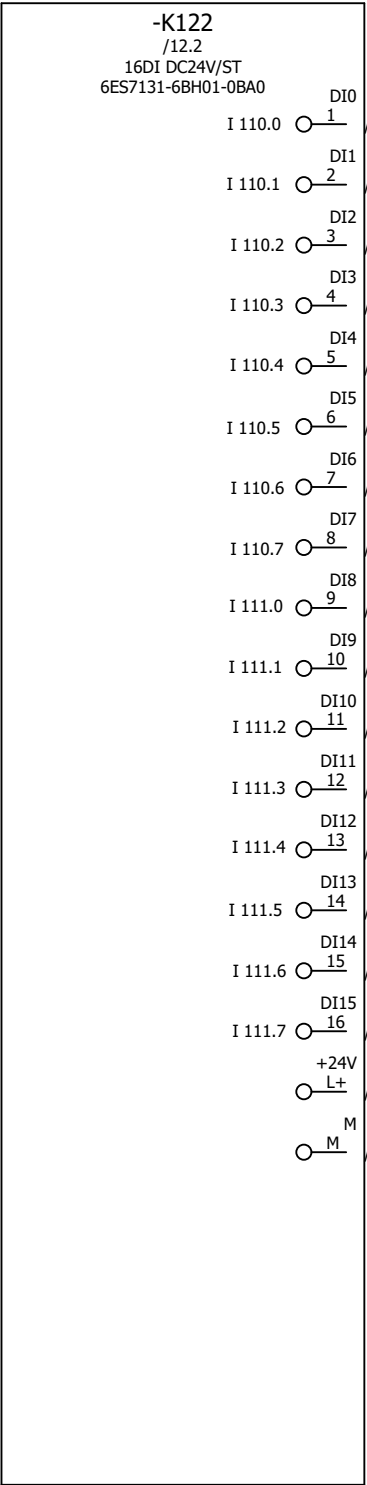
Power supply
24V DC
DO Card 1
Node-4

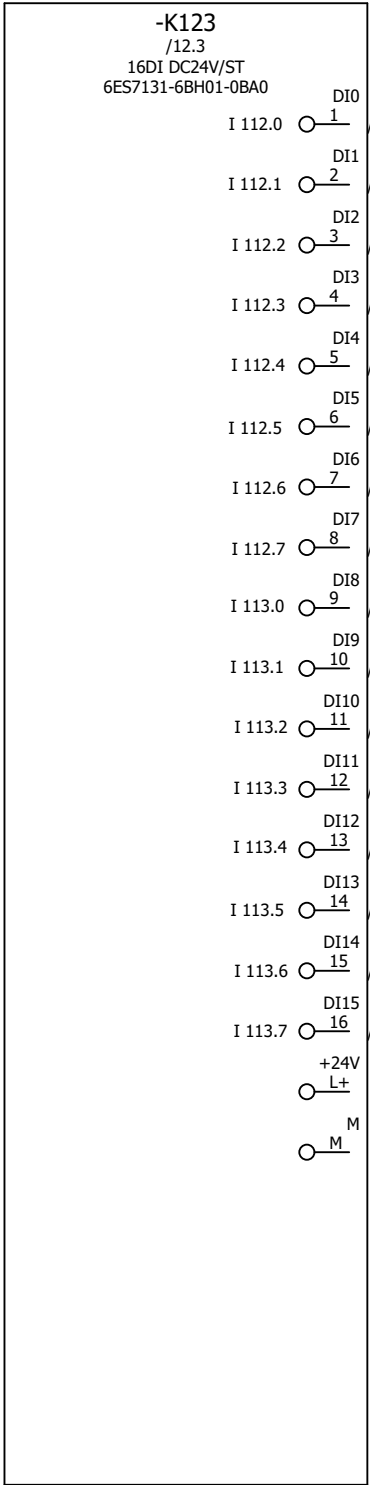


Power supply
24V DC
AI Card 1
Node-4

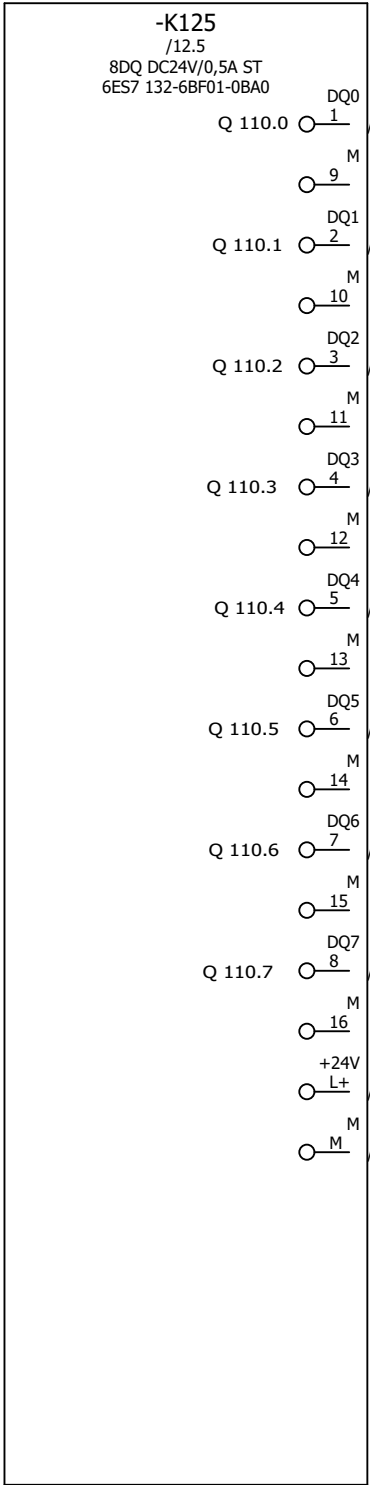


Power supply
24V DC
AO Card 1
Node-4





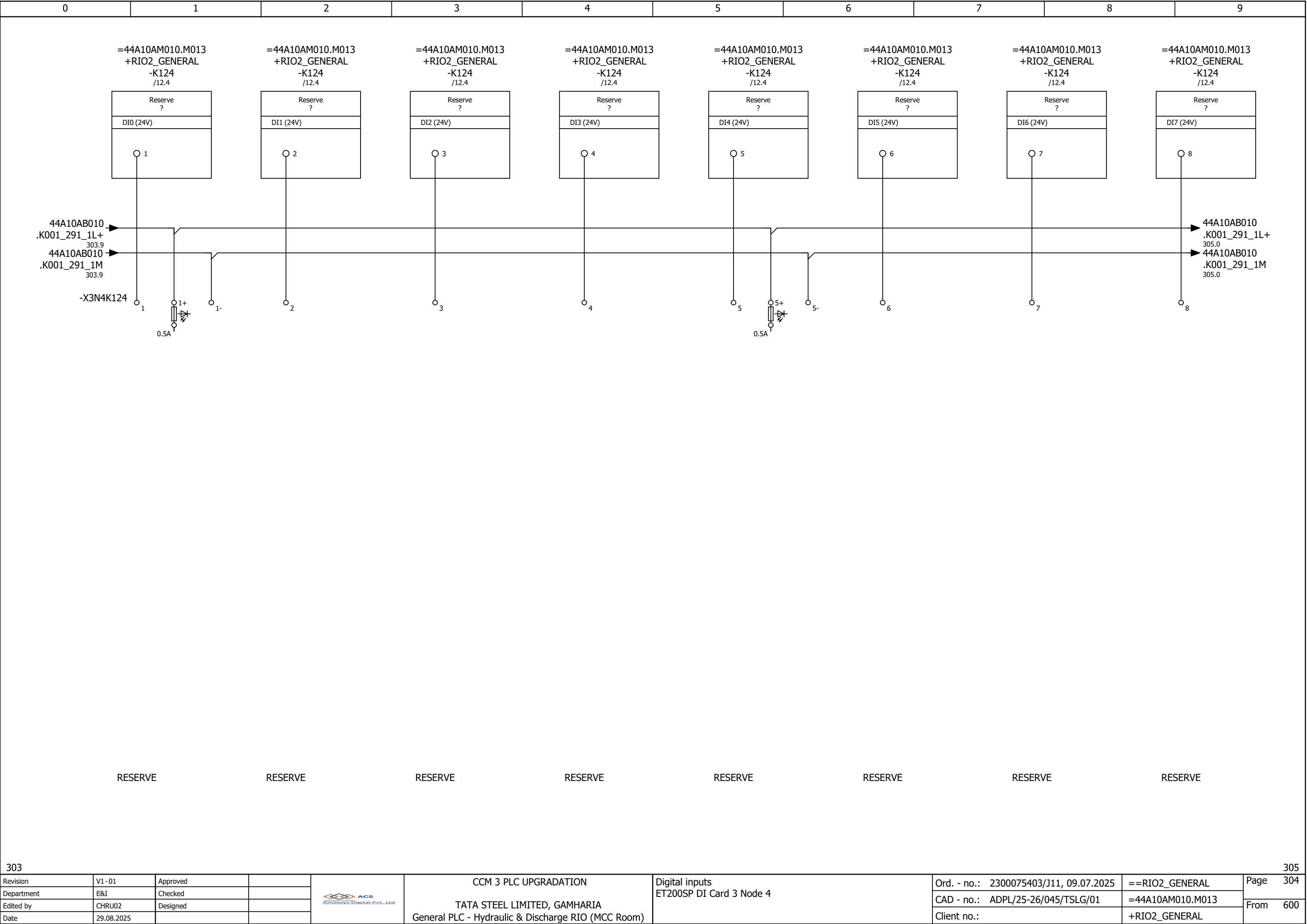
0	1	2	3	4	5	6	7	8	9	
-K124 /12.4 16DI DC24V/ST 6ES7131-6BH01-0BA0 DI0 I 114.0 1 /304.1 Spare Spare DI1 I 114.1 2 /304.2 Spare Spare DI2 I 114.2 3 /304.3 Spare Spare DI3 I 114.3 4 /304.4 Spare Spare DI4 I 114.4 5 /304.5 Spare Spare DI5 I 114.5 6 /304.6 Spare Spare DI6 I 114.6 7 /304.7 Spare Spare DI7 I 114.7 8 /304.9 Spare Spare DI8 I 115.0 9 /305.1 Spare Spare DI9 I 115.1 10 /305.2 Spare Spare DI10 I 115.2 11 /305.3 Spare Spare DI11 I 115.3 12 /305.4 Spare Spare DI12 I 115.4 13 /305.5 Spare Spare DI13 I 115.5 14 /305.6 Spare Spare DI14 I 115.6 15 /305.7 Spare Spare DI15 I 115.7 16 /305.9 Spare Spare +24V L+ M M										
71									73	
Revision	V1 -01	Approved		ACS Automation & Control Systems Digital Pvt. Ltd.	CCM 3 PLC UPGRADATION TATA STEEL LIMITED, GAMHARIA General PLC - Hydraulic & Discharge RIO (MCC Room)	Overview ET200SP DI Cards	Ord. - no.: 2300075403/J11, 09.07.2025	==RIO2_GENERAL	Page 72	
Department	E&I	Checked					CAD - no.: ADPL/25-26/045/TSLG/01	=44A10AM010.M013	From 600	
Edited by	CHRU02	Designed					Client no.: +RIO2_GENERAL			
Date	29.08.2025									



F10PW1_OS_LCS42_FD_FWD	Feeder forward lamp LCS42
F10PW1_OS_LCS42_FD_REV	Feeder reverse lamp LCS42
TLS_OS_PLS_OPN_LCS42	Transfer car hoist pliers opened LCS42
TLS_OS_PLS_CLS_LCS42	Transfer car hoist pliers closed LCS42
X70_VV_BYP_CLS	Main H.U. Pump by-pass valve closing
X30_VV_BYP_OPN	Ladle Turret & Tundish Car H.U. Pump by-pass valve opening
X80_VV_BYP_OPN	Run out area H.U. Pump by-pass valve opening
Spare	Spare

0	1	2	3	4	5	6	7	8	9	
-K127 /12.7 AI 4XU/I 2-WIRE SIMATIC ET 200SP U0+ 1 U0- 5 U1+ 2 U1- 6 U2+ 3 U2- 7 U3+ 4 U3- 8 I0+ 13 UV0 9 I1+ 14 UV1 10 I2+ 15 UV2 11 I3+ 16 UV3 12 +24V L+ M M PIW 772 /500.1 PIW 774 /500.3 PIW 776 /500.5 PIW 778 /500.7 /500.8 /54.2 /54.2 Spare Spare Spare Spare Spare										
74									76	
Revision	V1 -01	Approved		ACS AMTESYS Digital Pvt. Ltd.	CCM 3 PLC UPGRADATION TATA STEEL LIMITED, GAMHARIA General PLC - Hydraulic & Discharge RIO (MCC Room)	Overview ET200SP AI Cards	Ord. - no.: 2300075403/J11, 09.07.2025	==RIO2_GENERAL	Page 75	
Department	E&I	Checked					CAD - no.: ADPL/25-26/045/TSLG/01	=44A10AM010.M013	From 600	
Edited by	CHRU02	Designed					Client no.: +RIO2_GENERAL			
Date	29.08.2025									

0	1	2	3	4	5	6	7	8	9	
-K128 /12.8 4AQ U/I ST 6ES7 135-6HD00-0BA1 Q0+ 1 PQW 772 /600.1 Q0- 5 /600.1 S0+ 9 S0- 13 Q1+ 2 PQW 774 /600.3 Q1- 6 /600.3 S1+ 10 S1- 14 Q2+ 3 PQW 776 /600.5 Q2- 7 /600.5 S2+ 11 S2- 15 Q3+ 4 PQW 778 /600.7 Q3- 8 /600.8 S3+ 12 S3- 16 +24V L+ /54.6 M M /54.6 Spare Spare Spare Spare Spare										
75									300	
Revision	V1 -01	Approved		ACS AMDESYS Digital Pvt. Ltd.	CCM 3 PLC UPGRADATION TATA STEEL LIMITED, GAMHARIA General PLC - Hydraulic & Discharge RIO (MCC Room)	Overview ET200SP AO Cards	Ord. - no.: 2300075403/J11, 09.07.2025	==RIO2_GENERAL	Page 76	
Department	E&I	Checked					CAD - no.: ADPL/25-26/045/TSLG/01	=44A10AM010.M013	From 600	
Edited by	CHRU02	Designed					Client no.: +RIO2_GENERAL			
Date	29.08.2025									



Revision	V1 -01	Approved	
Department	E&I	Checked	
Edited by	CHRU02	Designed	
Date	29.08.2025		



CCM 3 PLC UPGRADATION

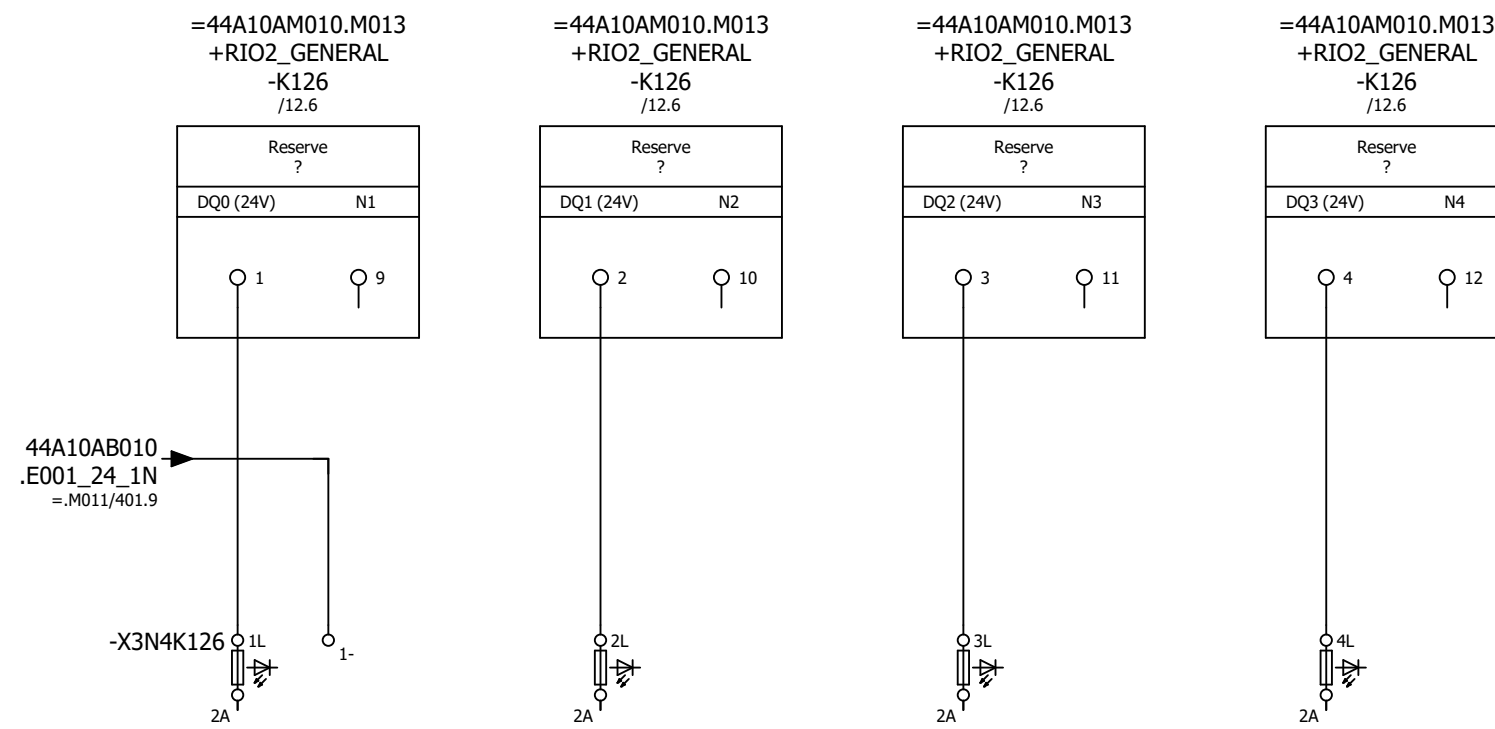
TATA STEEL LIMITED, GAMHARIA

General PLC - Hydraulic & Discharge RIO (MCC Room)

Digital inputs

ET200SP DI Card 3 Node 4

Ord. - no.:	2300075403/J11, 09.07.2025	==RIO2_GENERAL	Page	304
CAD - no.:	ADPL/25-26/045/TSLG/01	=44A10AM010.M013	From	600
Client no.:		+RIO2_GENERAL		

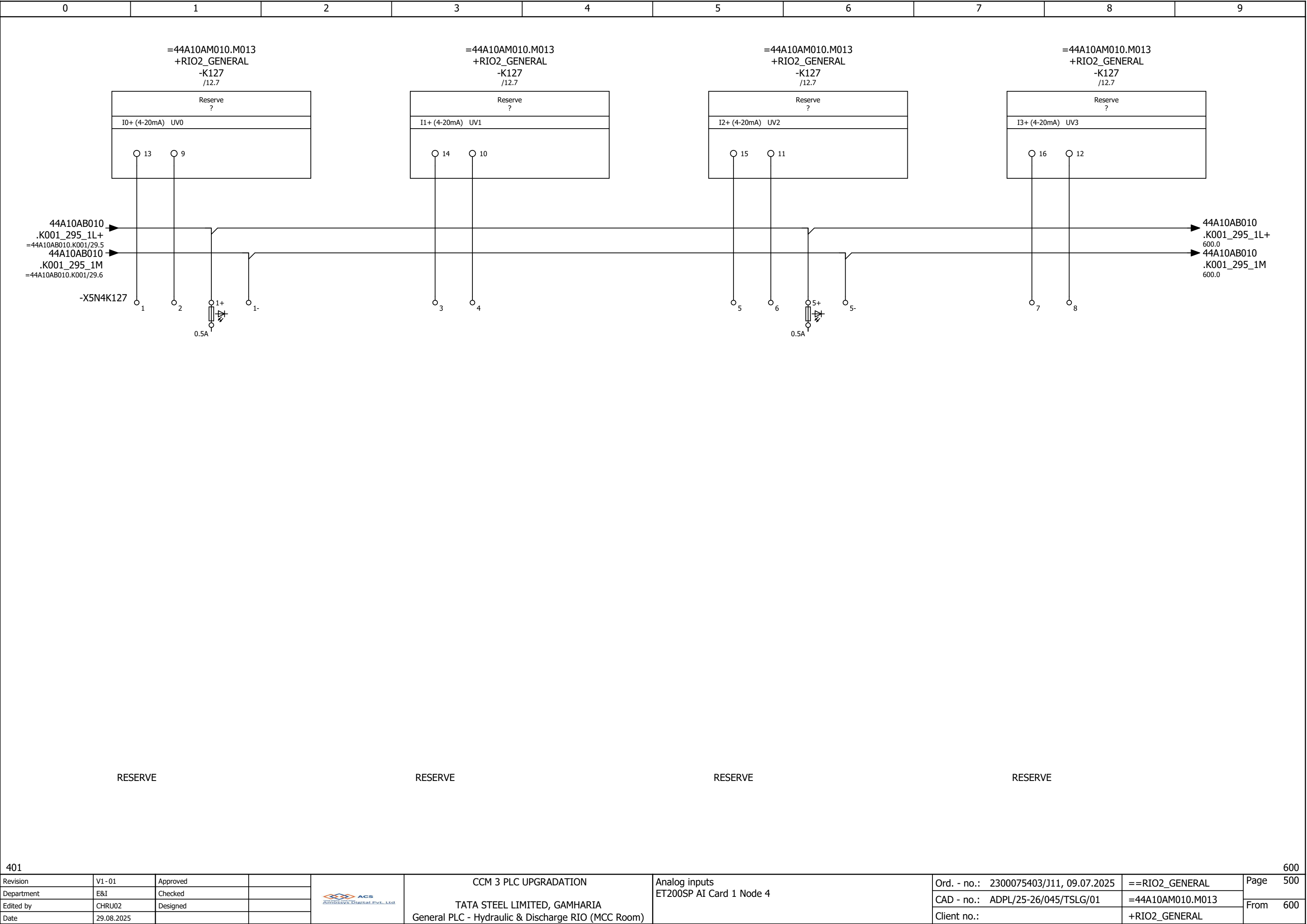


RESERVE

RESERVE

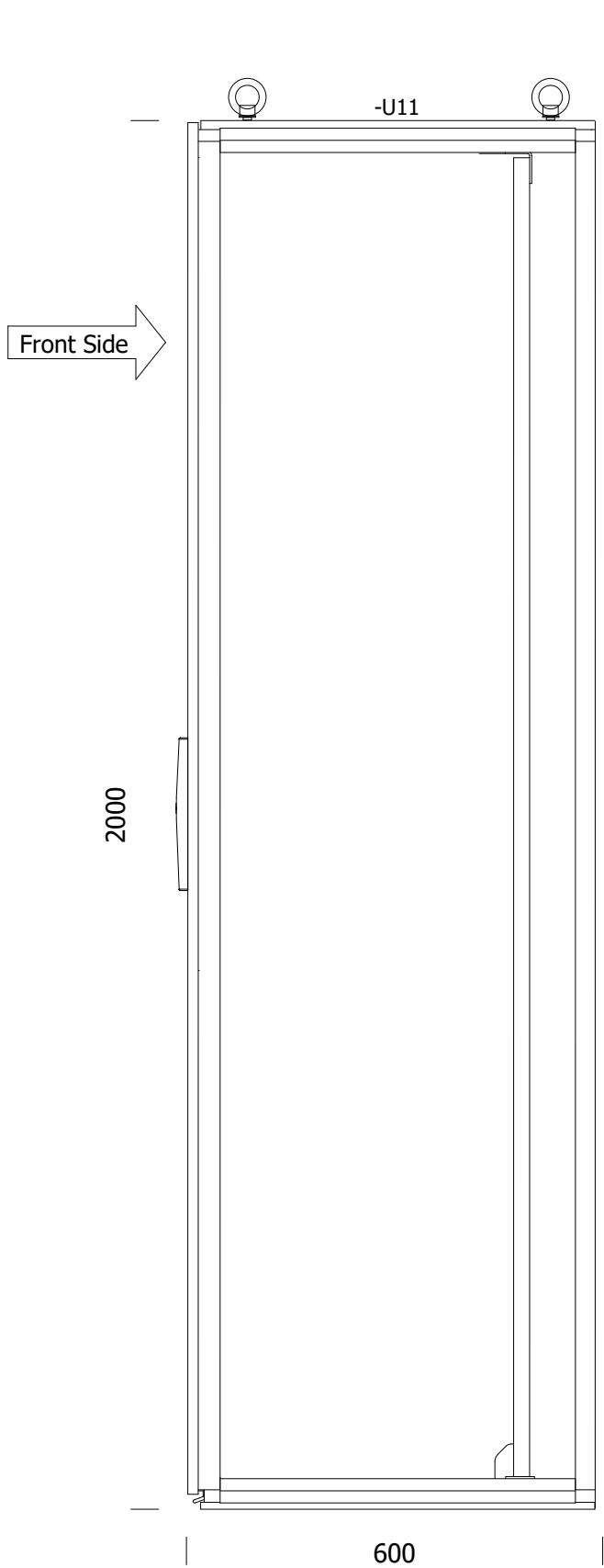
RESERVE

RESERVE

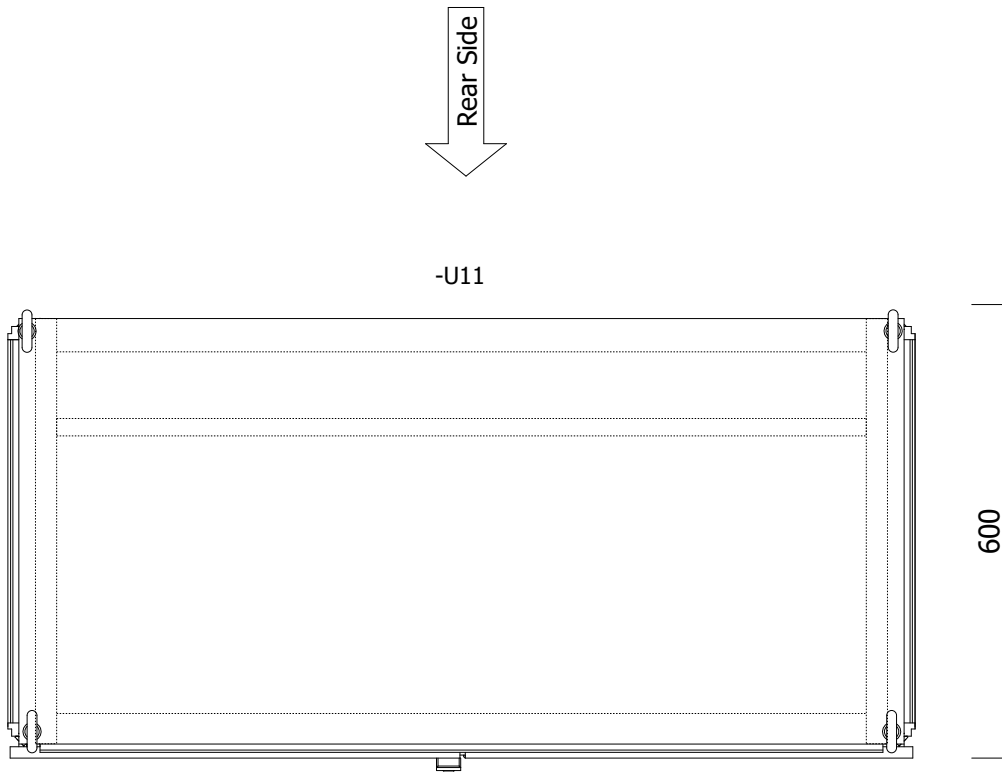




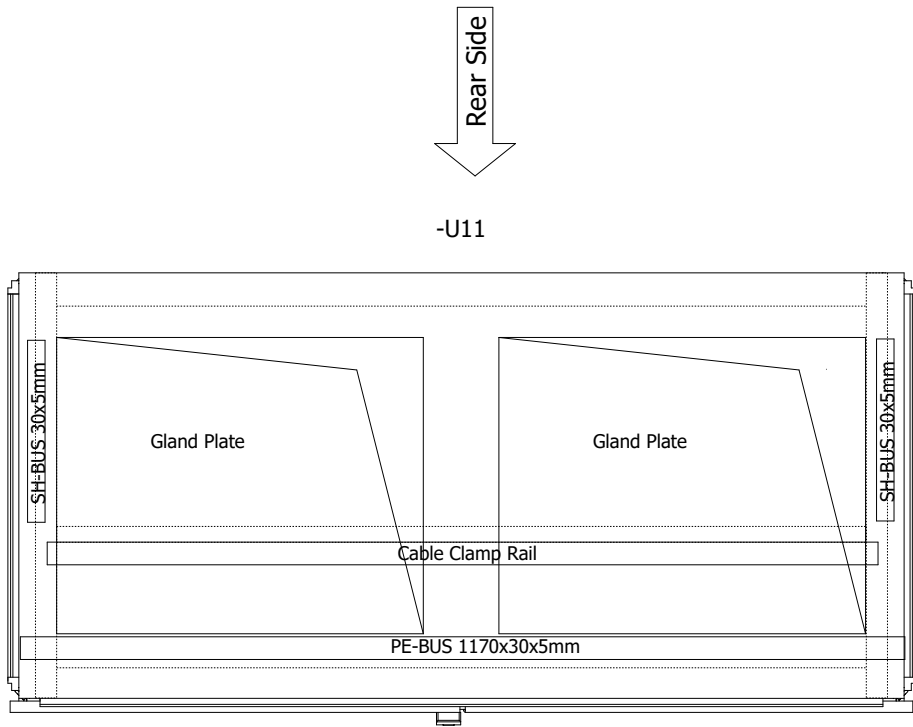
Front View



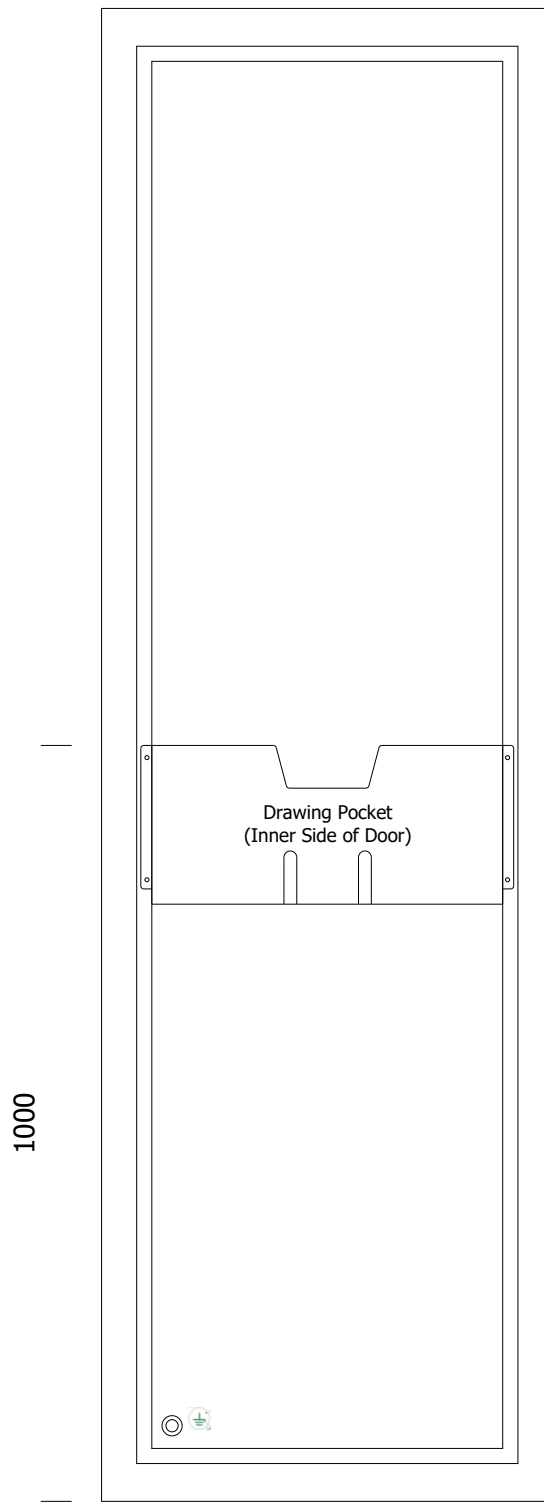
Right Side View



Top View



Bottom View



Front Right Door Inner Side View

Cabinet make	Rittal
Dimensions(WxHxD)	1200 x 2000 x 600
Protection Class	IP 54
Enclosure Frame, Doors and Roof	2.0mm Sheet steel
Mounting plate	2.0 mm Galvanized sheet steel
Gland plate	3.0 mm Galvanized sheet steel
Colour Shade	RAL 7035 & RAL 7022 For Base
Scale	A3=M1:10
Access Type	Front only

Enclosure legend

+RIO2_GENERAL		
Item number	Description	Device tag
Front Mounting Plate		
1	LED Batten	=44A10AB010.E003-E15
2	Position switch	=44A10AB010.E003-B15
3	Power supply	=44A10AB010.K001-T13
4	Power supply	=44A10AB010.K001-T15
5	Redundancy module	=44A10AB010.K001-T14
6	Circuit breaker	=44A10AB010.E003-F11
7	Circuit breaker	=44A10AB010.E001-F11
8	Circuit breaker	=44A10AB010.E001-F13
9	Circuit breaker	=44A10AB010.E001-F15
10	Circuit breaker	=44A10AB010.K001-F201
11	Circuit breaker	=44A10AB010.K001-F203
12	Circuit breaker	=44A10AB010.K001-F205
13	Circuit breaker	=44A10AB010.K001-F207
14	Circuit breaker	=44A10AB010.K001-F211
15	Circuit breaker	=44A10AB010.K001-F213
16	Circuit breaker	=44A10AB010.K001-F215
17	Circuit breaker	=44A10AB010.K001-F217
18	Circuit breaker	=44A10AB010.K001-F221
19	Circuit breaker	=44A10AB010.K001-F223
20	Circuit breaker	=44A10AB010.K001-F225
21	Circuit breaker	=44A10AB010.K001-F227
22	Circuit breaker	=44A10AB010.K001-F231
23	Circuit breaker	=44A10AB010.K001-F233
24	Circuit breaker	=44A10AB010.K001-F235
25	Circuit breaker	=44A10AB010.K001-F237
26	Circuit breaker	=44A10AB010.K001-F291

27	Circuit breaker	=44A10AB010.K001-F293
28	Circuit breaker	=44A10AB010.K001-F295
29	Circuit breaker	=44A10AB010.K001-F297
30	Universal switch socket	=44A10AB010.E003-S12
31	Base unit	=44A10AM010.M010-K122
32	Input module	=44A10AM010.M010-K122
33	Base unit	=44A10AM010.M010-K123
34	Input module	=44A10AM010.M010-K123
35	Base unit	=44A10AM010.M010-K124
36	Input module	=44A10AM010.M010-K124
37	Base unit	=44A10AM010.M010-K125
38	Input module	=44A10AM010.M010-K125
39	Base unit	=44A10AM010.M010-K126
40	Output module	=44A10AM010.M010-K126
41	Input module	=44A10AM010.M011-K122
42	Base unit	=44A10AM010.M011-K122
43	Input module	=44A10AM010.M011-K123
44	Base unit	=44A10AM010.M011-K123
45	Input module	=44A10AM010.M011-K124
46	Base unit	=44A10AM010.M011-K124
47	Input module	=44A10AM010.M011-K125
48	Base unit	=44A10AM010.M011-K125
49	Input module	=44A10AM010.M011-K126
50	Base unit	=44A10AM010.M011-K126
51	Base unit	=44A10AM010.M011-K127
52	Digital output module	=44A10AM010.M011-K127
53	Base unit	=44A10AM010.M011-K128
54	Digital output module	=44A10AM010.M011-K128
55	Output module	=44A10AM010.M012-K122

Enclosure legend

+RIO2_GENERAL		
Item number	Description	Device tag
Front Mounting Plate		
56	Base unit	=44A10AM010.M012-K122
57	Base unit	=44A10AM010.M012-K123
58	Output module	=44A10AM010.M012-K123
59	Base unit	=44A10AM010.M012-K124
60	Output module	=44A10AM010.M012-K124
61	Base unit	=44A10AM010.M012-K125
62	Output module	=44A10AM010.M012-K125
63	Input module	=44A10AM010.M013-K122
64	Base unit	=44A10AM010.M013-K122
65	Input module	=44A10AM010.M013-K123
66	Base unit	=44A10AM010.M013-K123
67	Input module	=44A10AM010.M013-K124
68	Base unit	=44A10AM010.M013-K124
69	Base unit	=44A10AM010.M013-K125
70	Output module	=44A10AM010.M013-K125
71	Digital output module	=44A10AM010.M013-K126
72	Base unit	=44A10AM010.M013-K126
73	Base unit	=44A10AM010.M013-K127
74	Analog input module	=44A10AM010.M013-K127
75	Base unit	=44A10AM010.M013-K128
76	Output module	=44A10AM010.M013-K128
77	PN bundle IM 155-6PN ST	=44A10AM010.M010-K120
78	PN bundle IM 155-6PN ST	=44A10AM010.M011-K120
79	PN bundle IM 155-6PN ST	=44A10AM010.M012-K120
80	PN bundle IM 155-6PN ST	=44A10AM010.M013-K120

Proposed cable gland								Cable no	Terminal strip							Cable no																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	</
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Terminal diagram

[illegible]

Terminal diagram

[illegible]

Revision	V1 -01	Approved			CCM 3 PLC UPGRADATION TATA STEEL LIMITED, GAMHARIA General PLC - Hydraulic & Discharge RIO (MCC Room)	Terminal diagram +RIO2_GENERAL-X5N4K127	Ord. - no.: 2300075403/J11, 09.07.2025	=RIO2_GENERAL	Page 31
Department	E&I	Checked					CAD - no.: ADPL/25-26/045/TSLG/01	=44A10AZ000.X001	From 32
Edited by	CHRU02	Designed					Client no.:	+RIO2_GENERAL	
Date	29.08.2025								

Terminal diagram

[illegible]

Parts list

Quantity	Designation / Description	Producer / Supplier	Order number Type number		Device tag M&C-Position
1 pc.	Circuit breaker	SIEMENS	5SL42067RC BETAGARD 5SL4,10KA		=44A10AB010.E001-F11
1 pc.	Circuit breaker	SIEMENS	5SL42067RC BETAGARD 5SL4,10KA		=44A10AB010.E001-F13
1 pc.	Circuit breaker	SIEMENS	5SL42107RC BETAGARD 5SL4,10KA		=44A10AB010.E001-F15
1 pc.	Position switch	SIEMENS	3SE5250-0BC05 SIRIUS 3SE5		=44A10AB010.E003-B15
1 pc.	LED Batten	REPUTED			=44A10AB010.E003-E15
1 pc.	Circuit breaker	SIEMENS	5SL42207RC BETAGARD 5SL4,10KA		=44A10AB010.E003-F11
1 pc.	Universal switch socket	REPUTED			=44A10AB010.E003-S12
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F201
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F203
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F205
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F207
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F211
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F213
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F215
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F217
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F221
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F223
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F225
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F227
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F231
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F233
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F235
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F237
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F291
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F293
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F295
1 pc.	Circuit breaker	SIEMENS	5SL41067RC BETAGARD 5SL4,10KA		=44A10AB010.K001-F297
1 pc.	Power supply	SIEMENS	6EP1336-1LB00 SITOP PSU100L		=44A10AB010.K001-T13
1 pc.	Redundancy module	SIEMENS	6EP4347-7RB00-0AX0 SITOP RED1200		=44A10AB010.K001-T14

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Parts list

Quantity	Designation / Description	Producer / Supplier	Order number Type number		Device tag M&C-Position
1 pc.	Power supply	SIEMENS	6EP1336-1LB00 SITOP PSU100L		=44A10AB010.K001-T15
1 pc.	PN bundle IM 155-6PN ST	SIEMENS	6ES7155-6AA02-0BN0 SIMATIC ET200SP IM155-6PN ST		=44A10AM010.M010-K120
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M010-K122
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M010-K122
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M010-K123
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M010-K123
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M010-K124
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M010-K124
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M010-K125
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M010-K125
1 pc.	Output module	SIEMENS	6ES7132-6BH01-0BA0 SIMATIC ET 200SP		=44A10AM010.M010-K126
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M010-K126
1 pc.	PN bundle IM 155-6PN ST	SIEMENS	6ES7155-6AA02-0BN0 SIMATIC ET200SP IM155-6PN ST		=44A10AM010.M011-K120
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M011-K122
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M011-K122
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M011-K123
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M011-K123
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M011-K124
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M011-K124
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M011-K125
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M011-K125
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M011-K126
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M011-K126
1 pc.	Digital output module	SIEMENS	6ES7132-6FD00-0BB1 SIMATIC ET 200SP		=44A10AM010.M011-K127
1 pc.	Base unit	SIEMENS	6ES7193-6BP20-0BB1 SIMATIC ET 200SP		=44A10AM010.M011-K127
1 pc.	Digital output module	SIEMENS	6ES7132-6FD00-0BB1 SIMATIC ET 200SP		=44A10AM010.M011-K128
1 pc.	Base unit	SIEMENS	6ES7193-6BP20-0BB1 SIMATIC ET 200SP		=44A10AM010.M011-K128
1 pc.	PN bundle IM 155-6PN ST	SIEMENS	6ES7155-6AA02-0BN0 SIMATIC ET200SP IM155-6PN ST		=44A10AM010.M012-K120
1 pc.	Output module	SIEMENS	6ES7132-6BH01-0BA0 SIMATIC ET 200SP		=44A10AM010.M012-K122

Parts list

Quantity	Designation / Description	Producer / Supplier	Order number Type number		Device tag M&C-Position
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M012-K122
1 pc.	Output module	SIEMENS	6ES7132-6BH01-0BA0 SIMATIC ET 200SP		=44A10AM010.M012-K123
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M012-K123
1 pc.	Output module	SIEMENS	6ES7132-6BH01-0BA0 SIMATIC ET 200SP		=44A10AM010.M012-K124
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M012-K124
1 pc.	Output module	SIEMENS	6ES7132-6BH01-0BA0 SIMATIC ET 200SP		=44A10AM010.M012-K125
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M012-K125
1 pc.	PN bundle IM 155-6PN ST	SIEMENS	6ES7155-6AA02-0BN0 SIMATIC ET200SP IM155-6PN ST		=44A10AM010.M013-K120
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M013-K122
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M013-K122
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M013-K123
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M013-K123
1 pc.	Input module	SIEMENS	6ES7131-6BH01-0BA0 SIMATIC ET200SP		=44A10AM010.M013-K124
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0BA0 DP ET200SP 193		=44A10AM010.M013-K124
1 pc.	Output module	SIEMENS	6ES7132-6BF01-0BA0 SIMATIC ET200SP 132		=44A10AM010.M013-K125
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M013-K125
1 pc.	Digital output module	SIEMENS	6ES7132-6FD00-0BB1 SIMATIC ET 200SP		=44A10AM010.M013-K126
1 pc.	Base unit	SIEMENS	6ES7193-6BP20-0BB1 SIMATIC ET 200SP		=44A10AM010.M013-K126
1 pc.	Analog input module	SIEMENS	6ES7134-6HD01-0BA1 SIMATIC ET 200SP		=44A10AM010.M013-K127
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M013-K127
1 pc.	Output module	SIEMENS	6ES7135-6HD00-0BA1 SIMATIC ET200SP		=44A10AM010.M013-K128
1 pc.	Base unit	SIEMENS	6ES7193-6BP00-0DA0 DP ET200SP 193		=44A10AM010.M013-K128